Laplace Transformation

$$\mathcal{L}\{f(t)\} = \int_0^\infty f(t)e^{-st}\,dt = Fs$$

$$f(t) = \frac{F(s)}{\mathcal{L}}$$

$$\downarrow = \mathcal{L}^{-1}\{F(s)\}$$

inverse laplace transformation

1. $\mathcal{L}\{any\ c\} = \frac{any\ c}{s}$
2. $\mathcal{L}\{t^n\} = \frac{n!}{s^{n+1}}$

Example -

$n = 0, 1, 2, 3, 4 \ldots$

$3! = 3 \times 2 \times 1$

$5! = 5 \times 4 \times 3 \times 2 \times 1$

$\mathcal{L}\{t^3\} = \frac{3!}{s^4}$

3. $\mathcal{L}\{e^{at}\} = \frac{1}{s-a}$

Example:

(**) $\mathcal{L}\{e^3\} = \frac{e^3}{s}$

(*) $\mathcal{L}\{e^{2t+6}\}$

$= \mathcal{L}\{e^{2t} \cdot e^6\}$

$= \frac{1}{s-2} \cdot e^6$

(*) $\mathcal{L}\{e^{-4t}\} = \frac{1}{s+4}$

4. $\mathcal{L}\{\sin at\} = \frac{a}{s^2+a^2}$
5. $\mathcal{L}\{\sinh at\} = \frac{a}{s^2-a^2}$
6. $\mathcal{L}\{\cos at\} = \frac{s}{s^2+a^2}$
7. $\mathcal{L}\{\cosh at\} = \frac{s}{s^2-a^2}$

Problem set - 1.1

(1) (a) $f(t) = e^{5t}$

$$\mathcal{L}\{e^{5t}\} = \frac{1}{s-a}$$

$$= \frac{1}{s-5}$$

(b) $f(t) = e^{-2t}$

$$\mathcal{L}\{e^{-2t}\} = \frac{1}{s-a}$$

$$= \frac{1}{s-(-2)}$$

$$= \frac{1}{s+2}$$

(c) $f(t) = \sin(\omega t + \theta)$

$$= \sin(\omega t)\cos\theta + \cos(\omega t)\sin\theta$$

$$\mathcal{L}\{\sin(\omega t+\theta)$$

$$= \cos\theta \cdot \mathcal{L}\{\sin(\omega t)\} + \sin\theta \cdot \mathcal{L}\{\cos(\omega t)\}$$

$$= \cos\theta \cdot \frac{\omega}{s^2+\omega^2} + \sin\theta \cdot \frac{s}{s^2+\omega^2}$$

$$= \frac{\omega\cos\theta + s\sin\theta}{s^2+\omega^2}$$

(d) $f(t) = 1.5\sin\left(3t - \frac{\pi}{2}\right) = 1.5\mathcal{L}\{\sin 3t \cos\pi/2 - \sin\frac{\pi}{2}\cos 3t\}$

$$= -1.5\cos(3t)$$

$$\mathcal{L}\{-1.5\cos(3t)\} = -1.5\,\frac{s}{s^2+9}$$

$$= \frac{-1.5s}{s^2+9}$$

$$= -\frac{3s}{2(s^2+9)}$$

(11) (e) $f(t) = e^{2t}\sinh t$

$\mathcal{L}\{e^{2t}\sinh t\}$

here,

$F(s) = \mathcal{L}\{\sinh t\}$

$= \frac{1}{s^2-1}$

So, $\mathcal{L}\{e^{2t}\sinh t\} = F(s-a)$

$= \frac{1}{(s-2)^2-1}$

(f) $f(t) = e^{-t}\sin 4t$

Here,

$F(s) = \mathcal{L}\{\sinh 4t\}$

$= \frac{4}{s^2-4^2}$

$= \frac{4}{s^2-16}$

So,

$\mathcal{L}\{e^{-t}\sinh 4t\} = F(s-a)$

$= \frac{4}{(s-(-1))^2-16}$

$= \frac{4}{(s+1)^2-16}$

(g) $f(t) = e^{2t}\cos 3t$

here,

$F(s) = \mathcal{L}\{\cos 3t\}$

$= \frac{s}{s^2+9}$

Now,

$\mathcal{L}\{e^{2t}\cos 3t\} = F(s-a)$

$= \frac{s-2}{(s-2)^2+9}$

(h) $f(t) = t^{10} e^{-7t}$

here,

$$F(s) = \mathcal{L}\{t^{10}\}$$

$$= \frac{10!}{s^{11}}$$

Now,

$$\mathcal{L}\{t^{10} e^{-7t}\} = F(s-a)$$

$$= \frac{10!}{(s-(-7))^{11}}$$

$$= \frac{10!}{(s+7)^{11}}$$

(i) $f(t) = \cosh 5t\, e^{3t}$

here,

$$F(s) = \mathcal{L}\{\cosh 5t\}$$

$$= \frac{s}{s^2-25}$$

Now,

$$\mathcal{L}\{\cosh 5t\, e^{3t}\} = F(s-a)$$

$$= \frac{s-3}{(s-3)^2-25}$$

(iii) (j) $f(t) = t\sin 2t$

here,

$F(s) = \mathcal{L}\{\sin 2t\}$

$= \frac{2}{s^2+4}$

Now,

$\mathcal{L}\{t\sin 2t\}$

$= (-1)^1 \frac{d^1}{ds^1}[F(s)]$

$= -\frac{d}{ds}\left(\frac{2}{s^2+4}\right)$

$= -\left\{\frac{-2.2s}{(s^2+4)^2}\right\}$

$= \frac{4s}{(s^2+4)^2}$

(k) $f(t) = t\cos 3t$

here,

$F(s) = \mathcal{L}\{\cos 3t\}$

$= \frac{s}{s^2+9}$

Now,

$\mathcal{L}\{t\cos 3t\}$

$= (-1)^1 \frac{d^1}{ds^1}[F(s)]$

$= -\frac{d}{ds}\left(\frac{s}{s^2+9}\right)$

$= -\frac{(s^2+9)\frac{d}{ds}(s) - s.\frac{d}{ds}(s^2+9)}{(s^2+9)^2}$

$= -\frac{s.2s - (s^2+9)}{(s^2+9)^2}$

$= \frac{2s^2}{(s^2+9)^2} - \frac{(s^2+9)}{(s^2+9)^2}$

$= \frac{2s^2 \quad s^2}{}$

2. (I) $\mathcal{L}\{e^{at}\}$

$f(t) = \mathcal{L}\{e^{at}\}$

Now,

$$\mathcal{L}\{f(t)\} = \int_0^\infty e^{-st} \cdot f(t)\, dt$$

$$= \int_0^\infty e^{-st} \cdot e^{at} \cdot dt$$

$$= \int_0^\infty e^{t(-s+a)} \cdot dt$$

$$= \frac{1}{-s+a} \left[e^{t(-s+a)}\right]_0^\infty$$

$$= \frac{1}{-s+a} [0 - 1]$$

$$= \frac{1}{s-a}$$

(II) $\mathcal{L}\{t^2 e^{at}\}$

$f(t) = \mathcal{L}\{t^2 e^{at}\}$

Now,

$$\mathcal{L}\{f(t)\} = \int_0^\infty e^{-st} \cdot f(t) \cdot dt$$

$$= \int_0^\infty e^{-st} \cdot t^2 e^{at} \cdot dt$$

$$= \int_0^\infty e^{-st} \cdot e^{at} \cdot t^2 dt$$

$$= \int_0^\infty e^{t(-s+a)} \cdot t^2 dt$$

$$= \int_0^\infty e^{-(s-a)t} \cdot t^2 dt$$

$$= \frac{2!}{(s-a)^3}$$

$$= \frac{2}{(s-a)^3}$$

(c) $\mathcal{L}\{f(t)\} = \int_0^\infty e^{-st} \sin 2t\, dt$

$= \left[\frac{e^{-st}(-s\sin 2t - 2\cos 2t)}{s^2+4}\right]_0^\infty$

$= 0 - \left(\frac{-2}{s^2+4}\right)$

$= \frac{2}{s^2+4}$

$\int e^{ax}\sin(bx)\,dx = \frac{e^{ax}(a\sin bx - b\cos bx)}{a^2+b^2}$

(d) $\mathcal{L}\{\cos at\}$

$\mathcal{L}\{f(t)\} = \mathcal{L}\{e^{-st}\cdot\cos at\cdot dt\}$

Now,

$\mathcal{L}\{f(t)\} = \int_0^\infty e^{-st}\cdot\cos at\cdot dt$

$= \left[\frac{e^{-st}(-s\cos at + a\sin at)}{a^2+s^2}\right]_0^\infty$

$= 0 - \left(\frac{-s}{s^2+a^2}\right)$

$= \frac{s}{s^2+a^2}$

(e) $\mathcal{L}\{\cos 3t\}$

$\mathcal{L}\{f(t)\} = \mathcal{L}\{e^{-st}\cos 3t\cdot dt\}$

Now,

$\mathcal{L}\{f(t)\} = \int_0^\infty e^{-st}\cdot\cos 3t\cdot dt$

$= \left[\frac{e^{-st}(-s\cos 3t + 3\sin 3t)}{s^2+9}\right]_0^\infty$

$= 0 - \left(\frac{-s}{s^2+9}\right)$

$= \frac{s}{s^2+9}$

$\int e^{ax}\cdot\cos(bx)\,dx = \frac{e^{ax}(a\cos bx + b\sin bx)}{a^2+b^2}$

② (11) (a) $f(t) = 3t + 12$

$$\mathcal{L}\{f(t)\} = \mathcal{L}\{3t + 12\}$$
$$= \mathcal{L}\{3t\} + \mathcal{L}\{12\}$$
$$= 3\mathcal{L}\{t\} + \mathcal{L}\{12\}$$
$$= 3 \cdot \frac{1}{s^{1+1}} + \frac{12}{s}$$
$$= \frac{3}{s^2} + \frac{12}{s}$$
$$= \frac{3 + 12s}{s^2}$$

(b) $f(t) = \sinh t\, e^{2t}$

We know,

$$\mathcal{L}\{e^{at} \cdot f(t)\} = F(s-a)$$
$$\therefore \mathcal{L}\{e^{2t} \cdot \sinh t\} = F(s-2) \quad \text{—①}$$

Now,

$$f(t) = \sinh t$$
$$F(s) = \mathcal{L}\{f(t)\}$$
$$= \mathcal{L}\{\sinh t\}$$
$$= \frac{1}{s^2 - 1^2}$$

From ① equation,

$$\mathcal{L}\{e^{2t} \cdot \sinh t\} = \frac{1}{(s-2)^2 - 1}$$

(c) $f(t) = (5-6t)^2$

$f(t) = 5^2 - 2.5.6t + (6t)^2$

$\therefore \mathcal{L}\{f(t)\} = \mathcal{L}\{25 - 60t + 36t^2\}$

$= \mathcal{L}\{25\} - \mathcal{L}\{60t\} + \mathcal{L}\{36t^2\}$

$= \frac{25}{s} - \frac{60}{s^2} + \frac{2.36}{s^3}$

$= \frac{25}{s} - \frac{60}{s^2} + \frac{72}{s^3}$

(d) $f(t) = e^{-t} \sinh 4t$

We know,

$\mathcal{L}\{e^{at}.f(t)\} = F(s-a)$

$\therefore \mathcal{L}\{e^{-t}.\sinh 4t\} = F(s+1)$ — ①

Now,

$f(t) = \sinh 4t$

$F(s) = \mathcal{L}\{f(t)\}$

$= \mathcal{L}\{\sinh 4t\}$

$= \frac{4}{s^2 - 4^2}$

from equation ①,

$\mathcal{L}\{e^{-t}.\sinh 4t\} = \frac{4}{(s+1)^2 - 16}$

(e) $f(t) = \cos^2 \omega t$

$\mathcal{L}\{f(t)\} = \mathcal{L}\{\cos^2 \omega t\}$ $[\because 2\cos^2\theta = 1 + \cos 2\theta]$

$= \mathcal{L}\{\frac{1}{2}(1 + \cos 2\omega t)\}$

$= \frac{1}{2}\mathcal{L}\{1\} + \frac{1}{2}\mathcal{L}\{\cos 2\omega t\}$

$= \frac{1}{2s} + \frac{s}{2(s^2 + 4\omega^2)}$

(f) $f(t) = t^5 e^{2t}$

We know,

$\mathcal{L}\{e^{at} \cdot f(t)\} = F(s-a)$

$\mathcal{L}\{e^{2t} \cdot t^5\} = F(s-2)$ —①

Now,

~~$\mathcal{L}\{e^{2t} \cdot t^5\}$~~

$f(t) = t^5 e^{2t}$

$= F(s-2)$

$= \mathcal{L}\{f(t-2)\}$

$= \mathcal{L}\{(t-2)^5\}$

$= \frac{5!}{(s-2)^6}$

(g) $f(t) = \cos \pi t$

$\mathcal{L}\{f(t)\} = \mathcal{L}\{\cos \pi t\}$

$= \frac{s}{s^2 + \pi^2}$

(h) $f(t) = \sin^2\left(\frac{1}{2}\pi t\right)$

$\Rightarrow \mathcal{L}\{f(t)\} = \mathcal{L}\{\sin^2\left(\frac{1}{2}\pi t\right)\}$ $[\because \sin^2\theta = \frac{1}{2}(1-\cos 2\theta)]$

$= \mathcal{L}\{\frac{1}{2}\left(1 - \cos 2 \times \frac{1}{2}\pi t\right)\}$

$= \mathcal{L}\{\frac{1}{2}(1 - \cos \pi t)\} = \frac{1}{2}\mathcal{L}\{1\} - \frac{1}{2}\mathcal{L}\{\cos \pi t\}$

$= \frac{1}{2s} - \frac{s}{2(s^2 + \pi^2)}$

(i) $f(t) = \frac{1 + \cos(2\omega t)}{2}$

$\Rightarrow \mathcal{L}\{f(t)\} = \mathcal{L}\frac{1 + \cos(2\omega t)}{2}$

$= \frac{1}{2}\mathcal{L}\{1\} + \frac{1}{2}\mathcal{L}\{\cos(2\omega t)\}$

$= \frac{1}{2} \cdot \frac{1}{s} + \frac{1}{2} \cdot \frac{s}{s^2 + 4\omega^2}$

$= \frac{1}{2}\left[\frac{1}{s} + \frac{s}{s^2 + 4\omega^2}\right]$

(j) $f(t) = 12te^{-3t}$

We know,

$$\mathcal{L}\{t^n . f(t)\} = (-1)^n \cdot \frac{d^n}{ds^n}[F(s)]$$

$$= (-1)^1 \times 12 \times \frac{d}{ds}\mathcal{L}\{e^{-3t}\}$$

$$= -12 \frac{d}{ds} \times \frac{1}{s+3}$$

$$= -12 \times \frac{-1}{(s+3)^2}$$

$$= \frac{12}{(s+3)^2}$$

(k) $f(t) = 5\cosh 2t - 3\sinh t$

$$\mathcal{L}\{f(t)\} = 5\mathcal{L}\{\cosh 2t\} - 3\mathcal{L}\{\sinh t\}$$

$$= 5 \times \frac{s}{s^2 - 2^2} - 3 \times \frac{1}{s^2 - 1^2}$$

$$= \frac{5s}{s^2 - 4} - \frac{3}{s^2 - 1}$$

(l) $f(t) = t^2 e^{-3t}$

We know,

$$\mathcal{L}\{e^{at} . f(t)\} = F(s-a)$$

$$\therefore \mathcal{L}\{e^{-3t} . t^2\} = F(s+3) \quad \text{—①}$$

Now,

$$F(s) = \mathcal{L}\{f(t)\}$$

$$= \mathcal{L}\{t^2\}$$

$$= \frac{2}{s^3}$$

from equation ①,

$$\mathcal{L}\{e^{-3t} t^2\} = \frac{2}{(s+3)^3}$$

(m) $f(t) = e^{-t}(\cos 4t - 2\sin 4t)$

$\Rightarrow f(t) = e^{-t}\cos 4t - e^{-t}\cdot 2\sin 4t$

We know,

$\mathcal{L}\{f(t)\cdot e^{at}\} = F(s-a)$

$\therefore \mathcal{L}\{e^{-t}(\cos 4t - 2\sin 4t)\} = F_1(s+1) - F_2(s+1)$ — ①

Now,

$F_1(s) = \mathcal{L}\{\cos 4t\} = \frac{s}{s^2+4^2}$

$F_2(s) = \mathcal{L}\{2\sin 4t\} = 2\times\frac{4}{s^2+4^2} = \frac{8}{s^2+4^2}$

Now, from equ ① →

$\mathcal{L}\{e^{-t}(\cos 4t - 2\sin 4t)\} = \frac{(s+1)}{(s+1)^2+16} - \frac{8}{(s+1)^2+4^2}$

$= \frac{(s+1-8)}{(s+1)^2+16}$

$= \frac{s-7}{(s+1)^2+16}$

(n) $f(t) = t\cos t + \sin t$ — ①

We know,

$\mathcal{L}\{t^n\cdot f(t)\} = (-1)^n\cdot\frac{d^n}{ds^n}[F(s)]$

$\therefore \mathcal{L}\{t\cdot\cos t\} = (-1)^1\frac{d}{ds}\mathcal{L}\{\cos t\}$

$= -\frac{d}{ds}\times\frac{s}{s^2+1}$

$= -\left[\frac{(s^2+1)\frac{d}{ds}\times s - s\times\frac{d}{ds}\times(s^2+1)}{(s^2+1)^2}\right]$

$= -\left[\frac{(s^2+1)-s\cdot 2s}{(s^2+1)^2}\right]$

$= \frac{2s^2-s^2-1}{(s^2+1)^2}$

$= \frac{s^2-1}{(s^2+1)^2}$

Again, $\mathcal{L}\{\sin t\} = \frac{1}{s^2+1}$

So, from the equ ① →

$\mathcal{L}\{t\cdot\cos t + \sin t\} = \frac{s^2-1}{(s^2+1)^2} + \frac{1}{s^2+1}$

(O) $f(t) = 2e^{-3t}\cos 4t$

$\Rightarrow \mathcal{L}\{e^{-3t}\cos 4t\} = \frac{s-(-3)}{(s-(-3))^2+4^2}$

$= \frac{s+3}{(s+3)^2+16}$

$\mathcal{L}\{2e^{-3t}\cos 4t\} = \frac{2(s+3)}{(s+3)^2+16}$

(P) $f(t) = e^t(t^2+\sin 3t)$

$\mathcal{L}\{f(t)\} = \mathcal{L}\{e^t.t^2\}$ & $\mathcal{L}\{e^t\sin 3t\} = F_1(s-1)+F_2(s-1)$ — ①

$F_1(s) = \mathcal{L}\{e^t.t^2\}$

$= \frac{2!}{(s-1)^3}$

$= \frac{2}{(s-1)^3}$

$F_2(s) = \mathcal{L}\{e^t\sin 3t\}$

$= \frac{3}{(s-1)^2+9}$

① →

$\mathcal{L}\{e^t(t^2+\sin 3t)\} = \frac{2}{(s-1)^3} + \frac{3}{(s-1)^2+9}$

(Q) $f(t) = 3e^{-4t}\sin 5\pi t$

$\Rightarrow \mathcal{L}\{e^{-4t}\sin(5\pi t)\} = \frac{5\pi}{(s-(-4))^2+(5\pi)^2}$

$= \frac{5\pi}{(s+4)^2+25\pi^2}$

$\therefore \mathcal{L}\{3e^{-4t}\sin 5\pi t\} = \frac{15\pi}{(s+4)^2+25\pi^2}$

(R) $f(t) = e^{-3t}(t+4)^2$

$(t+4)^2 = t^2+8t+16$

$\mathcal{L}\{(t+4)^2\} = \mathcal{L}\{t^2\} + \mathcal{L}\{8t\} + \mathcal{L}\{16\}$

$= \frac{2}{s^3} + \frac{8}{s^2} + \frac{16}{s}$

$\therefore \mathcal{L}\{e^{-3t}(t+4)^2\} = \frac{2}{(s+3)^3} + \frac{8}{(s+3)^2} + \frac{16}{(s+3)}$

① (iv) (l) $f(t) = \begin{cases} 1-t ; & 0<t<1 \\ 0 ; & t>1 \end{cases}$

Soln:

$$\mathcal{L}\{f(t)\}$$

$$= \int_0^\infty e^{-st} f(t)\,dt$$

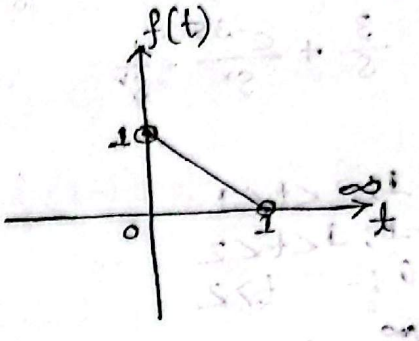
$$= \int_0^1 e^{-st}(1-t)\,dt + \int_1^\infty e^{-st} \cdot 0\,dt$$

$$= \left((1-1)\frac{e^{-s}}{-s} + \frac{e^{-s}}{s^2}\right) - \left(\frac{e^0}{-s} + \frac{e^0}{s^2}\right)$$

$$= \frac{e^{-s}}{s^2} + \frac{1}{s} - \frac{1}{s^2}$$

$$= \frac{e^{-s} + s - 1}{s^2}$$

+	$1-t$	e^{-st}
−	-1	$\frac{e^{-st}}{-s}$
+	0	$\frac{e^{-st}}{s^2}$



(m) $f(t) = \begin{cases} 1 ; & 0<t<2 \\ 0 ; & t>2 \end{cases}$

$$\Rightarrow \mathcal{L}\{f(t)\} = \int_0^\infty e^{-st} \cdot f(t)\,dt$$

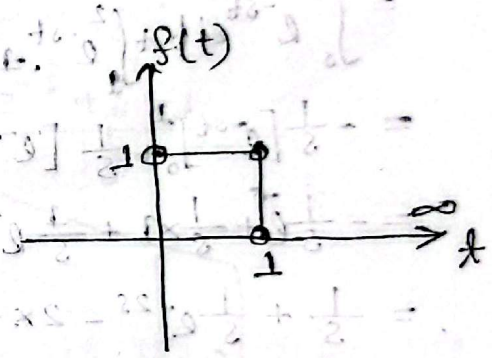
$$= \int_0^2 e^{-st} \cdot 1\,dt + \int_2^\infty e^{-st} \cdot 0\,dt$$

$$= -\frac{1}{s}\left[e^{-st}\right]_0^2 + 0$$

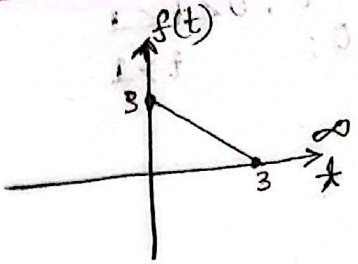
$$= -\frac{1}{s}\left[e^{-2s} - e^0\right]$$

$$= \frac{1}{s}\left[1 - e^{-2s}\right]$$

$$= \frac{1}{s} - \frac{e^{-2s}}{s}$$



(n) $f(t) = \begin{cases} 3-t ; & 0<t<3 \\ 0 ; & t>3 \end{cases}$



$\Rightarrow \mathcal{L}\{f(t)\} = \int_0^\infty e^{-st} \cdot f(t) \cdot dt$

$= \int_0^3 e^{-st}(3-t)\,dt + \int_3^\infty e^{-st} \cdot 0 \cdot dt$

$= \left[(3-t)\cdot\frac{-e^{-st}}{s} + \frac{e^{-st}}{s^2}\right]_0^3 + 0$

$= \frac{3}{s} + \frac{e^{-3s}}{s^2} - \frac{1}{s^2}$

+	3-t	e^{-st}
−	−1	$\frac{e^{-st}}{-s}$
+	0	$\frac{e^{-st}}{s^2}$

(o) $f(t) = \begin{cases} 1 ; & 0<t<1 \\ -1 ; & 1<t<2 \\ 0 ; & t>2 \end{cases}$

$\Rightarrow \mathcal{L}\{f(t)\} = \int_0^\infty e^{-st} \cdot f(t) \cdot dt$

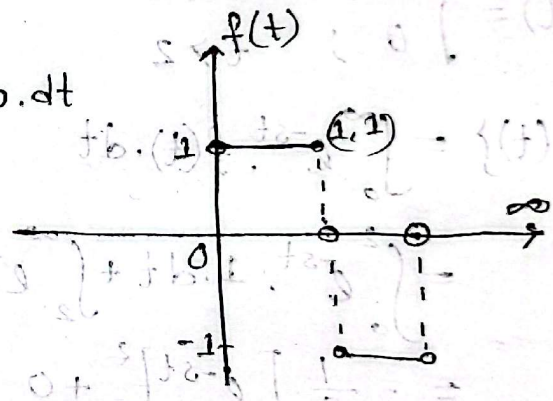
$= \int_0^1 e^{-st}\,dt + \int_1^2 e^{-st} \cdot -1 \cdot dt + \int_2^\infty e^{-st} \cdot 0 \cdot dt$

$= -\frac{1}{s}\left[e^{-st}\right]_0^1 + \frac{1}{s}\left[e^{-st}\right]_1^2 + 0$

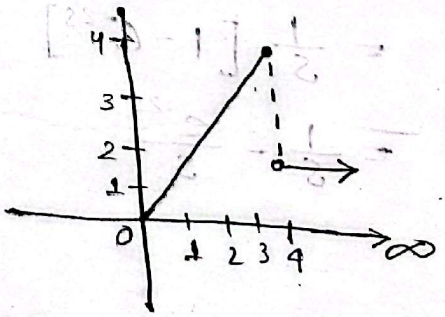
$= -\frac{1}{s}e^{-s} + \frac{1}{s} \times 1 + \frac{1}{s}e^{-2s} - \frac{1}{s}e^{-s}$

$= \frac{1}{s} + \frac{1}{s}e^{-2s} - 2 \times \frac{1}{s}e^{-s}$

$= \frac{1}{s} + \frac{1}{s}e^{-2s} - \frac{2}{s}e^{-s}$



(p) $f(t) = \begin{cases} t ; & 0 \le t \le 4 \\ 1 ; & t>4 \end{cases}$



$\Rightarrow \mathcal{L}\{f(t)\} = \int_0^\infty e^{-st} \cdot f(t)\,dt$

$= \int_0^4 e^{-st} \cdot t \cdot dt + \int_4^\infty e^{-st} \cdot 1 \cdot dt$

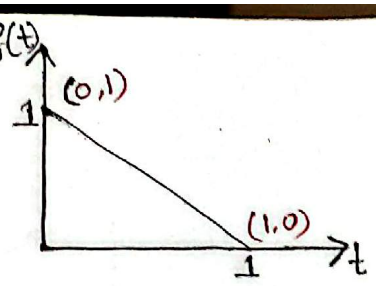
$= \left[t \times -\frac{1}{s}e^{-st} - \frac{1}{s^2}e^{-st}\right]_0^4 + \left[-\frac{1}{s}e^{st}\right]_4^\infty$

$= -\frac{1}{s}\left[4e^{-4s} - 0 \times e^0\right] - \frac{1}{s^2}\left[e^{-4s} - e^0\right] - \frac{1}{s}\left[e^{-\infty s} - e^{-4s}\right]$

$= -\frac{4e^{-4s}}{s} - \frac{e^{-4s}}{s^2} + \frac{1}{s^2} + \frac{1}{s}e^{-4s}$

$= \frac{1}{s^2} - \frac{3e^{-4s}}{5} - \frac{e^{-4s}}{s^2}$

2. (iii) (a)



We know,

$$\frac{x-x_1}{x_1-x_2} = \frac{y-y_1}{y_1-y_2}$$

$$\Rightarrow \frac{x-0}{0-1} = \frac{y-1}{1-0}$$

$$\Rightarrow \frac{x}{-1} = \frac{y-1}{1}$$

$$\Rightarrow -y+1 = x$$

$$\therefore y = 1-x$$

Now,

$$f(t) = \begin{cases} 1-t ; & 0<t<1 \\ 0 ; & t>1 \end{cases}$$

We know,

$$\therefore \mathcal{L}\{f(t)\} = \int_0^\infty e^{-st} \cdot f(t)\,dt$$

$$= \int_0^1 e^{-st} \cdot (1-t)\,dt + \int_1^\infty e^{-st} \cdot 0 \cdot dt$$

$$= (1-t)\int e^{-st}\,dt - \int \left\{\frac{d}{dt}(1-t) \cdot \int e^{-st}\,dt\right\} dt + 0$$

$$= \left[(1-t)x - \frac{1}{s}e^{-st}\right]_0^1 - \left[\int -1x - \frac{1}{s}e^{-st}\,dt\right]$$

$$= \frac{1}{s} - \left[-\frac{1}{s^2}e^{-st}\right]_0^1$$

$$= \frac{1}{s} + \frac{1}{s^2}\left[e^{-st}\right]_0^1$$

$$= \frac{1}{s} + \frac{e^{-s}}{s^2} - \frac{1}{s^2}$$

$$\mathcal{L}\{f(t)\} = \int_0^\infty e^{-st} f(t)\,dt$$

$$= \int_0^1 e^{-st}(1-t)\,dt + \int_1^\infty e^{-st} \cdot 0\,dt$$

$$= \left[\left\{(1-t)\frac{e^{-st}}{-s}\right\} - \left\{(-1)\frac{e^{-st}}{s^2}\right\}\right]_0^1$$

$$= \left[\left\{(1-t)\frac{e^{-st}}{-s}\right\} + \frac{e^{-st}}{s^2}\right]_0^1$$

$$= \left[\left\{(1-1)\frac{e^{-s}}{-s} + \frac{e^{-s}}{s^2}\right\} - \left\{(1-0)\frac{e^0}{-s} + \frac{e^0}{s^2}\right\}\right]$$

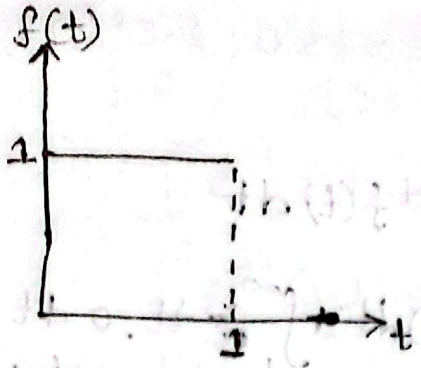
$$= \left[0 + \frac{e^{-s}}{s^2} - \left(\frac{1}{-s} + \frac{1}{s^2}\right)\right]$$

$$= \frac{e^{-s}}{s^2} + \frac{1}{s} - \frac{1}{s^2}$$

$$= \frac{e^{-s} + s - 1}{s^2}$$

+	$1-t$	e^{-st}
−	-1	$\frac{e^{-st}}{-s}$
+	0	$\frac{e^{-st}}{s^2}$

(b)



$\therefore f(t) = \begin{cases} 1 ; & 0 < t < 1 \\ 0 ; & t > 1 \end{cases}$

We know,

$\mathcal{L}\{f(t)\} = \int_0^\infty e^{-st} \cdot f(t) \cdot dt$

$= \int_0^1 e^{-st} \cdot 1 \cdot dt + \int_1^\infty e^{-st} \cdot 0 \cdot dt$

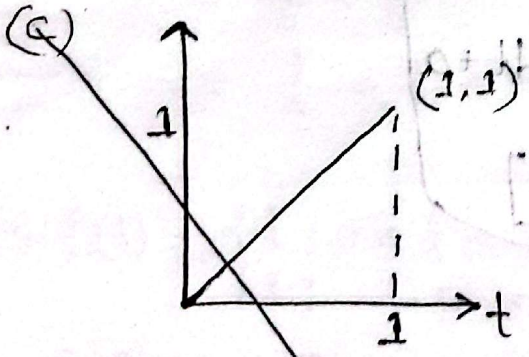
$= 1 \times -\frac{1}{s} \left[e^{-st}\right]_0^1 + 0$

$= -\frac{1}{s} \left[e^{-s\times 1} - e^0\right]$

$= -\frac{1}{s} \left[e^{-s} - 1\right]$

$= \frac{1}{s} - \frac{e^{-s}}{s}$

(c)



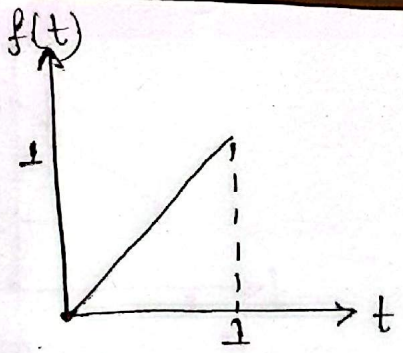
$\therefore f(t) = \begin{cases} t ; & 0 < t < 1 \\ 0 ; & t > 1 \end{cases}$

Now,

$\mathcal{L}\{f(t)\} = \int_0^1 e^{-st} \cdot t \cdot dt + \int_1^\infty e^{-st} \cdot 0 \cdot dt$

$= t \int_0^1 e^{-st} \cdot dt - \int_0^\infty 2 \frac{d}{dt} \int e^{-st} \cdot dt$

(c)



$$f(t) = \begin{cases} t\ ; & 0<t<1 \\ 0\ ; & t>1 \end{cases}$$

$$\mathcal{L}\{f(t)\} = \int_0^\infty e^{-st} f(t)\,dt$$

$$= \int_0^1 e^{-st}.t.dt + \int_1^\infty e^{-st}.0.dt$$

$$= \left[t.-\frac{1}{s}e^{-st} - \frac{1}{s^2}e^{-st}\right]_0^1 + 0$$

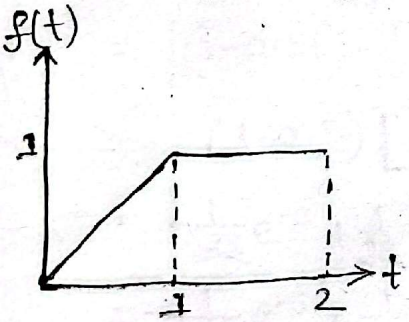
$$= -\frac{1}{s}\left[te^{-st}\right]_0^1 - \frac{1}{s^2}\left[e^{-st}\right]_0^1$$

$$= -\frac{1}{s}\left[e^{-s}\right] - \frac{1}{s^2}\left[e^{-s} - e^{0}\right]$$

$$= -\frac{e^{-s}}{s} - \frac{e^{-s}}{s^2} + \frac{1}{s^2}$$

$$= \frac{1}{s^2} - \frac{e^{-s}}{s^2} - \frac{e^{-s}}{s}$$

(d)



$$f(t) = \begin{cases} t\ ; & 0<t<1 \\ 1\ ; & 1<t<2 \\ 0\ ; & t>2 \end{cases}$$

$$\mathcal{L}\{f(t)\} = \int_0^\infty e^{-st} f(t)\,dt$$

$$= \int_0^1 e^{-st}.t.dt + \int_1^2 e^{-st}.1.dt + \int_2^\infty e^{-st}.0.dt$$

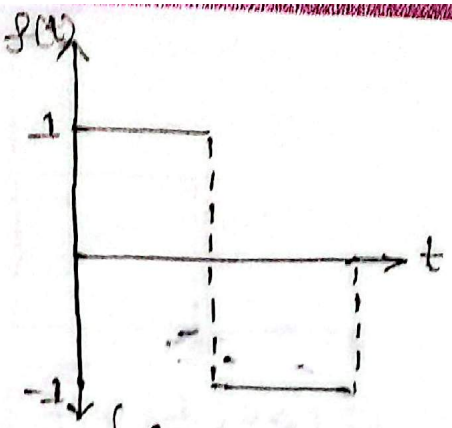
$$= \left[t.-\frac{1}{s}e^{-st} - \frac{1}{s^2}e^{-st}\right]_0^1 + -\frac{1}{s}\left[e^{-st}\right]_1^2 + 0$$

$$= -\frac{1}{s}\left[te^{-st}\right]_0^1 - \frac{1}{s^2}\left[e^{-st}\right]_0^1 - \frac{1}{s}\left[e^{-st}\right]_1^2$$

$$= -\frac{1}{s}e^{-s} - \frac{1}{s^2}\left[e^{-s} - e^{0}\right] - \frac{1}{s}\left[e^{-2s} - e^{-s}\right]$$

$$= -\frac{e^{-s}}{s} - \frac{e^{-s}}{s^2} + \frac{1}{s^2} - \frac{e^{-2s}}{s} + \frac{e^{-s}}{s} = \frac{1}{s^2} - \frac{e^{-s}}{s^2} - \frac{e^{-2s}}{s}$$

(e)



$$f(t) = \begin{cases} 1; & 0<t<1 \\ -1; & 1<t<2 \\ 0; & t>2 \end{cases}$$

$$\mathcal{L}\{f(t)\} = \int_0^\infty e^{-st} \cdot f(t)\,dt$$

$$= \int_0^1 e^{-st} \cdot 1 \cdot dt + \int_1^2 e^{-st} \cdot -1\,dt + \int_2^\infty e^{-st} \cdot 0 \cdot dt$$

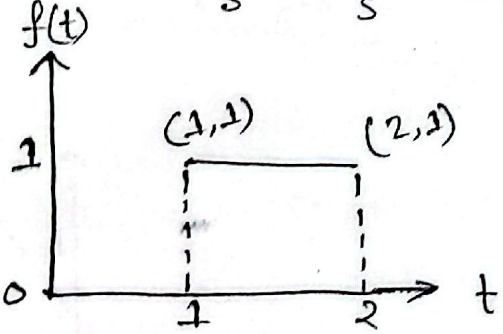
$$= -\frac{1}{s}\left[e^{-st}\right]_0^1 + \frac{1}{s}\left[e^{-st}\right]_1^2 + 0$$

$$= -\frac{1}{s}\left[e^{-s} - e^{0}\right] + \frac{1}{s}\left[e^{-2s} - e^{-s}\right]$$

$$= -\frac{e^{-s}}{s} + \frac{1}{s} + \frac{e^{-2s}}{s} - \frac{e^{-s}}{s}$$

$$= \frac{1}{s} + \frac{e^{-2s}}{s} - \frac{2e^{-s}}{s}$$

(f)



We know,

$$f(t) = \begin{cases} 0; & 0<t<1 \\ 1; & 1<t<2 \\ 0; & t>2 \end{cases}$$

$$\mathcal{L}\{f(t)\} = \int_0^\infty e^{-st} \cdot f(t)\,dt$$

$$= \int_0^1 e^{-st} \cdot 0 \cdot dt + \int_1^2 e^{-st} \cdot 1\,dt + \int_2^\infty e^{-st} \cdot 0\,dt$$

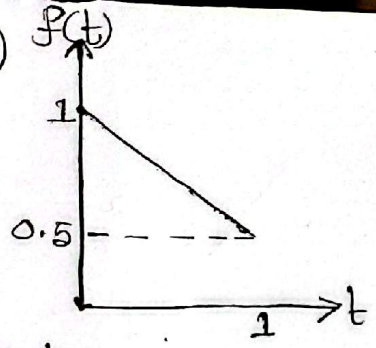
$$= 0 + -\frac{1}{s}\left[e^{-st}\right]_1^2 + 0$$

$$= -\frac{1}{s}\left[e^{-2s} - e^{-s}\right]$$

$$= -\frac{e^{-2s}}{s} + \frac{e^{-s}}{s}$$

$$= \frac{e^{-s}}{s} - \frac{e^{-2s}}{s}$$

(g)



We know,

$$\frac{x-x_1}{x_1-x_2} = \frac{y-y_1}{y_1-y_2}$$

$$\Rightarrow \frac{x-0}{0-1} = \frac{y-1}{1-0.5}$$

$$\Rightarrow \frac{x}{-1} = \frac{y-1}{0.5}$$

$$\therefore y = 1 - 0.5x$$

$$f(t) = \begin{cases} 1-0.5t & ; 0<t<1 \\ 0 & ; t>1 \end{cases}$$

$$\mathcal{L}\{f(t)\} = \int_0^1 (1-0.5t)e^{-st}\,dt + \int_1^\infty e^{-st}\cdot 0\,dt$$

$$= (1-0.5t)\int_0^1 e^{-st}\,dt - \int_0^1 \left\{\frac{d}{dt}(1-0.5t)\int e^{-st}\,dt\right\}dt$$

$$= \left[(1-0.5t)\cdot\frac{1}{s}\cdot e^{-st}\right]_0^1 - \int_0^1 -0.5\times -\frac{1}{s}e^{-st}\,dt$$

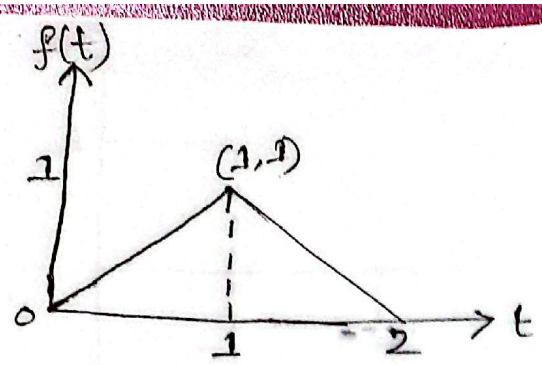
$$= (1-0.5)\left\{-\frac{1}{s}(e^{-s}-e^0)\right] + \frac{1}{2s^2}\left[e^{-st}\right]_0^1$$

$$= -\frac{1}{2s}e^{-s} + \frac{1}{2s} + \frac{1}{2s^2}\left[e^{-s}-e^0\right]$$

$$= -\frac{1}{2s}e^{-s} + \frac{1}{2s} - \frac{e^{-s}}{2s^2} + \frac{1}{2s^2}$$

$$= \frac{1}{2s} - \frac{e^{-s}}{2s} + \frac{e^{-s}}{2s^2} - \frac{1}{2s^2}$$

(h)



We know,

$$\frac{x-x_1}{x_1-x_2} = \frac{y-y_1}{y_1-y_2}$$

$$\Rightarrow \frac{x-1}{1-2} = \frac{y-1}{1-0}$$

$$\Rightarrow x-1 = -y+1$$

$$\therefore y = 2-x$$

$$f(t) = \begin{cases} t\ ; & 0<t<1 \\ 2-t\ ; & 1<t<2 \\ 0\ ; & t>2 \end{cases}$$

$$\mathcal{L}\{f(t)\} = \int_0^1 t\cdot e^{-st}\,dt + \int_1^2 (2-t)\,e^{-st}\,dt + \int_2^\infty 0\cdot e^{-st}\,dt$$

$$= \left[-\frac{t}{s}e^{-st} - \frac{1}{s^2}e^{-st}\right]_0^1 + \left[(2-t)\cdot\frac{1}{s}e^{-st}\right]_1^2 - \int_1^2 -1\times\frac{1}{s}e^{-st}\,dt$$

$$= -\frac{1}{s}e^{-s} - \frac{1}{s^2}[e^{-s}-e^0] + \frac{1}{s}e^{-s} - \left[-\frac{1}{s^2}e^{-st}\right]_1^2$$

$$= -\frac{1}{s}e^{-s} - \frac{1}{s^2}e^{-s} + \frac{1}{s^2} + \frac{1}{s}e^{-s} + \frac{1}{s^2}e^{-2s} - \frac{1}{s^2}e^{-s}$$

$$= \frac{e^{-2s}}{s^2} - \frac{2e^{-s}}{s^2} + \frac{1}{s^2}$$

$$= \frac{e^{-2s}-2e^{-s}+1}{s^2}$$

Problem set 1.2 (Unit step function/Heaviside function)

① (a) $f(t) = t\,u(t-1)$

We know,

$$\mathcal{L}\{f(t)u(t-a)\} = e^{-as}.\mathcal{L}\{f(t+a)\}$$

Now,

$$\begin{aligned}\mathcal{L}\{tu(t-1)\} &= e^{-s}\mathcal{L}\{f(t+1)\}\\ &= e^{-s}\mathcal{L}\{t+1\}\\ &= e^{-s}\mathcal{L}\{(t+1)\}\\ &= e^{-s}\left(\frac{1!}{s^2}+\frac{1}{s}\right)\\ &= \frac{e^{-s}}{s^2}+\frac{e^{-s}}{s}\end{aligned}$$

(b) $f(t) = (t-1)\,u(t-1)$

We know,

$$\mathcal{L}\{f(t).u(t-a)\} = e^{-as}.\mathcal{L}\{f(t+a)\}$$

Now,

$$\begin{aligned}\mathcal{L}\{(t-1).u(t-1)\} &= e^{-s}\mathcal{L}\{f(t+1)\}\\ &= e^{-s}\mathcal{L}\{f(t+1-1)\}\\ &= e^{-s}\mathcal{L}\{t\}\\ &= e^{-s}.\frac{1!}{s^2}\\ &= \frac{e^{-s}}{s^2}\end{aligned}$$

(c) $f(t) = (t-1)^2 u(t-1)$

We know,

$$\mathcal{L}\{f(t).u(t-a)\} = e^{-as}.\mathcal{L}\{f(t+a)\}$$

Now,

$$\mathcal{L}\{(t-1)^2.u(t-1)\} = e^{-s}.\mathcal{L}\{f(t+1)\}$$
$$= e^{-s}.\mathcal{L}\{(t-1+1)^2\}$$
$$= e^{-s}.\frac{2!}{s^3}$$
$$= \frac{2e^{-s}}{s^3}$$

(d) $f(t) = e^{-2t} u(t-3)$

We know,

$$\mathcal{L}\{f(t).u(t-a)\} = e^{-as}.\mathcal{L}\{f(t+a)\}$$

Now,

$$\mathcal{L}\{e^{-2t}.u(t-3)\} = e^{-3s}\mathcal{L}\{f(t+3)\}$$
$$= e^{-3s}.\mathcal{L}\{e^{-2k+3}\}$$
$$= e^{-3s}.\mathcal{L}\{e^{-2t}.e^{-6}\}$$
$$= e^{-3s}.e^{-6}.\mathcal{L}\{e^{-2t}\}$$
$$= e^{-3s-6} \times \frac{1}{s+2}$$
$$= \frac{e^{-3s-6}}{s+2}$$

(e) $f(t) = 4\cos t\, u(t-\pi)$

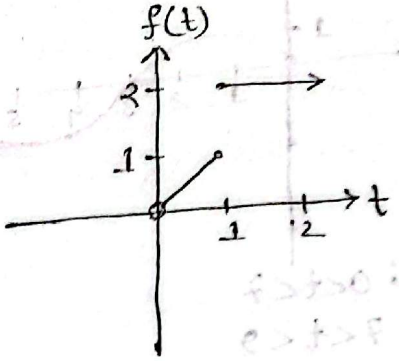
We know,

$$\mathcal{L}\{f(t).u(t-a)\} = e^{-as}.\mathcal{L}\{f(t+a)\}$$

Now,

$$\mathcal{L}\{u\cos t.u(t-\pi)\} = e^{-\pi s}.\mathcal{L}\{f(t+\pi)\}$$
$$= e^{-\pi s}.\mathcal{L}\{u\cos(t+\pi)\}$$
$$= 4e^{-\pi s}\mathcal{L}\{\cos t.\cos\pi - \sin t.\sin\pi\}$$
$$= 4e^{-\pi s}\cos\pi \times \frac{s}{s^2+1} - 4\pi e^{-\pi s}\sin\pi\,\frac{1}{s^2+1}$$
$$= \frac{-4e^{-\pi s}.s - 4\pi e^{-\pi s}\sin\pi}{s^2+1}$$
$$= \frac{-4e^{-\pi s} - 4e^{-\pi s}\times 0}{s^2+1} = -\frac{4se^{-\pi s}}{s^2+1}$$

2. (a) $f(t) = \begin{cases} t; & 0<t<1 \\ 2; & t>1 \end{cases}$



$f(t) = 2u(t-1) + t[u(t-0) - u(t-1)]$

$= 2u(t-1) + t^2 u - ut \cdot (t-1)$

$f(t) = u(t-1)\{(2-t)\} + t \cdot u(t)$

$\therefore F(s) = \mathcal{L}\{(2-t)u(t-1)\} + \mathcal{L}\{t \cdot u(t)\}$

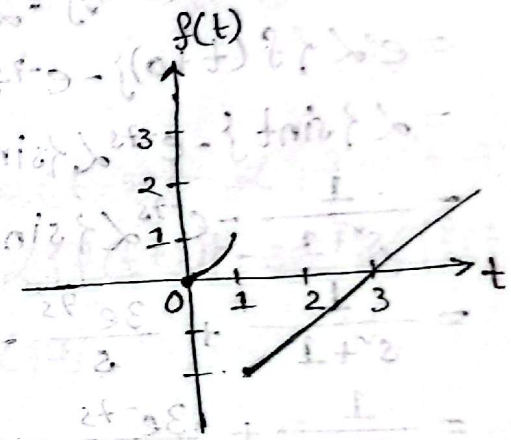
$= e^{-s}\mathcal{L}\{f(t+1)\} + e^{0}\mathcal{L}\{ft\}$

$= e^{-s}\mathcal{L}\{(2-t-1)\} + \mathcal{L}\{t\}$

$= e^{-s} \times \left(-\frac{1}{s^2} + \frac{1}{s}\right) + \frac{1}{s^2}$

$= \frac{-e^{-s} + e^{-s} + 1}{s^2}$

(b) $f(t) = \begin{cases} t^2; & 0 \le t < 1 \\ t-3; & t \ge 1 \end{cases}$



$\therefore f(t) = (t-3)[u(t-1)] + t^2[u(t-0) - u(t-1)]$

$= (t-3)[u(t-1)] + t^2 u(t) - t^2[u(t-1)]$

$= u[t-1]\{(t-3) - t^2\} + t^2 u(t)$

Now,

$F(s) = \mathcal{L}\{(t-3-t^2)u(t-1)\} + \mathcal{L}\{t^2 u(t+0)\}$

$= e^{-s}\mathcal{L}\{f(t+1)\} + e^{0}\mathcal{L}\{f(t+0)\}$

$= e^{-s}\mathcal{L}\{(t+1) - (t+1)^2 - 3\} + \mathcal{L}\{t^2\}$

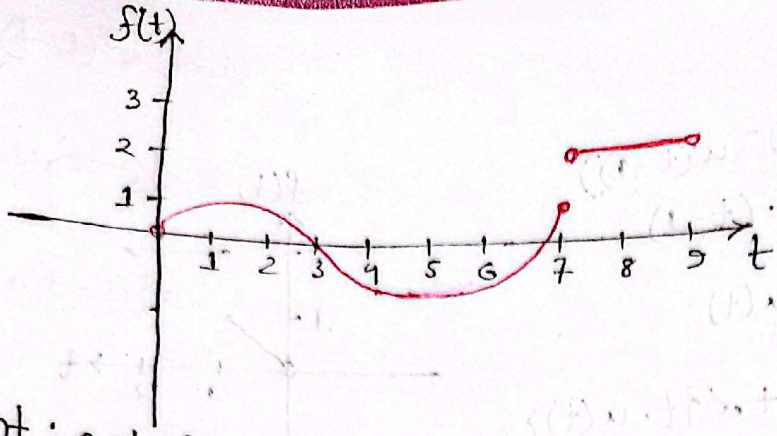
$= e^{-s}\mathcal{L}\{t+1-3-(t^2+2t+1)\} + \frac{2}{s^3}$

$= e^{-s}\mathcal{L}\{t+1-3-t^2-2t-1\} + \frac{2}{s^3}$

$= e^{-s}\mathcal{L}\{-t-3-t^2\} + \frac{2}{s^3}$

$= \frac{2}{s^3} - \left(\frac{e^{-s}}{s^2} + \frac{3e^{-s}}{s} + \frac{2e^{-s}}{s^3}\right)$

3. (a)



$$f(t) = \begin{cases} \sin t ; & 0<t<7 \\ 3 ; & 7<t<9 \end{cases}$$

$\therefore f(t) = \sin t\,[u(t-0) - u(t-7)] + 3[u(t-7) - u(t-9)]$

$= \sin t \cdot u(t-0) - \sin t \cdot u(t-7) + 3u(t-7) - 3u(t-9)$

Now,

$F(s) = \mathcal{L}\{\sin t \cdot u(t-0)\} - \mathcal{L}\{\sin t\, u(t-7)\} + 3\mathcal{L}\{u(t-7)\} - 3\mathcal{L}\{1\,u(t-9)\}$

$= e^{0}\mathcal{L}\{f(t+0)\} - e^{-7s}\mathcal{L}\{f(t+7)\} + 3e^{-7s}\mathcal{L}\{f(t+7)\} - 3e^{-9s}\mathcal{L}\{f(t+9)\}$

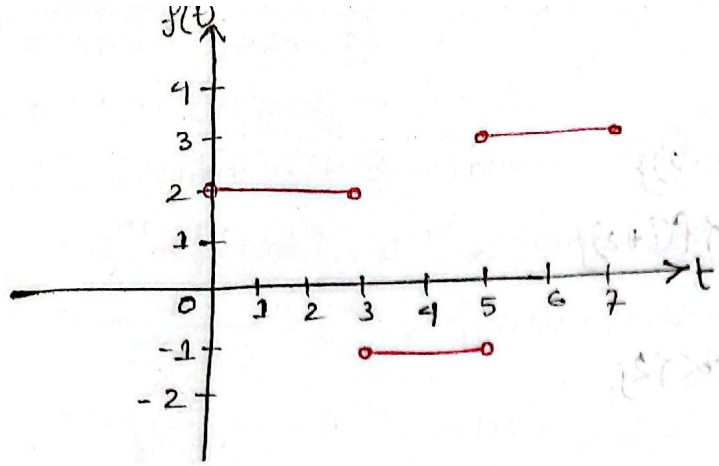
$= \mathcal{L}\{\sin t\} - e^{-7s}\mathcal{L}\{\sin(t+7)\} + 3e^{-7s}\mathcal{L}\{1\} - 3e^{-9s}\mathcal{L}\{1\}$

$= \frac{1}{s^2+1} - e^{-7s}\mathcal{L}\{\sin t \cdot \cos 7 + \cos t \cdot \sin 7\} + 3e^{-7s} \times \frac{1}{s} - 3e^{-9s} \times \frac{1}{s}$

$= \frac{1}{s^2+1} + \frac{3e^{-7s}}{s} - \frac{3e^{-9s}}{s} - e^{-7s} \cdot \cos 7\,\mathcal{L}\{\sin t\} - e^{-7s} \cdot \sin 7\,\mathcal{L}\{\cos t\}$

$= \frac{1}{s^2+1} + \frac{3e^{-7s}}{s} - \frac{3e^{-9s}}{s} - \frac{e^{7s}\cos t}{s^2+1} - \frac{se^{-7s} \cdot \sin t}{s^2+1}$

(b)



$$f(t)=\begin{cases}2 ; 0<t<3\\ -1 ; 3<t<5\\ 3 ; 5<t<7\end{cases}$$

$\therefore f(t) = 2[u(t-0)-u(t-3)] - 1[u(t-3)-u(t-5)] + 3[u(t-5)-u(t-7)]$

$= 2u(t) - 2u(t-3) - u(t-3) + u(t-5) + 3u(t-5) - 3u(t-7)$

$= 2u(t) - 3u(t-3) + 4u(t-5) - 3u(t-7)$

Now,

$F(s) = 2\mathcal{L}\{1\cdot u(t)\} - 3\mathcal{L}\{1\cdot u(t-3)\} + 4\mathcal{L}\{1\cdot u(t-5)\} - 3\mathcal{L}\{1\cdot u(t-7)\}$

$= 2\times e^{0}\mathcal{L}\{f(t)\} - 3e^{-3s}\mathcal{L}\{f(t+3)\} + 4e^{-5s}\mathcal{L}\{f(t+5)\} - 3e^{-7s}\mathcal{L}\{f(t+7)\}$

$= 2\mathcal{L}\{1\} - 3e^{-3s}\mathcal{L}\{1\} + 4e^{-5s}\mathcal{L}\{1\} - 3e^{-7s}\mathcal{L}\{1\}$

$= \frac{2}{s} - \frac{3e^{-3s}}{s} + \frac{4e^{-5s}}{s} - \frac{3e^{-7s}}{s}$

5.(a) $f(t) = t \quad (0<t<2)$

$\Rightarrow f(t) = t[u(t-0) - u(t-2)]$

$\therefore \mathcal{L}\{f(t)\} = \mathcal{L}\{t \cdot u(t-0)\} - \mathcal{L}\{t \cdot u(t-2)\}$

$= e^{0s}\mathcal{L}\{f(t+0)\} - e^{-2s}\mathcal{L}\{f(t+2)\}$

$= \mathcal{L}\{t\} - e^{-2s}\mathcal{L}\{t+2\}$

$= \mathcal{L}\{t\} - e^{-2s}\mathcal{L}\{t\} - e^{-2s}\mathcal{L}\{2\}$

$= \frac{1}{s^2} - \frac{e^{-2s}}{s^2} - \frac{2e^{-2s}}{s}$

(b) $f(t) = t^2 \quad (t > 3/2)$

$\Rightarrow f(t) = t^2[u(t-3/2)]$

$\therefore \mathcal{L}\{f(t)\} = \mathcal{L}\{(t^2)\, u(t-3/2)\}$

$= e^{-3/2s}\mathcal{L}\{f(t+3/2)^2\}$

$= e^{-3/2s}\mathcal{L}\left[t^2 + 2 \cdot t \cdot 3/2 + \frac{9}{4}\right]$

$= e^{-3/2s}\left[\frac{2}{s^3} + \frac{3}{s^2} + \frac{9}{4s}\right]$

(c) $f(t) = t-2 \quad (t>2)$

$\Rightarrow f(t) = (t-2)[u(t-2)]$

$\therefore \mathcal{L}\{f(t)\} = \mathcal{L}\{(t-2)\, u(t-2)\}$

$= e^{-2s}\mathcal{L}\{f(t+2)\}$

$= e^{-2s}\mathcal{L}\{t+2-2\}$

$= e^{-2s}\mathcal{L}\{t\}$

$= e^{-2s} \times \frac{1}{s^2}$

(d) $f(t) = \sinh t \quad (0<t<2)$

$\Rightarrow f(t) = \sinh t\,[u(t-0) - u(t-2)]$

$\therefore \mathcal{L}\{f(t)\} = \mathcal{L}\{\sinh t\, u(t-0) - \sinh t\, u(t-2)\}$

$= e^{0s}\mathcal{L}\{\sinh t\} - e^{-2s}.\mathcal{L}\{\sinh(t+2)\}$

$= \frac{1}{s^2-1^2} - e^{-2s}.\mathcal{L}\{(\sinh t.\cosh 2 + \cosh t \sinh 2)\}$

$= \frac{1}{s^2-1^2} - e^{-2s}.\cosh 2 \times \frac{1}{s^2-1^2} - e^{-2s}.\sinh 2 \times \frac{s}{s^2-1^2}$

(e) $f(t) = \cos 4t \quad (0<t<\pi)$

$\Rightarrow f(t) = \cos 4t\,[u(t-0) - u(t-\pi)]$

$\Rightarrow f(t) = \cos 4t\, u(t-0) - \cos 4t\, u(t-\pi)$

$\therefore \mathcal{L}\{f(t)\} = \mathcal{L}\{\cos 4t\, u(t-0) - \cos 4t\, u(t-\pi)\}$

$= e^{-0s}\mathcal{L}\{f(t+0)\} - e^{-\pi s}\mathcal{L}\{f(t+\pi)\}$

$= \mathcal{L}\{\cos 4t\} - e^{-\pi s}\mathcal{L}\{\cos 4(t+\pi)\}$

$= \frac{s}{s^2+4^2} - e^{-\pi s}\mathcal{L}[\cos 4t.\cos 4\pi - \sin 4t.\sin 4\pi]$

$= \frac{s}{s^2+4^2} - e^{-\pi s}\{\cos 4\pi \times \frac{s}{s^2+4^2} + e^{-\pi s}\sin 4\pi \times \frac{4}{s^2+4^2}$

(f) $f(t) = \sin t \quad (\pi/2 < t < \pi)$

$\Rightarrow f(t) = \sin t\,\{u(t-\pi/2) - u(t-\pi)\}$

$\therefore \mathcal{L}\{f(t)\} = \mathcal{L}\{\sin u(t-\pi/2) - \sin u(t-\pi)\}$

$= e^{-\pi/2 s}\mathcal{L}\{f(t+\frac{\pi}{2})\} - e^{-\pi s}\mathcal{L}\{f(t+\pi)\}$

$= e^{-\frac{\pi}{2}s}\mathcal{L}\{\sin(t+\frac{\pi}{2})\} - e^{-\pi s}\mathcal{L}\{\sin(t+\pi)\}$

$= e^{-\frac{\pi}{2}s}\mathcal{L}\{\sin t\cos\frac{\pi}{2} + \cos t \sin\frac{\pi}{2}\} - e^{-\pi s}\mathcal{L}\{\sin t\cos\pi + \cos t.\sin\pi\}$

$= e^{-\frac{\pi}{2}s}\cos\frac{\pi}{2} \times \frac{1}{s^2+1^2} + e^{-\frac{\pi}{2}}\sin\frac{\pi}{2} \times \frac{s}{s^2+1^2} - e^{-\pi s}\cos\pi \times \frac{1}{s^2+1^2} \sin\pi \frac{s}{s^2+1}$

$= 0 + e^{-\frac{\pi}{2}}.\frac{s}{s^2+1} + e^{-\pi s}.\frac{1}{s^2+1} + 0$

$= e^{-\frac{\pi}{2}}.\frac{s}{s^2+1} + e^{-\pi s}\frac{1}{s^2+1}$

(g) $f(t) = e^t \ (0 < t < \frac{\pi}{2})$

$\Rightarrow f(t) = e^t [u(t-0) - u(t-\frac{\pi}{2})]$

$\mathcal{L}\{f(t)\} = \mathcal{L}\{e^t u(t-0) - e^t u(t-\frac{\pi}{2})$

$= e^{0s} \mathcal{L}\{f(t+0) - e^{-\frac{\pi}{2}s} \mathcal{L}\{f(t+\frac{\pi}{2})\}$

$= \mathcal{L}\{e^t\} - e^{-\frac{\pi}{2}s} \mathcal{L}\{e^{t+\frac{\pi}{2}}\}$

$= \frac{1}{s-1} - e^{-\frac{\pi}{2}s} \cdot e^{\frac{\pi}{2}} \times \frac{1}{s-1}$

(h) $f(t) = 16t^2 u(t-\frac{1}{4})$

$\Rightarrow \mathcal{L}\{f(t)\} = \mathcal{L}\{16t^2 u(t-\frac{1}{4})\}$

$= e^{-\frac{1}{4}s} \cdot \mathcal{L}\{f(t+\frac{1}{4})^2\}$

$= e^{-\frac{1}{4}s} \cdot \mathcal{L}\{16(t+\frac{1}{4})^2\}$

$= 16 e^{-\frac{1}{4}s} \mathcal{L}\{t^2 + 2 \times \frac{1}{4} t + (\frac{1}{16})\}$

$= 16 e^{-\frac{1}{4}s} \left(\frac{2}{s^3} + \frac{1}{2} \times \frac{1}{s^2} + \frac{1}{16s}\right)$

$= e^{-\frac{1}{4}s} \left(\frac{32}{s^3} + \frac{8}{s^2} + \frac{1}{s}\right)$

(i) $f(t) = \sin \pi t \ (2 < t < 4)$

$\Rightarrow f(t) = \sin \pi t \ [u(t-2) - u(t-4)]$

$\therefore \mathcal{L}\{f(t)\} = \{\sin \pi t \ u(t-2) - \sin \pi t \cdot u(t-4)\}$

$= e^{2s} \mathcal{L}\{f(t+2)\} - e^{-4s} \mathcal{L}\{f(t+4)\}$

$= e^{-2s} \mathcal{L}\{\sin \pi (t+2)\} - e^{-4s} \mathcal{L}\{\sin \pi (t+4)\}$

$= e^{-2s} \mathcal{L}\{(\sin \pi t \cdot \cos \pi 2 + \cos \pi t \sin \pi 2\} - e^{-4s} \mathcal{L}\{\sin \pi t \cdot \cos \pi 4 + \cos \pi t \sin \pi 4\}$

$= e^{-2s} \cdot \cos \pi 2 \cdot \frac{1}{s^2+1^2} + e^{-2s} \cdot \sin \pi 2 \cdot \frac{s}{s^2+1^2} - e^{-4s} \cos \pi 4 \cdot \frac{1}{s^2+1^2} - e^{-4s} \sin \pi 4 \cdot \frac{s}{s^2+1^2}$

(j) $f(t) = e^{\frac{t}{2}} u(t-3)$

$\Rightarrow \mathcal{L}\{f(t)\} = \mathcal{L}\{e^{t/2} \cdot u(t-3)\}$

$= e^{-3s} \cdot \mathcal{L}\{e^{\frac{t+3}{2}}\}$

$= e^{-3s} \cdot \mathcal{L}\{e^{\frac{t}{2} + \frac{3}{2}}\}$

$= e^{-3s} \cdot e^{\frac{3}{2}} \mathcal{L}\{e^{t/2}\}$

$= e^{-3s} \cdot e^{3/2} \cdot \frac{1}{s - \frac{1}{2}}$

(k) $f(t) = e^{-\pi t} (2 < t < 4)$

$\Rightarrow f(t) = e^{-\pi t} [u(t-2) - u(t-4)]$

$\therefore \mathcal{L}\{f(t)\} = \mathcal{L}\{e^{-\pi t} u(t-2) - e^{-\pi t} u(t-4)\}$

$= e^{-2s} \mathcal{L}\{f(t+2)\} - e^{-4s} \mathcal{L}\{f(t+4)\}$

$= e^{-2s} \mathcal{L}\{e^{-\pi(t+2)}\} - e^{-4s} \mathcal{L}\{e^{-\pi(t+4)}\}$

$= e^{-2s} e^{-2\pi} \mathcal{L}\{e^{-\pi t}\} - e^{-4s} \cdot e^{-4\pi} \mathcal{L}\{e^{-\pi t}\}$

$= e^{-2s} \cdot e^{-2\pi} \times \frac{1}{s+\pi} - e^{-4s} e^{-4\pi} \times \frac{1}{s+\pi}$

(l) $f(t) = u(t-2\pi) \sin t$

We know,

$\mathcal{L}\{f(t) \cdot u(t-a)\} = e^{-as} \cdot \mathcal{L}\{(t+a)\}$

Now,

$\mathcal{L}\{u(t-2\pi) \sin t\}$

$= e^{-2\pi s} \mathcal{L}\{f(t+2\pi)\}$

$= e^{-2\pi s} \mathcal{L}\{\sin(t+2\pi)\}$

$= e^{-2\pi s} \mathcal{L}\{\sin t \cdot \cos 2\pi + \cos t \cdot \sin 2\pi\}$

$= e^{-2\pi s} \cdot \cos 2\pi \times \frac{1}{s^2+1^2} + e^{-2\pi s} \cdot \sin 2\pi \frac{s}{s^2+1^2}$

$= \frac{e^{-2\pi s}}{s^2+1} + 0$

$= \frac{e^{-2\pi s}}{s^2+1}$

Inverse Laplace

1. $\mathcal{L}^{-1}\left\{\frac{c}{s}\right\} = c$
2. $\mathcal{L}^{-1}\left\{\frac{1}{s^{n+1}}\right\} = \frac{t^n}{n!}$
3. $\mathcal{L}^{-1}\left\{\frac{1}{s-a}\right\} = e^{at}$
4. $\mathcal{L}^{-1}\left\{\frac{s}{s^2+a^2}\right\} = \cos at$
5. $\mathcal{L}^{-1}\left\{\frac{a}{s^2+a^2}\right\} = \sin at$
6. $\mathcal{L}^{-1}\left\{\frac{s}{s^2-a^2}\right\} = \cosh at$
7. $\mathcal{L}^{-1}\left\{\frac{a}{s^2-a^2}\right\} = \sinh at$
8. $\mathcal{L}^{-1}\left\{\frac{1}{s^2+a^2}\right\} = \frac{1}{a}\sin at$
9. $\mathcal{L}^{-1}\left\{\frac{1}{s^2-a^2}\right\} = \frac{1}{a}\sinh at$

Problem set-2.1

1. (a) $F(s) = \frac{1}{s-5}$

$$\mathcal{L}^{-1}\{f(t)\} = \mathcal{L}^{-1}\left\{\frac{1}{s-5}\right\}$$

$$= e^{5t}$$

(b) $F(s) = \frac{1}{s^5}$

$$\mathcal{L}^{-1}\{f(t)\} = \mathcal{L}^{-1}\left\{\frac{1}{s^5}\right\}$$

$$= \frac{t^4}{24}$$

(c) $F(s) = \frac{s^3 - 5s^2 + 6}{s^4}$

$$\mathcal{L}^{-1}\{f(t)\} = \mathcal{L}^{-1}\left\{\frac{s^3 - 5s^2 + 6}{s^4}\right\}$$

$$= \mathcal{L}^{-1}\left\{\frac{s^3}{s^4}\right\} - \mathcal{L}^{-1}\left\{\frac{5s^2}{s^4}\right\} + \mathcal{L}^{-1}\left\{\frac{6}{s^4}\right\}$$

$$= \mathcal{L}^{-1}\left\{\frac{1}{s}\right\} - 5\mathcal{L}^{-1}\left\{\frac{1}{s^2}\right\} + 6\mathcal{L}^{-1}\left\{\frac{1}{s^4}\right\}$$

$$= 1 - 5t + 6 \times \frac{t^3}{6}$$

$$= t^3 - 5t + 1$$

(d) $F(s) = \frac{2+4s}{s^2+25}$

$$\mathcal{L}^{-1}\{f(t)\} = \mathcal{L}^{-1}\left\{\frac{2+4s}{s^2+25}\right\}$$

$$= \mathcal{L}^{-1}\left\{\frac{2}{s^2+25}\right\} + \mathcal{L}^{-1}\left\{\frac{4s}{s^2+25}\right\}$$

$$= \frac{2}{5}\mathcal{L}^{-1}\frac{5}{s^2+25} + 4\mathcal{L}^{-1}\left\{\frac{s}{s^2+25}\right\}$$

$$= \frac{2}{5}\sin(5t) + 4\cos(5t)$$

(e) $F(s) = \frac{3}{s^2+4}$

$\mathcal{L}^{-1}\{f(t)\} = \mathcal{L}^{-1}\left\{\frac{3}{s^2+2^2}\right\}$

$= \frac{3}{2}\mathcal{L}^{-1}\left\{\frac{2}{s^2+2^2}\right\}$

$= \frac{3}{2}\sin(2t)$

(f) $F(s) = \frac{3}{s^2-1}$

$\mathcal{L}^{-1}\{f(t)\} = \mathcal{L}^{-1}\left\{\frac{3}{s^2-4}\right\}$

$= \frac{3}{2}\mathcal{L}^{-1}\left\{\frac{2}{s^2-2^2}\right\}$

$= \frac{3}{2}\sinh(2t)$

First Translation Property

(g) $F(s) = \frac{1}{(s-3)^4}$

$\mathcal{L}^{-1}\left\{\frac{1}{(s-3)^4}\right\}$

$= e^{3t}\mathcal{L}^{-1}\left\{\frac{1}{s^4}\right\}$

$= e^{3t} \times \frac{t^3}{6}$

(h) $F(s) = \frac{3}{(s+2)^2+9}$

$= \frac{3}{(s+2)^2+3^2}$

$\mathcal{L}^{-1}\left\{\frac{3}{(s+2)^2+3^2}\right\}$

$= e^{-2t}\sin(3t)$

(i) $F(s) = \frac{s-2}{(s-2)^2-16}$

$= \frac{s-2}{(s-2)^2-4^2}$

$\mathcal{L}^{-1}\left\{\frac{s-2}{(s-2)^2-4^2}\right\}$

$= e^{2t}\cos(4t)$

(j) $F(s) = \frac{s}{s^2+4s-9}$

$= \frac{s}{s^2+2.s.2+2^2-9-4}$

$= \frac{s}{(s+2)^2-13}$

$= \frac{(s+2)-2}{(s+2)^2-(\sqrt{13})^2}$

$\mathcal{L}^{-1}\left\{\frac{(s+2)-2}{(s+2)^2-(\sqrt{13})^2}\right\}$

$= \mathcal{L}^{-1}\left\{\frac{(s+2)}{(s+2)^2-(\sqrt{13})^2}\right\} - \mathcal{L}^{-1}\left\{\frac{2}{(s+2)^2-(\sqrt{13})^2}\right\}$

$= e^{-2t}\cosh(\sqrt{13}t) - \frac{2}{\sqrt{13}}e^{-2t}\sinh(\sqrt{13}t)$

$= e^{-2t}\left[\cosh(\sqrt{13}t) - \frac{2}{\sqrt{13}}\sinh(\sqrt{13}t)\right]$

(k) $F(s) = \frac{5s-7}{s^2-6s+25}$

$= \frac{5s-7}{s^2-2.s.3+3^2+25-9}$

$= \frac{5s-7}{(s-3)^2+16}$

$= \frac{5(s-3)+15-7}{(s-3)^2+16}$

$= \frac{5(s-3)+8}{(s-3)^2+4^2}$

$\mathcal{L}^{-1}\left\{\frac{5(s-3)}{(s-3)^2+4^2}\right\} + \mathcal{L}^{-1}\left\{\frac{8}{(s-3)^2+4^2}\right\}$

$\mathcal{L}^{-1}\left\{\frac{5(s-3)}{(s-3)^2+4^2}\right\} + 2\mathcal{L}^{-1}\left\{\frac{4}{(s-3)^2+4^2}\right\}$

$= 5e^{3t}\cos(4t) + 2e^{3t}\sin(4t)$

(l) $F(s) = \frac{s}{s^2-6s+10}$

$= \frac{s}{s^2-2.s.3+3^2+10+1}$

$= \frac{s}{(s-3)^2+1}$

$= \frac{(s-3)+3}{(s-3)^2+1^2}$

$\mathcal{L}^{-1}\left\{\frac{(s-3)}{(s-3)^2+1}\right\} + 3\mathcal{L}^{-1}\left\{\frac{1}{(s-3)^2+1}\right\}$

$= e^{3t}\cos(t) + 3e^{3t}\sin(t)$

Using partial fraction

(m) $F(s) = \frac{s+1}{s(s-2)(s+3)} = \frac{A}{s} + \frac{B}{s-2} + \frac{C}{s+3}$ — (1)

Multiply through by denominator,

$s+1 = A(s-2)(s+3) + B(s)(s+3) + C(s)(s-2)$

Substitute convenient s values,

$s=0;$

$0+1 = A(-2)(3) + B(0)(3) + C(0)(-2)$

$\Rightarrow 1 = -6A$

$\therefore A = -\frac{1}{6}$

$s=2;$

$2+1 = A(2-2)(2+3) + B(2)(2+3) + C(2)(2-2)$

$\Rightarrow 3 = 10B$

$\therefore B = \frac{3}{10}$

$s=-3;$

$-3+1 = A(-3-2)(-3+3) + B(-3)(-3+3) + C(-3)(-3-2)$

$\Rightarrow -2 = 15C$

$\therefore C = -\frac{2}{15}$

Put the value equation number ①,

$$\frac{s+1}{s(s-2)(s+3)} = -\frac{1}{6}\cdot\frac{1}{s} + \frac{3}{10}\cdot\frac{1}{s-2} - \frac{2}{15}\cdot\frac{1}{s+3}$$

$$\mathcal{L}^{-1}\left\{\frac{s+1}{s(s-2)(s+3)}\right\} = \mathcal{L}^{-1}\left\{-\frac{1}{6}\cdot\frac{1}{s} + \frac{3}{10}\cdot\frac{1}{s-2} - \frac{2}{15}\cdot\frac{1}{s+3}\right\}$$

$$= -\frac{1}{6}\mathcal{L}^{-1}\left\{\frac{1}{s}\right\} + \frac{3}{10}\mathcal{L}^{-1}\left\{\frac{1}{s-2}\right\} - \frac{2}{15}\mathcal{L}^{-1}\left\{\frac{1}{s+3}\right\}$$

$$= -\frac{1}{6} + \frac{3}{10}e^{2t} - \frac{2}{15}e^{-3t}$$

(n) $F(s) = \dfrac{6}{(s+2)(s-4)} \equiv \dfrac{A}{s+2} + \dfrac{B}{s-4}$ —①

$\Rightarrow 6 = A(s-4) + B(s+2)$

$s = 4$;

$6 = A(4-4) + B(4+2)$

$\Rightarrow 6 = 6B$

$\therefore B = 1$

$s = -2$

$6 = A(-2-4) + B(0)$

$\Rightarrow 6 = -6A$

$\therefore A = -1$

put the value equ. number ①,

$$\frac{6}{(s+2)(s-4)} \equiv \frac{-1}{s+2} + \frac{1}{s-4}$$

$$\mathcal{L}^{-1}\left\{\frac{6}{(s+2)(s-4)}\right\} = \mathcal{L}^{-1}\left\{\frac{-1}{s+2} + \frac{1}{s-4}\right\}$$

$$= -e^{-2t} + e^{4t}$$

$$= e^{4t} - e^{-2t}$$

(c) $F(s) = \frac{6s-17}{s^2-5s+6}$

$= \frac{6s-17}{s^2-3s-2s+6}$

$= \frac{6s-17}{s(s-3)-2(s-3)}$

$= \frac{6s-17}{(s-3)(s-2)}$

Now,

$\frac{6s-17}{(s-3)(s-2)} \equiv \frac{A}{s-2} + \frac{B}{s-3}$ — (1)

$6s-17 = A(s-3) + B(s-2)$

$s \neq 2;$

$6(2)-17 = A(2-3) + B(2-2)$

$\Rightarrow 12-17 = -A$

$\therefore A = 5$

$s = 3;$

$6(3)-17 = A(3-3) + B(3-2)$

$\Rightarrow 18-17 = B$

$\therefore B = 1$

put the value equ number (1),

$\frac{6s-17}{(s-2)(s-3)} = \frac{5}{s-2} + \frac{1}{s-3}$

$\mathcal{L}^{-1}\left\{\frac{6s-17}{(s-2)(s-3)}\right\} = \mathcal{L}^{-1}\left\{\frac{5}{s-2} + \frac{1}{s-3}\right\}$

$= 5e^{2t} + e^{3t}$

(P) $F(s) = \frac{s}{(s+1)^2} = \frac{A}{s+1} + \frac{B}{(s+1)^2}$ — (i)

multiply $(s+1)^2$ from both side,

$s = A(s+1) + B$ — (ii)

$s = As + A + B$ — (iii)

if $s+1 = 0$

$\therefore s = -1$

put in equ (ii) $\Rightarrow$

$-1 = B$

$\therefore B = -1$

equating like term,

$As = s$

$\therefore A = 1$

put the value of equ (i) $\Rightarrow$

$$\frac{A}{s+1} + \frac{B}{(s+1)^2} = \frac{1}{s+1} + \frac{-1}{(s+1)^2}$$

$$= \mathcal{L}^{-1}\left\{\frac{1}{s+1}\right\} - \mathcal{L}^{-1}\left\{\frac{1}{(s+1)^2}\right\}$$

$$= e^{-t} - e^{-t}.t$$

(9) $F(s) = \frac{7s^2+19s-9}{(s-1)^2(s-2)} = \frac{A}{s-1} + \frac{B}{(s-1)^2} + \frac{C}{s-2}$

$7s^2+19s-9 = A(s-1)(s-2)+B(s-2)+C(s-1)^2$

$s=2;$

$7(4)+19(2)-9 = A(2-1)(0)+B(0)+C(2-1)^2$

$\Rightarrow 28+28-9 = C(1)^2$

$\therefore C = 47$

$s=1;$

$7+19-9 = A(0)(-1)+B(-1)+C(0)^2$

$\therefore B = -12$

set convenient value $s=0;$

$7(0)^2+19(0)-9 = A(-1)(-2)+B(-2)+C(-1)^2$

$\Rightarrow -9 = 2A+(-12)(-2)+47$

$\Rightarrow 2A = -80$

$\therefore A = -40$

$$\frac{7s^2+19s-9}{(s-1)^2(s-2)} = \frac{-40}{s-1} + \frac{-12}{(s-1)^2} + \frac{47}{(s-2)}$$

$$= \mathcal{L}^{-1}\left\{\frac{-40}{s-1} - \frac{12}{(s-1)^2} + \frac{47}{(s-2)}\right\}$$

$$= -40e^t - 12te^t + 47e^{2t}$$

2. (a) $\frac{s+2}{s^2+4}$

$$F(s) = \frac{s}{s^2+4} + \frac{2}{s^2+4}$$

$$\mathcal{L}^{-1}\{f(t)\} = \mathcal{L}^{-1}\left\{\frac{s}{s^2+4}\right\} + \mathcal{L}^{-1}\left\{\frac{2}{s^2+4}\right\}$$

$$= \mathcal{L}^{-1}\left\{\frac{s}{s^2+2^2}\right\} + \frac{2}{2}\mathcal{L}^{-1}\left\{\frac{2}{s^2+2^2}\right\}$$

$$= \cos(2t) + \sin(2t)$$

(b) $\frac{3s+4}{s^2+4s+5}$

$$= \frac{3s+4}{(s^2+4s+4)+1}$$

$$= \frac{3s+4}{(s+2)^2+1}$$

$$F(s) = \frac{3s+4}{(s+2)^2+1}$$

~~$\mathcal{L}^{-1}\{f(t)\} = \mathcal{L}^{-1}\left\{\frac{3s+4}{(s+2)^2+1}\right\}$~~

~~$= \mathcal{L}^{-1}\left\{\frac{3s}{(s+2)^2+1^2}\right\} + \mathcal{L}^{-1}\left\{\frac{1}{(s+2)^2+1^2}\right\}$~~

$$= \mathcal{L}^{-1}\left\{\frac{3(s+2)-2}{(s+2)^2+1^2}\right\} + 4\mathcal{L}^{-1}\left\{\frac{1}{(s+2)^2+1^2}\right\}$$

~~$= 3e^{-2t}$~~

$$\mathcal{L}^{-1}\{f(t)\} = \mathcal{L}^{-1}\left\{\frac{3s+4}{(s+2)^2+1}\right\}$$

$$= \mathcal{L}^{-1}\left\{\frac{3(s+2)-2}{(s+2)^2+1}\right\}$$

$$= \mathcal{L}^{-1}\left\{\frac{3(s+2)}{(s+2)^2+1}\right\} - \mathcal{L}^{-1}\left\{\frac{2}{(s+2)^2+1}\right\}$$

$$= 3e^{-2t}\cos t - 2e^{-2t}\sin t$$

(c) $\frac{5s+1}{s^2-25}$

$\Rightarrow \frac{5s+1}{(s-5)(s+5)} \equiv \frac{A}{s-5} + \frac{B}{s+5}$ —— (1)

$\Rightarrow 5s+1 = A(s+5) + B(s-5)$

$\Rightarrow 5s+1 = As+5A+Bs-5B$

$\Rightarrow 5s+1 = (A+B)s + (5A-5B)$

$A+B = 5 \quad \therefore A = 5-B$

$5A-5B = 1$

$\Rightarrow A-B = \frac{1}{5}$

$\Rightarrow (5-B)-B = \frac{1}{5}$

$\Rightarrow 5-B-B = \frac{1}{5}$

$\Rightarrow 5-2B = \frac{1}{5}$

$\Rightarrow -2B = \frac{1}{5}-5$

$\Rightarrow -2B = \frac{1-25}{5}$

$\Rightarrow -2B = \frac{-24}{5}$

$\therefore B = \frac{12}{5}$

$A + \frac{12}{5} = 5$

$\Rightarrow A = 5 - \frac{12}{5}$

$\Rightarrow A = \frac{25-12}{5}$

$\therefore A = \frac{13}{5}$

$\frac{5s+1}{s^2-25} = \frac{\frac{13}{5}}{s-5} + \frac{\frac{12}{5}}{s+5}$

$= \mathcal{L}^{-1}\left\{\frac{13/5}{s-5}\right\} + \mathcal{L}^{-1}\left\{\frac{\frac{12}{5}}{s+5}\right\}$

$= \frac{13}{5}e^{5t} + \frac{12}{5}e^{-5t}$

(c) $\frac{5s+1}{s^2-25}$

$\Rightarrow 5\mathcal{L}^{-1}\left\{\frac{s}{s^2-25} + \frac{1}{s^2-25}\right\}$

$\Rightarrow 5\cosh(5t) + \frac{1}{5}\sinh(5t)$

(d) $\frac{3s}{s^2-2s+2}$

$= \frac{3s}{(s-1)^2+1}$

$= \frac{3(s-1)+3}{(s-1)^2+1}$

$\mathcal{L}^{-1}\left\{\frac{3(s-1)}{(s-1)^2+1} + \frac{3\cdot 1}{(s-1)^2+1}\right\}$

$= 3e^t\cos t + 3e^t\sin t$

$= 3e^t(\cos t + \sin t)$

(e) $\frac{s+1}{s^4+9s^2}$

$\Rightarrow \frac{s+1}{s^2(s^2+9)} \equiv \frac{A}{s} + \frac{B}{s^2} + \frac{Cs+D}{s^2+9}$

$\Rightarrow s+1 = As(s^2+9) + B(s^2+9) + (Cs+D)s^2$

$\Rightarrow s+1 = As^3 + 9As + Bs^2 + 9B + Cs^3 + Ds^2$

$\Rightarrow s+1 = (A+C)s^3 + (B+D)s^2 + 9As + 9B$

$A+C=0$

$B+D=0$

$9A = 1$

$A = 1/9$

$9B = 1$

$\therefore B = 1/9$

$C = -1/9$

$D = -1/9$

$\frac{s+1}{s^2(s^2+9)} = \frac{\frac{1}{9}}{s} + \frac{1/9}{s^2} - \frac{1/9\,s}{s^2+9} - \frac{1/9}{(s^2+9)}$

$= \mathcal{L}^{-1}\left\{\frac{1}{9s}\right\} + \mathcal{L}^{-1}\left\{\frac{1}{9s^2}\right\} - \mathcal{L}^{-1}\left\{\frac{s}{9(s^2+9)}\right\} - \mathcal{L}^{-1}\left\{\frac{1}{9(s^2+9)}\right\}$

$= \frac{1}{9} + \frac{t}{9} - \frac{1}{9}\cos(3t) - \frac{1}{27}\sin(3t)$

$= \frac{1}{27}(3t - \sin(3t) - 3\cos(3t) + 3)$

(f) $\frac{w\cos\theta + s\sin\theta}{s^2+w^2}$

$= \frac{w\cos\theta}{s^2+w^2} + \frac{s\sin\theta}{s^2+w^2}$

$= \mathcal{L}^{-1}\left\{\frac{w\cos\theta}{s^2+w^2}\right\} + \mathcal{L}^{-1}\left\{\frac{s\sin\theta}{s^2+w^2}\right\}$

$= \cos\theta \sin(wt) + \sin\theta \cos(wt)$

$= \sin(wt+\theta)$

(g) $\frac{1}{(s+\sqrt{2})(s-\sqrt{3})} \equiv \frac{A}{s+\sqrt{2}} + \frac{B}{s-\sqrt{3}}$

$\Rightarrow 1 = A(s-\sqrt{3}) + B(s+\sqrt{2})$

$\Rightarrow 1 = As - \sqrt{3}A + Bs + \sqrt{2}B$

$\Rightarrow 1 = (A+B)s + (-A\sqrt{3} + B\sqrt{2})$

$A+B = 0$

$\therefore A = -B$

$-A\sqrt{3} + B\sqrt{2} = 1$

$\Rightarrow B\sqrt{3} + B\sqrt{2} = 1$

$\Rightarrow B(\sqrt{2}+\sqrt{3}) = 1$

$\therefore B = \frac{1}{\sqrt{3}+\sqrt{2}}$

$\therefore A = -\frac{1}{\sqrt{3}+\sqrt{2}}$

$\frac{1}{(s+\sqrt{2})(s-\sqrt{3})} = -\frac{1}{\sqrt{3}+\sqrt{2}} \cdot \frac{1}{s+\sqrt{2}} + \frac{1}{\sqrt{3}+\sqrt{2}} \cdot \frac{1}{s-\sqrt{3}}$

$= -\frac{1}{\sqrt{3}+\sqrt{2}} \mathcal{L}^{-1}\left\{\frac{1}{s+\sqrt{2}}\right\} + \frac{1}{\sqrt{3}+\sqrt{2}} \mathcal{L}^{-1}\left\{\frac{1}{s-\sqrt{3}}\right\}$

$= -\frac{1}{\sqrt{3}+\sqrt{2}} e^{-\sqrt{2}t} + \frac{1}{\sqrt{3}+\sqrt{2}} e^{\sqrt{3}t}$

(h) $\frac{6(s+1)}{s^4}$

$= \frac{6s+6}{s^4}$

$= \mathcal{L}^{-1}\frac{6s}{s^4} + \mathcal{L}^{-1}\frac{6}{s^4}$

$= 6\left(\frac{t^2}{2!}\right) + 6\left(\frac{t^3}{3!}\right)$

$= 6\left(\frac{t^2}{2!} + \frac{t^3}{3!}\right)$

(i) $\frac{12}{s^4} - \frac{228}{s^6}$

$= \mathcal{L}^{-1}\left\{\frac{12}{s^4}\right\} - \mathcal{L}^{-1}\left\{\frac{228}{s^6}\right\}$

$= 12\,\frac{t^3}{3!} - 228\,\frac{t^5}{5!}$

(j) $\frac{\pi}{(s+\pi)^2}$

$= \mathcal{L}^{-1}\left\{\frac{\pi}{(s+\pi)^2}\right\}$

$= \pi t e^{-\pi t}$

(k) $\frac{4s+32}{s^2+16}$

$= \mathcal{L}^{-1}\left\{\frac{4s}{s^2-4^2}\right\} + \mathcal{L}^{-1}\left\{\frac{32}{s^2-4^2}\right\}$

$= 4\cosh(4t) + 8\mathcal{L}^{-1}\left\{\frac{4}{s^2-4^2}\right\}$

$= 4\cosh(4t) + 8\sinh(4t)$

(l) $\frac{6}{(s+1)^3}$

$= \mathcal{L}^{-1}\left\{\frac{6}{(s+1)^3}\right\}$

$= 6\left(\frac{t^2}{2}\right)e^{-t}$

$= 3t^2e^{-t}$

(m) $\frac{s+10}{s^2-s-2}$

$= \frac{s+10}{\left(s-\frac{1}{2}\right)^2 - \left(\frac{3}{2}\right)^2}$

$= \frac{\left(s-\frac{1}{2}\right)+\frac{21}{2}}{\left(s-\frac{1}{2}\right)^2 - \left(\frac{3}{2}\right)^2}$

$= \frac{\left(s-\frac{1}{2}\right)}{\left(s-\frac{1}{2}\right)^2 - \left(\frac{3}{2}\right)^2} + \frac{\frac{21}{2}}{\left(s-\frac{1}{2}\right)^2 - \left(\frac{3}{2}\right)^2}$

$= e^{t/2}\cosh\left(\frac{3}{2}t\right) + 7e^{t/2}\sin\left(\frac{3}{2}t\right)$

(n) $\frac{21}{(s+\sqrt{2})^4}$

$= \mathcal{L}^{-1}\left\{\frac{21}{(s+\sqrt{2})^4}\right\}$

$= 21\left(\frac{t^3}{3!}\right)e^{-\sqrt{2}t}$

$= \frac{21}{6}t^3e^{-\sqrt{2}t}$

$= \frac{7}{2}t^3e^{-\sqrt{2}t}$

(o) $\frac{s+1}{s^2(s-3)} \equiv \frac{A}{s} + \frac{B}{s^2} + \frac{C}{s-3}$

$s+1 = As(s-3) + B(s-3) + Cs^2$

$\Rightarrow s+1 = As^2 - 3As + Bs - 3B + Cs^2$

$\Rightarrow s+1 = (A+C)s^2 + (-3A+B)s - 3B$

$A+C=0$ | $-3B=1$

$\Rightarrow A=-C$ | $\therefore B=-\frac{1}{3}$

$-3A+B=1$

$\Rightarrow -3A-\frac{1}{3}=1$

$\Rightarrow -3A=\frac{4}{3}$

$\therefore A=-\frac{4}{9}$

$\therefore C=\frac{4}{9}$

$$\frac{s+1}{s^2(s-3)} = -\frac{4}{9}\frac{1}{s} - \frac{1}{3}\frac{1}{s^2} + \frac{4}{9}\frac{1}{s-3}$$

$$= -\frac{4}{9}\mathcal{L}^{-1}\left\{\frac{1}{s}\right\} - \frac{1}{3}\mathcal{L}^{-1}\left\{\frac{1}{s^2}\right\} + \frac{4}{9}\mathcal{L}^{-1}\left\{\frac{1}{s-3}\right\}$$

$$= -\frac{4}{9} - \frac{1}{3}t + \frac{4}{9}e^{3t}$$

$$= \frac{4}{9}e^{3t} - \frac{1}{3}t - \frac{4}{9}$$

$$= \frac{1}{9}(4e^{3t} - 3t - 4)$$

$$= \frac{1}{9}(-3t + 4e^{3t} - 4)$$

(P) $\dfrac{4}{s^2-2s-3}$

$$= \frac{4}{(s-1)^2 - 4}$$

$$= \frac{4}{(s-1)^2 - 2^2}$$

$$= \mathcal{L}^{-1}\left\{\frac{4}{(s-1)^2 - 2^2}\right\}$$

$$= 2e^t \sinh(2t)$$

(9) $\dfrac{7.5}{s^2-2s-8}$

$$= \frac{7.5}{s^2 - 2s.1 + 1^2 - 8 - 1}$$

$$= \frac{7.5}{(s-1)^2 - 9}$$

$$= \frac{7.5}{(s-1)^2 - 3^2}$$

$$= 2.5\,\mathcal{L}^{-1}\left\{\frac{3}{(s-1)^2 - 3^2}\right\}$$

$$= 2.5\, e^t \sinh(3t)$$

$s^2+10\pi s+24\pi^2$

$$= \frac{\pi}{(s+5\pi)^2-\pi^2}$$

$$= \mathcal{L}^{-1}\left\{\frac{\pi}{(s+5\pi)^2-\pi^2}\right\}$$

$$= e^{-5\pi t}\sinh(\pi t)$$

(s) $\dfrac{1/16}{s^2+s+1/2}$

$$= \frac{1/16}{(s+1/2)^2+\frac{1}{4}}$$

$$= \mathcal{L}^{-1}\left\{\frac{1/16}{(s+1/2)^2+(\frac{1}{2})^2}\right.$$

$$= \frac{1}{8}\mathcal{L}^{-1}\left\{\frac{1/2}{(s+1/2)^2+(1/2)^2}\right.$$

$$= \frac{1}{8}e^{-t/2}\sin(t/2)$$

(t) $\dfrac{2s-1}{s^2-6s+18}$

$$= \frac{2s-1}{(s-3)^2+9}$$

$$= \frac{2(s-3)+5}{(s-3)^2+9}$$

$$= \frac{2(s-3)}{(s-3)^2+9} + \frac{5}{(s-3)^2+9}$$

$$= 2\mathcal{L}^{-1}\left\{\frac{(s-3)}{(s-3)^2+3^2}\right\} + \frac{5}{3}\mathcal{L}^{-1}\left\{\frac{3}{(s-3)^2+3^2}\right\}$$

$$= 2e^{3t}\cos(3t) + \frac{5}{3}e^{3t}\sin(3t)$$

(iv) $\frac{s-6.25}{s^2-6.25}$

$= \frac{s-6.25}{s^2-(2.5)^2}$

$= \frac{s}{s^2-(2.5)^2} - \frac{6.25}{s^2-(2.5)^2}$

$= \cosh(2.5t) - 2.5\sinh(2.5t)$

(v) $\frac{3}{s^2+s/4}$

$= \frac{3}{s(s+\frac{1}{4})} \equiv \frac{A}{s} + \frac{B}{s+1/4}$

$\Rightarrow 3 = A(s+1/4) + B(s)$

$\Rightarrow 3 = As + 1/4\,A + Bs$

$\Rightarrow 3 = (A+B)s + \frac{A}{4}$

$A+B=0$

$\therefore A = -B$

$\frac{A}{4} = 3$

$\therefore A = 12$

$\therefore B = -12$

$\frac{3}{s(s+1/4)} = \frac{12}{s} - \frac{12}{s+1/4}$

$= \mathcal{L}^{-1}\left\{\frac{12}{s}\right\} - \mathcal{L}^{-1}\left\{\frac{12}{s+1/4}\right\}$

$= 12 - 12e^{-t/4}$

$= 12(1-e^{-t/4})$

$= 3(4-4e^{-t/4})$

(iv) $\frac{20}{s^2-2\pi s^2}$

$= \frac{20}{s^2(s-2\pi)} \equiv \frac{A}{s} + \frac{B}{s^2} + \frac{C}{s-2\pi}$

$\Rightarrow 20 = A\,s(s-2\pi) + B(s-2\pi) + Cs^2$

$\Rightarrow 20 = As^2 - A2\pi s + Bs - B2\pi + Cs^2$

$\Rightarrow 20 = (A+C)s^2 + (-2\pi A+B)s + 2\pi B$

$A+C=0$

$\therefore A = -C$

$-2\pi A+B=0$

$-2\pi B = 20$

$\therefore B = -\frac{10}{\pi}$

$-2\pi A - \frac{10}{\pi} = 0$

$\Rightarrow A = -\frac{5}{\pi^2}$

$\therefore C = \frac{5}{\pi^2}$

$\frac{20}{s^2(s-2\pi)} = -\frac{5}{\pi^2}\cdot\frac{1}{s} - \frac{10}{\pi}\frac{1}{s^2} + \frac{5}{\pi^2}\frac{1}{s-2\pi}$

$= -\frac{5}{\pi^2}\mathcal{L}^{-1}\left\{\frac{1}{s}\right\} - \frac{10}{\pi}\mathcal{L}^{-1}\left\{\frac{1}{s^2}\right\} + \frac{5}{\pi^2}\mathcal{L}^{-1}\left\{\frac{1}{s-2\pi}\right\}$

$= -\frac{5}{\pi^2} - \frac{10}{\pi}t + \frac{5}{\pi^2}e^{2\pi t}$

$= \frac{5(-2\pi t + e^{2\pi t} - 1)}{\pi^2}$

(x) $\frac{1}{s^3 + as^2}$

$= \frac{1}{s^2(s+a)} \equiv \frac{A}{s} + \frac{B}{s^2} + \frac{C}{s+a}$

$\Rightarrow 1 = As(s+a) + B(s+a) + Cs^2$

$\Rightarrow 1 = As^2 + Aas + Bs + Ba + Cs^2$

$\Rightarrow 1 = (A+C)s^2 + (aA+B)s + Ba$

$A + C = 0$

$aA + B = 0$

$Ba = 1$

$\therefore B = \frac{1}{a}$

$aA + B = 0$

$\Rightarrow aA + \frac{1}{a} = 0$

$\therefore A = -\frac{1}{a^2}$

$\therefore C = \frac{1}{a^2}$

$\frac{1}{s^2(s-a)} = -\frac{1}{a^2} \cdot \frac{1}{s} + \frac{1}{a} \cdot \frac{1}{s^2} + \frac{1}{a^2} \cdot \frac{1}{s+a}$

$= -\frac{1}{a^2} \mathcal{L}^{-1}\left\{\frac{1}{s}\right\} + \frac{1}{a} \mathcal{L}^{-1}\left\{\frac{1}{s^2}\right\} + \frac{1}{a^2} \mathcal{L}^{-1}\left\{\frac{1}{s+a}\right\}$

$= -\frac{1}{a^2} + \frac{t}{a} + \frac{1}{a^2} e^{-at}$

$= \frac{at + e^{-at} - 1}{a^2}$

(2) $\frac{1}{s^4 - s^2}$

$= \frac{1}{s^2(s^2-1)}$

$= \frac{1}{s^2(s-1)(s+1)} \equiv \frac{A}{s} + \frac{B}{s^2} + \frac{C}{s-1} + \frac{D}{s+1}$

$\Rightarrow 1 = As(s-1)(s+1) + B(s-1)(s+1) + Cs^2(s+1) + Ds^2(s-1)$

$s = 0;$

$1 = B(-1)(1)$

$\therefore B = -1$

$s = 1$

$1 = C(1)^2(2)$

$\therefore C = 1/2$

$s = -1$

$1 = D(-1)(-2)$

$\therefore D = -1/2$

$s = 2$

$1 = 6A + 3B + 12C - 4D$

$\Rightarrow 1 = 6A - 3 + 6 + 2$

$\therefore A = -2/3$

$$\frac{1}{s^4 - s^2} = -\frac{2}{3}\cdot\frac{1}{s} - \frac{1}{s^2} + \frac{1}{2}\frac{1}{s-1} - \frac{1}{2}\frac{1}{s+1}$$

$$= -\frac{2}{3}\mathcal{L}^{-1}\left\{\frac{1}{s}\right\} - \mathcal{L}^{-1}\left\{\frac{1}{s^2}\right\} + \frac{1}{2}\mathcal{L}^{-1}\left\{\frac{1}{s-1}\right\} - \frac{1}{2}\mathcal{L}^{-1}\left\{\frac{1}{s+1}\right\}$$

$$= -\frac{2}{3} - t + \frac{1}{2}e^t - \frac{1}{2}e^{-t}$$

(2) $\frac{2s+1}{(s^2+1)(s-2)} = \frac{As+B}{s^2+1} + \frac{C}{s-2}$

$\Rightarrow 2s+1 = (As+B)(s-2) + C(s^2+1)$

$\Rightarrow 2s+1 = As^2 - 2As + Bs - 2B + Cs^2 + C$

$\Rightarrow 2s+1 = (A+C)s^2 + (-2A+B)s - 2B + C$

$A+C=0$

$\therefore C = -A$

$-2A+B=2$

$\Rightarrow -2A = 2-B$

$\therefore A = 4-B$

$-2B+C=1$

$\Rightarrow -2B = 1-C$

$\therefore B = \frac{1-C}{-2}$

$A = 4 - \frac{1-C}{-2}$

$= \frac{-8+1-C}{-2}$

$A+C=0$

$\therefore C = -A$

$-2A+B=2$

$-2B+C=1$

$\Rightarrow 2B+(-A)=1$

$\Rightarrow 2B+A=-1$

$\therefore B = \frac{-1-A}{2}$

$-2A + \frac{-1-A}{2} = 2$

$\Rightarrow -2A - \frac{1}{2} - \frac{A}{2} = 2$

$\Rightarrow -\frac{5}{2}A - \frac{1}{2} = 2$

$\Rightarrow -5/2\, A = 2 + 1/2$

$\therefore A = -1$

$\therefore C = -A$

$= 1$

$\therefore B = \frac{-1-(-1)}{2}$

$= 0$

$\frac{2s+1}{(s^2+1)(s-2)} = \frac{-s}{s^2+1} + \frac{1}{s-2}$

$= \mathcal{L}^{-1}\left\{\frac{-s}{s^2+1}\right\} + \mathcal{L}^{-1}\left\{\frac{1}{s-2}\right\}$

$= -\cos t + e^{2t}$

$= e^{2t} - \cos t$

Problem set-2.2

1. (a) $F(s) = 3\left(\frac{e^{-5s}}{s}\right)$

$\mathcal{L}^{-1}\{f(t)\} = 3\mathcal{L}^{-1}\left\{\frac{e^{-5s}}{s}\right\}$

$= 3\mathcal{L}^{-1}\left\{e^{-5s} \cdot \frac{1}{s}\right\}$

$= 3\,u(t-5) \cdot 1$

$= 3\,u(t-5)$

$F(s) = \frac{1}{s}$

$f(t) = 1$

$f(t-a) = t-5$

$\boxed{\mathcal{L}\{u(t-a)\} = \frac{e^{-as}}{s}}$

(b) $F(s) = 4\left(\frac{e^{-3s}}{s^2}\right)$

$= 4\mathcal{L}^{-1}\left\{\frac{e^{-3s}}{s^2}\right\}$

$= 4\mathcal{L}^{-1}\left\{e^{-3s} \cdot \frac{1}{s^2}\right\}$

$= 4(t-3)\,u(t-3)$

$F(s) = \frac{1}{s^2}$

$f(t) = t$

$f(t-a) = t-3$

(c) $F(s) = \frac{se^{-\pi s}}{s^2+25}$

$= e^{-\pi s}\mathcal{L}^{-1}\left\{\frac{s}{s^2+25}\right\}$

$= e^{-\pi s} \cdot \cos(5t)$

$= u(t-\pi) \cdot \cos[5(t-\pi)]$

$\mathcal{L}\{f(t)u(t-a)\} = e^{-as}\mathcal{L}\{f(t+a)\}$

$u(t-a)f(t-a) = \mathcal{L}^{-1}\{e^{-as}F(s)\}$

(d) $F(s) = \frac{2(e^{-3s} - 3e^{-4s})}{s}$

$= 2\left(\frac{e^{-3s}}{s} - 3\frac{e^{-4s}}{s}\right)$

~~$= 2u(t-3)$~~

$= 2\mathcal{L}^{-1}\left\{\frac{e^{-3s}}{s}\right\} - 6\mathcal{L}^{-1}\left\{\frac{e^{-4s}}{s}\right\}$

$= 2\,u(t-3) - 6\,u(t-4)$

(e) $F(s) = \frac{5(e^{-\pi s} + e^{-2\pi s})}{s^2+25}$

$= \mathcal{L}^{-1}\left\{\frac{e^{-\pi s} \cdot 5}{s^2+25}\right\} + \mathcal{L}^{-1}\left\{\frac{e^{-2\pi s} \cdot 5}{s^2+25}\right\}$

$= \mathcal{L}^{-1}\left\{e^{-\pi s} \cdot \frac{5}{s^2+5^2}\right\} + \mathcal{L}^{-1}\left\{e^{-2\pi s} \cdot \frac{5}{s^2+5^2}\right\}$

$= f(t-\pi)\,u(t-\pi) + f(t-2\pi)\,u(t-2\pi)$

$= u(t-\pi)\sin[5(t-\pi)] + u(t-2\pi)\sin[5(t-2\pi)]$

Dirac's delta function

(f) $F(s) = 1$

$\Rightarrow \mathcal{L}^{-1}\{1\} = \delta(t)$

$\therefore f(t) = \delta(t)$

(g) $F(s) = e^{-3s}$

$\Rightarrow \mathcal{L}^{-1}\{e^{-3s}\} = \delta(t-3)$

$\therefore f(t) = \delta(t-3)$

(h) $F(s) = 25e^{-2s}$

$\mathcal{L}^{-1}\mathcal{L}\{f(t)\} = \mathcal{L}^{-1}\{25e^{-2s}\}$

$\therefore f(t) = 25\,\delta(t-2)$

$\mathcal{L}\{f(t)\,\delta(t-a)\} = e^{-as} f(a)$

2. (a) $\frac{e^{-3s}}{(s-1)^3}$

Here,

$F_1(s) = \frac{1}{(s-1)^3}$

$\therefore \mathcal{L}^{-1}\{F_1(s)\} = e^t \cdot \frac{t^2}{2}$

$\therefore \mathcal{L}^{-1}\{F(s)\} = f(t-3) \cdot u(t-3)$

$= e^{t-3} \cdot \frac{(t-3)^2}{2} \cdot u(t-3)$

$= \frac{e^{t-3}(t-3)^2}{2} \cdot u(t-3)$

(b) $\frac{e^{-3s}}{s^4}$

Here,

$F_1(s) = \frac{1}{s^4}$

$\mathcal{L}^{-1}\{F_1(s)\} = \mathcal{L}^{-1}\{\frac{1}{s^4}\}$

$= e^3/3! \cdot$

$= \frac{1}{6} t^3$

$= f(t)$

Now,

$\mathcal{L}^{-1}\{F(s)\} = f(t-3)\, u(t-3)$

$= \frac{1}{6}(t-3)^3 u(t-3)$

(c) $\left(\frac{s+1}{s^2}\right) e^{-s}$

$F_1(s) = \frac{s+1}{s^2}$

$\mathcal{L}^{-1}\{F_1(s)\} = \mathcal{L}^{-1}\{\frac{s+1}{s^2}\}$

$= \mathcal{L}^{-1}\{\frac{s}{s^2}\} + \mathcal{L}^{-1}\{\frac{1}{s^2}\}$

$= 1 + t$

Now,

$\mathcal{L}^{-1}\{F(s)\} = f(t-1)\, u(t-1)$

$= (1 + t - 1)\, u(t-1)$

$= t\, u(t-1)$

(d) $\frac{6(1-e^{-\pi s})}{s^2+9}$

here,

$F_1(s) = \frac{6}{s^2+9}$

$= 6\left(\frac{1}{s^2+3^2}\right)$

$\mathcal{L}^{-1}\{F_1(s)\} = 6\mathcal{L}^{-1}\left\{\frac{1}{s^2+3^2}\right\}$

$= 6 \times \frac{1}{3}\sin 3t$

$= 2\sin 3t = f(t)$

Now,

$\mathcal{L}^{-1}\{F(s)\} = \mathcal{L}^{-1}\left\{\frac{6}{s^2+9}\right\} - \mathcal{L}^{-1}\left\{\frac{6e^{-\pi s}}{s^2+9}\right\}$

$= 2\sin 3t - 2\sin 3(t-\pi)u(t-\pi)$

(e) $\frac{2(e^{-s}-e^{-3s})}{s^2-4}$

$F(s) = \frac{2e^{-s}-2e^{-3s}}{s^2-4}$

$F(s) = \frac{2e^{-s}}{s^2-2^2} - \frac{2e^{-3s}}{s^2-2^2}$

here,

$f_1(s) = \frac{2}{s^2-2^2}$

$\therefore \mathcal{L}^{-1}\{F_1(s)\} = \mathcal{L}^{-1}\left\{\frac{2}{s^2-2^2}\right\}$

$= \sinh 2t$

$= f_1 t$

here,

$F_2(s) = \frac{2}{s^2-2^2}$

$\therefore \mathcal{L}^{-1}\{F_2(s)\} = \mathcal{L}^{-1}\left\{\frac{2}{s^2-2^2}\right\}$

$= \sinh 2t$

$= f_2 t$

Now,

$\mathcal{L}^{-1}\{F(s)\} = f(t-1)\,u(t-1) - f(t-3)\,u(t-3)$

$= \sinh 2(t-1)\,u(t-1) - \sinh 2(t-3)\,u(t-3)$

(g) $\left(\frac{2s-10}{s^3}\right)e^{-5s}$

here,

$$F_1(s) = \frac{2s-10}{s^3}$$

$$= \frac{2s}{s^3} - \frac{10}{s^3}$$

$$= \frac{2}{s^2} - \frac{10}{s^3}$$

$$\mathcal{L}^{-1}\{F_1(s)\} = \mathcal{L}^{-1}\left\{\frac{2}{s^2}\right\} - \mathcal{L}^{-1}\left\{\frac{10}{s^3}\right\}$$

$$= 2t - 5t^2$$

$$= f_1(t)$$

Now,

$$\mathcal{L}^{-1}\{F(s)\} = \mathcal{L}^{-1}\left\{\frac{2s-10}{s^3}\right\}$$

$$= u(t-5)\, f_1(t-5)$$

$$= u(t-5)\left[2(t-5) - 5(t-5)^2\right]$$

Problem set-3.1

1. (a) $y'(t) = 3 \; ; \; y(0) = 2$

$\Rightarrow \mathcal{L}\{y'(t)\} = \mathcal{L}\{3\}$

$\Rightarrow sY(s) - y(0) = \frac{3}{s}$

$\Rightarrow sY(s) - 2 = \frac{3}{s}$

$\Rightarrow sY(s) = \frac{3}{s} + 2$

$\Rightarrow Y(s) = \frac{3+2s}{s^2}$

Now,

$$\mathcal{L}^{-1}\{Y(s)\} = \mathcal{L}^{-1}\left\{\frac{2s+3}{s^2}\right\}$$

$$= \mathcal{L}^{-1}\left\{\frac{2s}{s^2} + \frac{3}{s^2}\right\}$$

$$= 2\mathcal{L}^{-1}\left\{\frac{1}{s}\right\} + 3\mathcal{L}^{-1}\left\{\frac{1}{s^2}\right\}$$

$$= 2 + 3t$$

(b) $y'(t) = 4t \; ; \; y(0) = 1$

$\Rightarrow \mathcal{L}^{-1}\{y'(t)\} = \mathcal{L}\{4t\}$

$\Rightarrow sY(s) - y(0) = \frac{4}{s^2}$

$\Rightarrow sY(s) - 1 = \frac{4}{s^2}$

$\Rightarrow sY(s) = \frac{4}{s^2} + 1$

$\Rightarrow Y(s) = \frac{4s}{s^3} + \frac{1}{s}$

Now,

$$\mathcal{L}^{-1}\{Y(s)\} = \mathcal{L}^{-1}\left\{\frac{4s}{s^3}\right\} + \mathcal{L}^{-1}\left\{\frac{1}{s}\right\}$$

$$= 4 \cdot \frac{t^2}{2} + 1$$

$$= 2t^2 + 1$$

(c) $y'(t) = 2t - 1 \; ; y(0) = 3$

$\Rightarrow \mathcal{L}\{y'(t)\} = \mathcal{L}\{2t - 1\}$

$\Rightarrow sY(s) - y(0) = \frac{2}{s^2} - \frac{1}{s}$

$\Rightarrow sY(s) - 3 = \frac{2}{s^2} - \frac{1}{s}$

$\Rightarrow sY(s) = \frac{2}{s^2} - \frac{1}{s} + 3$

$\Rightarrow Y(s) = \frac{2}{s^3} - \frac{1}{s^2} + \frac{3}{s}$

Now,

$\mathcal{L}^{-1}\{Y(s)\} = \mathcal{L}^{-1}\{\frac{2}{s^3}\} - \mathcal{L}^{-1}\{\frac{1}{s^2}\} + \mathcal{L}^{-1}\{\frac{3}{s}\}$

$= 2\frac{t^2}{2} - t + 3$

$= t^2 - t + 3$

(d) $y'(t) = t^2 \; ; y(0) = 4$

$\Rightarrow \mathcal{L}\{y'(t)\} = \mathcal{L}\{t^2\}$

$\Rightarrow sY(s) - y(0) = \frac{2}{s^3}$

$\Rightarrow sY(s) - 4 = \frac{2}{s^3}$

$\Rightarrow sY(s) = \frac{2}{s^3} + 4$

$\Rightarrow Y(s) = \frac{2}{s^4} + \frac{4}{s}$

Now,

$\mathcal{L}^{-1}\{Y(s)\} = \mathcal{L}^{-1}\{\frac{2}{s^4}\} + \mathcal{L}^{-1}\{\frac{4}{s}\}$

$= 2 \cdot \frac{t^3}{3} + 4$

$= \frac{2}{3}t^3 + 4$

(e) $y'(t) = e^{2t}$; $y(0) = 2$

$\Rightarrow \mathcal{L}\{y'(t)\} = \mathcal{L}\{e^{2t}\}$

$\Rightarrow sY(s) - y(0) = \frac{1}{s-2}$

$\Rightarrow sY(s) - 2 = \frac{1}{s-2}$

$\Rightarrow sY(s) = \frac{1}{s-2} + 2$

$\Rightarrow Y(s) = \frac{1}{s(s-2)} + \frac{2}{s}$

Now,

~~$\mathcal{L}^{-1}\{Y(s)\} = \mathcal{L}^{-1}\left\{\frac{1}{s(s-2)}\right\} + \mathcal{L}^{-1}\left\{\frac{2}{s}\right\}$~~

using partial fraction,

$\frac{1}{s(s-2)} \equiv \frac{A}{s} + \frac{B}{s-2}$

$\Rightarrow 1 = A(s-2) + Bs$

$\Rightarrow 1 = As - 2A + Bs$

$\Rightarrow 1 = (A+B)s - 2A$

$A + B = 0$

$\therefore B = -A$

$-2A = 1$

$\therefore A = -\frac{1}{2}$

$\therefore B = \frac{1}{2}$

$\frac{1}{s(s-2)} = -\frac{1}{2}\cdot\frac{1}{s} + \frac{1}{2}\cdot\frac{1}{s-2}$

$Y(s) = -\frac{1}{2}\frac{1}{s} + \frac{1}{2}\frac{1}{s-2} + \frac{2}{s}$

$= \frac{3}{2}\frac{1}{s} + \frac{1}{2}\frac{1}{s-2}$

$\mathcal{L}^{-1}\{Y(s)\} = \mathcal{L}^{-1}\left\{\frac{3}{2}\frac{1}{s}\right\} + \mathcal{L}^{-1}\left\{\frac{1}{2}\frac{1}{s-2}\right\}$

$= \frac{3}{2} + \frac{1}{2}e^{2t}$

~~(e) $y'(t) - 2y'(t) = \cos t$; $y(0) =$~~

~~$\Rightarrow \mathcal{L}\{y'(t)$~~

(f) $y'(t) + y(t) = 2$; $y(0) = 0$

$\mathcal{L}\{y'(t)\} + \mathcal{L}\{y(t)\} = \mathcal{L}\{2\}$

$\Rightarrow sY(s) - y(0) + Y(s) = \frac{2}{s}$

$\Rightarrow sY(s) - 0 .$

(f) $y'(t) + y(t) = 2$; $y(0) = 0$

$\mathcal{L}\{y'(t)\} = \mathcal{L}\{2\}$; ~~$\mathcal{L}\{y(t)\} = Y(s)$ ①~~

$\Rightarrow sY(s) - y(0) = \frac{2}{s}$

$\Rightarrow sY(s) = \frac{2}{s}$

$\Rightarrow Y(s) = \frac{2}{s^2}$

(f) $y'(t) + y(t) = 2$; $y(0) = 0$

here,

$\mathcal{L}\{y'(t)\} + \mathcal{L}\{y(t)\} = \mathcal{L}\{2\}$

$\Rightarrow sY(s) - y(0) + Y(s) = \frac{2}{s}$

$\Rightarrow (s+1)Y(s) = \frac{2}{s}$

$\Rightarrow Y(s) = \frac{2}{s(s+1)}$

Using partial fraction,

$\frac{2}{s(s+1)} \equiv \frac{A}{s} + \frac{B}{s+1}$

$\Rightarrow 2 = A(s+1) + Bs$

$\Rightarrow 2 = As + A + Bs$

$\Rightarrow 2 = (A+B)s + A$

$A + B = 0$

$\therefore A = -B$

$\therefore A = 2$

$\therefore B = -2$

$\therefore \frac{2}{s(s+1)} = \frac{2}{s} + \frac{2}{s+1}$

$= \mathcal{L}^{-1}\{\frac{2}{s}\} - \mathcal{L}^{-1}\{\frac{2}{s+1}\}$

$= 2 - 2e^{-t}$

$= 2(1 - e^{-t})$

(g) $y''(t) = 5 ; y(0) = 1 ; y'(0) = 2$

$\Rightarrow \mathcal{L}\{y''(t)\} = \mathcal{L}\{5\}$

$\Rightarrow s^2Y(s) - sy(0) - y'(0) = \frac{5}{s}$

$\Rightarrow s^2Y(s) - 1(s) - 2 = \frac{5}{s}$

$\Rightarrow s^2Y(s) = \frac{5}{s} + s + 2$

$\Rightarrow Y(s) = \frac{5}{s^3} + \frac{s}{s^2} + \frac{2}{s^2}$

$\mathcal{L}^{-1}\{Y(s)\} = \mathcal{L}^{-1}\{\frac{5}{s^3}\} + \mathcal{L}^{-1}\{\frac{s}{s^2}\} + \mathcal{L}^{-1}\{\frac{2}{s^2}\}$

$= 5\frac{t^2}{2} + 1 + 2t$

(h) $y''(t) - 2y'(t) = \cos t ; y(0) = 0 ; y'(0) = 1$

$\Rightarrow \mathcal{L}\{y''(t)\} - \mathcal{L}\{2y'(t)\} = \mathcal{L}\{\cos t\}$

$\Rightarrow s^2Y(s) - sy(0) - y'(0) - 2sY(s) - 2y(0) = \frac{s}{s^2+1}$

$\Rightarrow s^2Y(s) - 1 - 2sy(s) = \frac{s}{s^2+1}$

$\Rightarrow (s^2 - 2s)Y(s) = \frac{s}{s^2+1} + 1$

$\Rightarrow Y(s) = \frac{\frac{s}{s^2+1} + 1}{s^2 - 2s}$

$\Rightarrow Y(s) = \frac{s}{(s^2+1)s(s-2)} + \frac{1}{s(s-2)}$

Partial fraction for $\frac{1}{s(s-2)}$;

$\frac{1}{s(s-2)} = \frac{A}{s} + \frac{B}{s-2}$

$\Rightarrow 1 = A(s-2) + Bs$

$\Rightarrow 1 = As - 2A + Bs$

$\Rightarrow 1 = (A+B)s - 2A$

$A + B = 0$
$\therefore A = -B$

$-2A = 1$
$\therefore A = -1/2$
$B = 1/2$

So,

$$\frac{1}{s(s-2)} = -\frac{1}{2s} + \frac{1}{2(s-2)}$$

for,

$$\frac{1}{(s^2+1)(s-2)} \equiv \frac{As+B}{s^2+1} + \frac{C}{s-2}$$

$$\Rightarrow 1 = (As+B)(s-2) + C(s^2+1)$$

$$\Rightarrow 1 = As^2 - 2As + Bs - 2B + Cs^2 + C$$

$$\Rightarrow 1 = (A+C)s^2 + (-2A+B)s - 2B + C$$

$$A + C = 0$$

$$-2A + B = 0$$

$$-2B = 0$$

$$\therefore B = 0$$

$$C = 1$$

$$A + C = 0$$

$$\Rightarrow A = -1$$

2.(a) $y'' - y = t$; $y(0) = 1$; $y'(0) = 1$

$\Rightarrow \mathcal{L}\{y''\} - \mathcal{L}\{y\} = \mathcal{L}\{t\}$

$\Rightarrow s^2Y(s) - sy(0) - y'(0) - Y(s) = \frac{1}{s^2}$

$\Rightarrow s^2Y(s) - s1 - 1 - Y(s) = \frac{1}{s^2}$

$\Rightarrow (s^2-1)Y(s) = \frac{1}{s^2} + s + 1$

$\therefore Y(s) = \frac{\frac{1}{s^2} + s + 1}{s^2 - 1}$

$= \frac{s+1}{s^2-1} + \frac{1}{s^2(s^2-1)}$

$= \frac{s+1}{(s+1)(s-1)} + \frac{1}{s^2(s+1)(s-1)}$

$= \frac{1}{(s-1)} + \frac{1}{s^2(s+1)(s-1)}$

$$\frac{1}{s^2(s+1)(s-1)} = \frac{A}{s} + \frac{B}{s^2} + \frac{C}{s+1} + \frac{D}{s-1}$$

$\Rightarrow 1 = As(s+1)(s-1) + B(s+1)(s-1) + Cs^2(s-1) + Ds^2(s+1)$

$\Rightarrow 1 = As(s^2 - s + s - 1) + B(s^2 - s + s - 1) + Cs^2(s-1) + Ds^2(s+1)$

$\Rightarrow 1 = As^3 - As + Bs^2 - B + Cs^3 - Cs^2 + Ds^3 + Ds^2$

$\Rightarrow 1 = (A + C + D)s^3 + (B - C + D)s^2 - As - B$

$A + C + D = 0$

$B - C + D = 0$

$-A = 0$

$\therefore A = 0$

$B = -1$

$A + C + D = 0$

$\Rightarrow C = -D$

$B + D + D = 0$

$\Rightarrow -1 + 2D = 0$

$\Rightarrow 2D = 1$

$\therefore D = \frac{1}{2}$

$C = -\frac{1}{2}$

$$\frac{1}{s^2(s-1)(s+1)} = -\frac{1}{s^2} + \frac{1}{2(s-1)} + \frac{1}{2(s+1)}$$

$$y(s) = \frac{1}{s-1} - \frac{1}{s^2} + \frac{1}{2(s-1)} + \frac{1}{2(s+1)}$$

$$= \frac{3}{2(s-1)} + \frac{1}{2(s+1)} - \frac{1}{s^2}$$

$$= \mathcal{L}^{-1}\left\{\frac{3}{2(s-1)}\right\} + \mathcal{L}^{-1}\left\{\frac{1}{2(s+1)}\right\} - \mathcal{L}\left\{\frac{1}{s^2}\right\}$$

$$= \frac{3}{2}e^{t} + \frac{1}{2}e^{-t} - t$$

~~Problem set: 3.2~~

~~1.(a) $x' = y$; $y' = 16x$; $x(0) = 0$, $y(0) = 4$~~

Problem set-4.1

1. (a) $\frac{(1+i)^2}{1-i}$

$(1+i)^2 = 1 + 2i + i^2$
$= 1 + 2i - 1$
$= 2i$

$\frac{(1+i)^2}{1-i} = \frac{2i}{1-i}$

$= \frac{2i(1+i)}{(1-i)(1+i)}$

$= \frac{2i+2i^2}{1^2 - i^2}$

$= \frac{2i-2}{2}$

$= \frac{2(i-1)}{2}$

$= i - 1$

$= -1 + i$

(b) (a) $Re\left\{\frac{1+\sqrt{3}i}{1-i}\right\}$

$\frac{1+\sqrt{3}i}{1-i}$

$= \frac{(1+\sqrt{3}i)(1+i)}{(1-i)(1+i)}$

$= \frac{1 + \sqrt{3}i^2 + \sqrt{3}i + i}{1^2 - i^2}$

$= \frac{(1-\sqrt{3}) + (1+\sqrt{3})i}{2}$

$= \frac{1-\sqrt{3}}{2} + \frac{1+\sqrt{3}}{2}i$

$Re\left\{\frac{1+\sqrt{3}i}{1-i}\right\} = \frac{1-\sqrt{3}}{2}$

(b) $\left|\frac{z}{\bar{z}}\right|$

We know,
$|z| = |\bar{z}|$

$\left|\frac{z}{\bar{z}}\right| = \left|\frac{z}{\bar{z}}\right| = 1$

(c) $Im\left\{\frac{z}{\bar{z}}\right\}$

$\frac{z}{\bar{z}} = \frac{a+ib}{a-ib}$

$= \frac{(a+ib)(a+ib)}{(a-ib)(a+ib)}$

$= \frac{a^2 + aib + iab + i^2b^2}{a^2 - i^2b^2}$

$= \frac{a^2 + aib + iab - b^2}{a^2 + b^2}$

$= \frac{(a^2 - b^2) + (2ab)i}{a^2 + b^2}$

$\therefore Im\left\{\frac{z}{\bar{z}}\right\} = \frac{2ab}{a^2+b^2}$

(2) (i) $z = -1+i$

$r = \sqrt{(-1)^2+1^2}$

$= \sqrt{2}$

$(-1,1)$ is in the 2nd quadrant,

$\theta = \pi - \tan^{-1}\left|\frac{y}{x}\right|$

$= \pi - \tan^{-1}\left|\frac{1}{-1}\right|$

$= \pi - \frac{\pi}{4}$

$= \frac{4\pi-\pi}{4}$

$= \frac{3\pi}{4}$

Polar equation,

$z = r(\cos\theta + i\sin\theta)$

$= \sqrt{2}\left(\cos\frac{3\pi}{4} + i\sin\frac{3\pi}{4}\right)$

(ii) $z = -3-\sqrt{3}i$

$r = \sqrt{(-3)^2+(-\sqrt{3})^2}$

$= \sqrt{9+3}$

$= \sqrt{12}$

$= 2\sqrt{3}$

$(-3,-\sqrt{3}) \rightarrow$ 3rd quadrant

$\theta = -\pi + \tan^{-1}\left|\frac{y}{x}\right|$

$= -\pi + \tan^{-1}\left|\frac{-\sqrt{3}}{-3}\right|$

$= -\pi + \frac{\pi}{6}$

$= \frac{-6\pi+\pi}{6}$

$= \frac{-5\pi}{6}$

Polar equation,

$z = 2\sqrt{3}\left(\cos\left(-\frac{5\pi}{6}\right) + i\sin\left(-\frac{5\pi}{6}\right)\right)$

(iii) $z = \frac{(1-i)^2}{1+i}$

$(1-i)^2 = 1-2i+i^2$

$= -2i$

$z_1 = \frac{-2i}{1+i}$

$= \frac{-2i(1-i)}{(1+i)(1-i)}$

$= \frac{-2i+2i^2}{2}$

$= \frac{-2i(1-i)}{2}$

$= -1-i$

$r = \sqrt{(-1)^2+(-1)^2}$

$= \sqrt{2}$

$(-1,-1) \rightarrow$ 3rd,

$\theta = -\pi + \tan^{-1}\left|\frac{y}{x}\right|$

$= -\pi + \tan^{-1}\left|\frac{-1}{-1}\right|$

$= -\pi + \frac{\pi}{4}$

$= \frac{-4\pi+\pi}{4}$

$= \frac{-3\pi}{4}$

Polar,

$z = \sqrt{2}\left(\cos\left(-\frac{3\pi}{4}\right) + i\sin\left(-\frac{3\pi}{4}\right)\right)$

(iv) $z = \left(\frac{1-i}{1+i}\right)^{18}$

$\frac{1-i}{1+i}$

$= \frac{(1-i)(1-i)}{(1+i)(1-i)}$

$= \frac{1-i-i+i^2}{2}$

$= \frac{-2i}{2}$

$= -i$

So,

$z = (-i)^{18}$

$(-i)^2 = -1$

$(-i)^{18} = (-1)^9 = -1$

$r = 1$

$\theta = \pi$

Polar,

$z = 1(\cos\pi + i\sin\pi)$

(d) $z = \sqrt{3}e^{i\frac{\pi}{3}}$

$z = \sqrt{3}\left(\cos\frac{\pi}{3} + i\sin\frac{\pi}{3}\right)$

$\cos\frac{\pi}{3} = \frac{1}{2}$

$\sin\frac{\pi}{3} = \frac{\sqrt{3}}{2}$

$z = \sqrt{3}\left(\frac{1}{2} + i\frac{\sqrt{3}}{2}\right)$

$= \frac{\sqrt{3}}{2} + i\frac{3}{2}$

$z = 2e^{i\frac{\pi}{4}}$

$z = 2\left(\cos\frac{\pi}{4} + i\sin\frac{\pi}{4}\right)$

$\cos\frac{\pi}{4} = \sin\frac{\pi}{4} = \frac{\sqrt{2}}{2}$

$z = 2\left(\frac{\sqrt{2}}{2} + i\frac{\sqrt{2}}{2}\right)$

$= \sqrt{2} + i\sqrt{2}$

(e) (i) $z = (-1-i)^4$

$(-1,-1) \rightarrow$ 3rd

$\theta = -\pi + \tan^{-1}\left|\frac{-1}{-1}\right|$

$= -\pi + \frac{\pi}{4}$

$= \frac{-4\pi + \pi}{4}$

$= \frac{-3\pi}{4}$

multiply by 4,

$\text{Arg}(z) = 4 \cdot \left(-\frac{3\pi}{4}\right)$

$= -3\pi$

Reduce to $(-\pi, \pi)$

$-3\pi + 2\pi$

$= -\pi$

∴ Principal argument $= -\pi$

(ii) $z = (-2+2\sqrt{3}i)^3$

$(-2, 2\sqrt{3}) \rightarrow 2nd$

$\theta = \pi - \tan^{-1}\left|\frac{y}{x}\right|$

$= \pi - \tan^{-1}\left|\frac{2\sqrt{3}}{-2}\right|$

$= \pi - \frac{\pi}{3}$

$= \frac{2\pi}{3}$

$\text{Arg}(z) = 3.\frac{2\pi}{3}$

$= 2\pi$

Reduce to $(-\pi, \pi]$

$2\pi - 2\pi$

$= 0$

Principal argument 0.

(iii) $z = \frac{(1+i)^3}{(1-i)}$

$(1+i) \rightarrow 1st$

$\theta = \tan^{-1}\left|\frac{1}{1}\right|$

$= \frac{\pi}{4}$

$(1+i)^3 = 3.\frac{\pi}{4}$

$= \frac{3\pi}{4}$

$(1-i) \rightarrow 4th,$

$\theta = -\tan^{-1}\left|\frac{y}{x}\right|$

$= -\tan^{-1}\left|\frac{-1}{1}\right|$

$= -\frac{\pi}{4}$

$\text{Arg}(z) = \frac{3\pi}{4} - (-\frac{\pi}{4})$

$= \frac{3\pi}{4} + \frac{\pi}{4}$

$= \frac{4\pi}{4}$

$= \pi$

(f) (i) $z^5 + 9 = 0$

$z^5 = -9$

~~Polar form: $9 = 9e^{i\pi}$~~

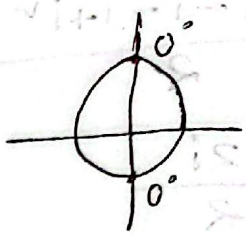
~~$z = \sqrt{9}e^{i\pi/2}$, $z = 3e^{i\pi/2}$~~

$\Rightarrow z = \sqrt{9}i = \pm 3i$

for, $z = 3i$

$r = 3$

$\theta = 0$



(ii) $z^3 - \sqrt{3} - i = 0$

$\Rightarrow z^3 = \sqrt{3} + i$

$\Rightarrow z = (\sqrt{3}+i)^{1/3}$

So, $r = 2$

$\theta = \tan^{-1}\left(\frac{1}{\sqrt{3}}\right)$

$= \frac{\pi}{6}$

So,

$z = \left(2e^{i\frac{\pi}{6}}\right)^{\frac{1}{3}}$

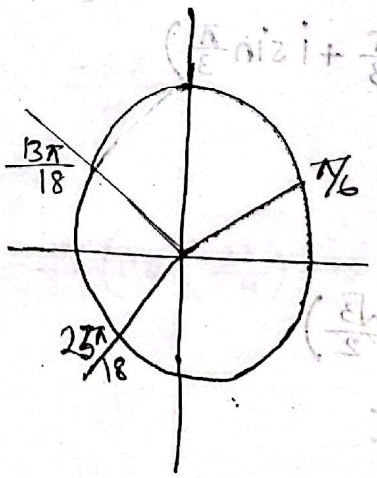
$= \left\{2e^{\left(i\frac{\pi}{6} + 2\pi n\right)}\right\}^{\frac{1}{3}}$

for,
$n = 0, 1, 2 \ldots$

$z_1 = \sqrt[3]{2}e^{i\frac{\pi}{18}}$

$z_2 = \sqrt[3]{2}e^{i\frac{13\pi}{18}}$

$z_3 = \sqrt[3]{2}e^{i\frac{25\pi}{18}}$



(iii) $z^3 = -i$

$\Rightarrow z = (0 - i)^{\frac{1}{3}}$

$r = 1$

$\theta = \frac{3\pi}{2}$ (direct in 3rd axis)

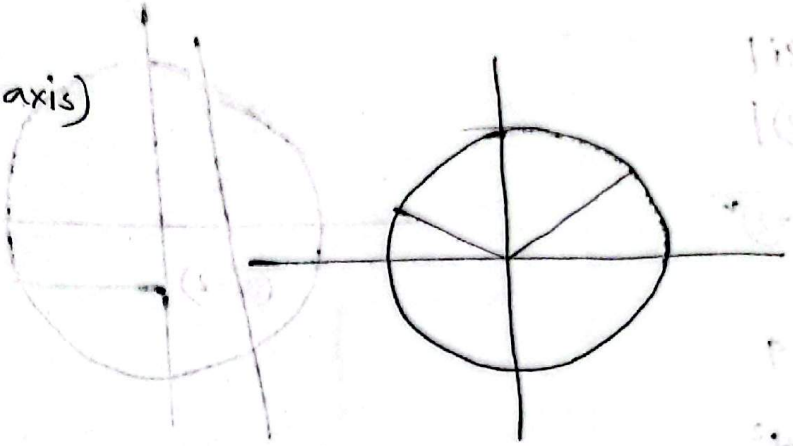
$z = e^{i(\frac{3\pi}{2} + 2\pi n)}$

So, $n = 0, 1, 2$

$z_1 = e^{i\frac{7\pi}{2}}$

$z_2 = e^{i\frac{11\pi}{2}}$

$z_3 = e^{i\frac{13\pi}{2}}$



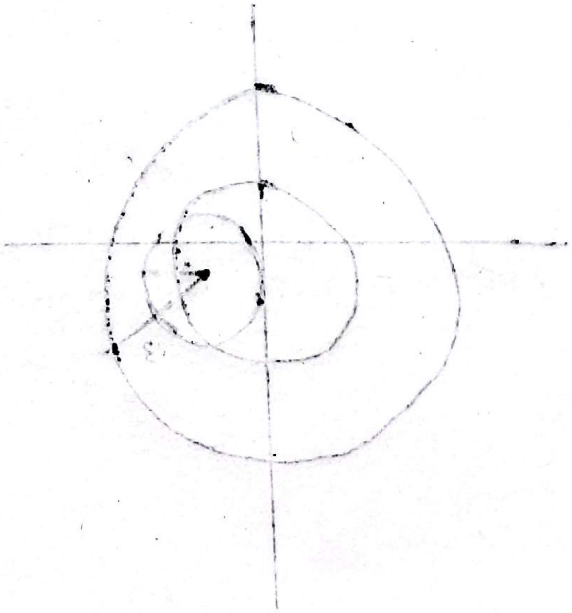
(iv) $z^4 - 1 = 0$

$\Rightarrow z^4 = 1$

$\Rightarrow z = (1 - 0)$

$\theta = 0$

So, $z = e^{i(0 + 2\pi n)}$



(8) (i) $|z+2i| > 4$

So,

$|z+2i|$

$= |x+iy+2i|$

$= |x+i(y+2)|$

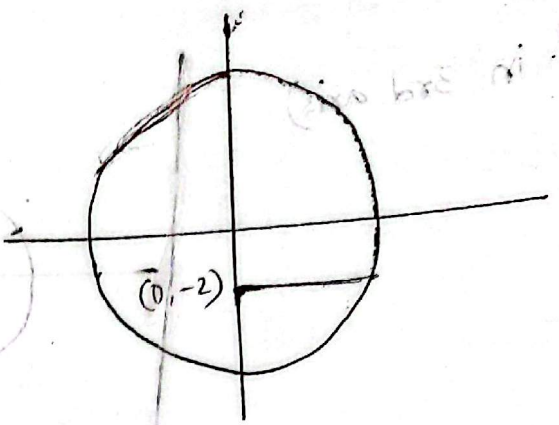
So, $r = \sqrt{x^2+(y+2)^2}$

Now,

$\sqrt{x^2+(y+2)^2} > 4$

$\therefore x^2+(y+2)^2 > 4^2$

Comparing with circle



(ii) $1 < |z-2+i| \leq 3$

So,

$|z-2+i|$

$\Rightarrow |x+iy-2+i|$

$\Rightarrow |(x-2)+i(y+1)|$

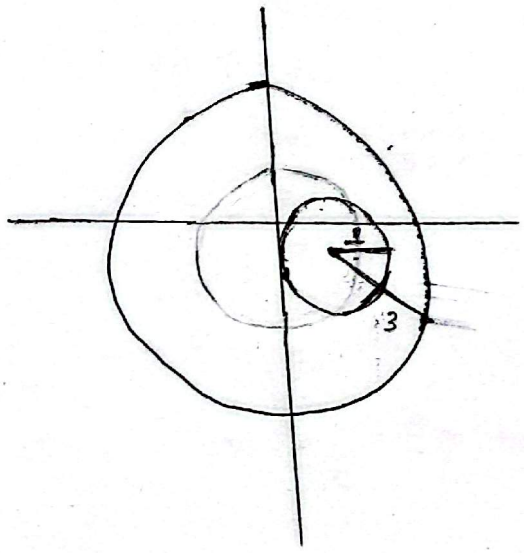
So,

$r = \sqrt{(x-2)^2+(y+1)^2}$

Now,

$1 < \sqrt{(x-2)^2+(y+1)^2} \leq 3$

$\Rightarrow 1 < (x-2)^2+(y+1)^2 \leq 3^2$



(iii) $\text{Im}\{z^2\} = 9$

So, $z^2 = (x+iy)^2$

$= x^2+i2xy-y^2$

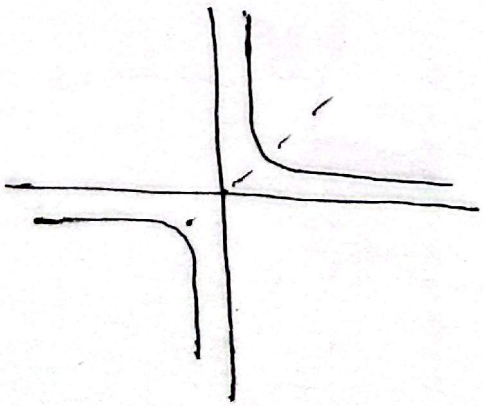
$= (x^2-y^2)+i2xy$

So, $\text{Im}\{z^2\} = 2xy$

So, $2xy = 9$

$xy = \frac{9}{2}$

$\therefore y = \frac{9}{2}x$



(iv) $|z-1| \leq 1$

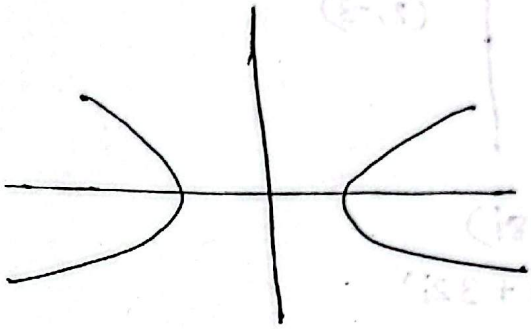
So,

$|x+iy-1| \leq 1$

$\Rightarrow (x-1)+iy \leq 1$

$\Rightarrow \sqrt{(x-1)+y^2} \leq 1$

$\Rightarrow (x-1)^2+(y-0)^2 \leq 1$



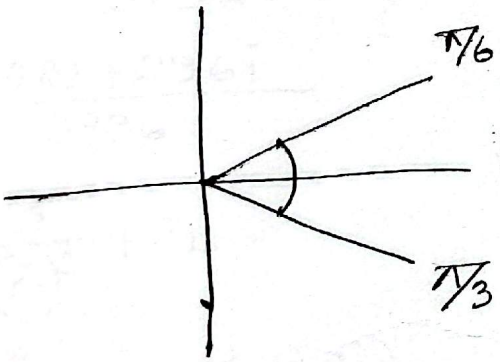
(v) $Re\{z^2\} < 4$

$\Rightarrow \{x^2+i2xy+i^2y^2\}\ Re < 4$

$\Rightarrow x^2-y^2 < 4$

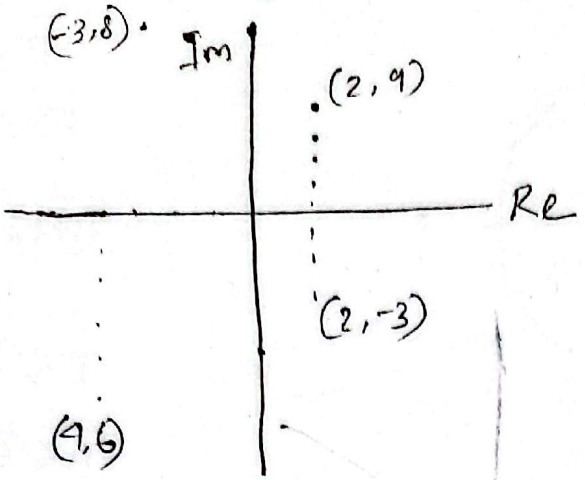
$\Rightarrow x^2-y^2 < 2^2$

(vi) $\frac{\pi}{6} \leq arg\, z \leq \frac{\pi}{3}$



2. (a) $z_1 = 2+4i$, $z_2 = -3+8i$, $z_3 = 2-3i$ and $z_4 = -4-6i$

(i) Locate them in Argand diagram.



(ii) $z_1 z_2 = (2+4i)(-3+8i)$

$= -6+16i-12i+32i^2$

$= -6+4i-32$

$= -38+4i$

(iii) $z_1 + z_2 = (2+4i)+(-3+8i)$

$= 2+4i-3+8i$

$= -1+12i$

(iv) $|z_1 \bar{z}_2 - z_3| = |(2+4i)(-3-8i)-(2-3i)|$

$= |(-6-16i-32i+12)-(2+3i)|$

$= |6-28i-32+3i|$

$= |-40-25i|$

$= \sqrt{(-40)^2+(-25)^2}$

$= 47.17$

(v) $\frac{z_3}{z_4} + \frac{z_1}{z_2}$

$= \frac{2-3i}{-4-6i} + \frac{2+9i}{-3+8i}$

$= \frac{(2-3i)(-4+6i)}{(-4-6i)(-4+6i)} + \frac{(2+9i)(-3-8i)}{(-3+8i)(-3-8i)}$

$= \frac{-8+12i+12i+18}{(-4)^2-(6i)^2} + \frac{-6-16i-12i+32}{(-3)^2-(8i)^2}$

$= \frac{10+24i}{16+36} + \frac{26-28i}{9+64}$

$= \frac{10+24i}{52} + \frac{26-28i}{73}$

$= \frac{73(10+24i)+52(26-28i)}{52\times 73}$

$= \frac{730+1752i+1352-1456i}{3796}$

$= \frac{2082+296i}{3796}$

(vi) $\frac{1}{(z_2)^3} + z_1$

$= \frac{1}{(2-3i)^3} + (2+9i)$

(b)(i) $(5-9i)+(2-4i)$

$= 7-13i$

(ii) $i(4-i)+4i(1+2i)$

$= 4i+1+4i-8$

$= -7+8i$

(iii) $3(4-i)-3(5+2i)$

$= 12-3i-15-6i$

$= -3-9i$

(iv) $i(5+7i)$

$= 5i-7$

(v) $\frac{i}{1+i}$

$= \frac{i(1-i)}{(1+i)(1-i)}$

$= \frac{i+1}{(1)^2-(i^2)}$

$= \frac{1}{2}+\frac{1}{2}i$

(vi) $\frac{(3-i)(2+3i)}{(1+i)}$

$= \frac{(6+9i-2i+3)(1-i)}{(1+i)(1-i)}$

$= \frac{(9+7i)(1-i)}{(1+1)}$

$= \frac{9-9i+7i+7}{2}$

$= 8-i$

(vii) $3i+\frac{1}{2-i}$

$= 3i+\frac{2+i}{(2-i)(2+i)}$

$= 3i+\frac{2+i}{4+1}$

$= \frac{2+i+15i}{5}$

$= \frac{2}{5}+\frac{16i}{5}$

(viii) $\frac{(4+5i)+2i^3}{(2+i)^2}$

$= \frac{4+5i-2i}{4+4i-1}$

$= \frac{4+3i}{3+4i}$

$= \frac{(4+3i)(3-4i)}{(3+4i)(3-4i)}$

$= \frac{12-16i+9i+12}{9+16}$

$= \frac{24-7i}{25}$

$= \frac{24}{25}-\frac{7}{25}i$

(ix) $\frac{10-5i}{6+2i} = \frac{(10-5i)(6-2i)}{(6+2i)(6-2i)} = \frac{60-20i-30i-10}{6^2+2^2}$

$= \frac{50-50i}{40} = \frac{5}{4}-\frac{5}{4}i$

(x) $\frac{2-4i}{3+5i}$

$= \frac{(2-4i)(3-5i)}{(3+5i)(3-5i)}$

$= \frac{6-10i-12i+20i^2}{3^2+5^2}$

$= \frac{6-10i-12i-20}{9+25}$

$= \frac{-14-22i}{34}$

$= -\frac{7}{17}-\frac{11}{17}i$

(x) $\frac{(5-4i)-(3+7i)}{(4+2i)+(2+3i)}$

$= \frac{5-4i-3-7i}{4+2i+2-3i}$

$= \frac{2-11i}{6-i}$

$= \frac{(2-11i)(6+i)}{(6-i)(6+i)}$

$= \frac{12+2i-66i-11i^2}{6^2+1^2}$

$= \frac{23-64i}{37}$

$= \frac{23}{37} - \frac{64i}{37}$

(xi) $\left(\frac{1}{2} - \frac{1}{4}i\right)\left(\frac{2}{3} + \frac{5}{3}i\right)$

$= \frac{3}{4} + \frac{2}{3}i$

(xii) $\frac{2-3i}{4+i}$

$= \frac{(2-3i)(4-i)}{(4+i)(4-i)}$

$= \frac{8-2i-12i-3}{16+1}$

$= \frac{5}{17} - \frac{14}{17}i$

(xiv) $\frac{(1+i)(1-2i)}{(2+i)(4-3i)}$

$= \frac{1-2i+i-2i^2}{8-6i+4i-3i^2}$

$= \frac{3-i}{11-2i}$

$= \frac{(3-i)(11+2i)}{(11-2i)(11+2i)}$

$= \frac{33+6i-11i-2i^2}{11^2+2^2}$

$= \frac{35-5i}{125}$

$= \frac{7}{25} - \frac{1}{25}i$

(c) let, $z = -6-5i$

(i) $Re\left(\frac{1}{z}\right)$

$= \frac{1}{-6-5i}$

$= \frac{-6+5i}{(-6-5i)(-6+5i)}$

$= \frac{-6+5i}{36+25}$

$= -\frac{6}{61}$

(ii) $Re(iz) = Re\{i(-6-5i)\}$

$= Re(-6i+5)$

$= 5$

(iii) $Re(z^2) = Re\{(-6-5i)^2\}$

$= Re(36-30i-25)$

$= 11$

(iv) $|z| = |-6-5i|$

$= \sqrt{36+25}$

$= \sqrt{61}$

(v) $\text{Im}(2z+4\bar{z}-4i)$

$= \text{Im}(-12-10i-24+20i-4i)$

$= \text{Im}(-34+6i)$

$= 6$

(vi) $(1+i)z$

$= (1+i)(-6-5i)$

$= -6-5i-6i+5$

$= -1-11i$

(vii) $\text{Im}(\bar{z}^2+z^2)$

$= \text{Im}\{(-6+5i)^2+(-6-5i)^2\}$

$= \text{Im}\{(36-60i-25)+(36+60i-25)\}$

$= \text{Im}(11-60i+11+60i)$

$= 60$

(viii) $\text{Im}(iz)$

$= \text{Im}(-6i+5)$

$= -6$

(d) (i) $z = (1-i)^2$

$= (1-2i+1)$

$= 2-2i$

So, $|z| = |2-2i|$

$= \sqrt{4}$

$= 2$

(ii) $z = \frac{2i}{3-4i} = \frac{2i(3+4i)}{(3-4i)(3+4i)}$

$= \frac{6i-8}{25}$

So, $|z| = \sqrt{\frac{64}{25^2}+\frac{36}{25^2}} = \sqrt{\frac{100}{625}} = \frac{10}{25} = \frac{2}{5}$

(iii) $z = i(2-i)$

$= 2i-i^2-4+i$

$= i-3$

$= (-3+i)$

So, $|z| = \sqrt{9+1}$

$= \sqrt{10}$

(iv) $z = \frac{1-2i}{1+i} + \frac{2-i}{1-i}$

$= \frac{(1-i)(2-2i)+(2-i)(1+i)}{(1+i)(1-i)}$

$= \frac{1-2i-i-2+2+2i-i+1}{1+1}$

$= \frac{2-2i}{2}$

$= \frac{2(1-i)}{2}$

$= 1-i$

So,

$|z| = \sqrt{2}$

(e) (i) $z = 1+i$

$\theta = \tan^{-1}\left(\frac{1}{1}\right)$

$= \frac{\pi}{4}$

(ii) $z = \frac{1-i}{(1+i)^2}$

$= \frac{1-i}{1+2i-1^2}$

$= \frac{(1-i)\,i}{2i \cdot i}$

$= \frac{i+1}{-2}$

$= -\frac{1}{2} - \frac{1}{2}i$

So, $\pi + \tan^{-1}\left(\frac{-1/2}{-1/2}\right)$

$= \pi + \tan^{-1}(1)$

$= \pi + \pi/4$

$= \frac{5\pi}{4}$

(iii) $z = -1+\sqrt{3}i$

$\theta = \pi - \tan^{-1}\left(\frac{\sqrt{3}}{-1}\right)$

$= \pi + (-\pi/3)$

$= \frac{2\pi}{3}$

(iv) $z = -5+5i$

$\theta = \pi + \tan^{-1}\left(\frac{5}{-5}\right)$

$= \pi - \pi/4$

$= \frac{3\pi}{4}$

(f) (i) $z = 2$

$r = 2$

$\theta = \tan^{-1}\left(\frac{0}{2}\right)$

$= 0$

So, $z = 2e^{i0}$

(ii) $z = \sqrt{3}+i$

$r = 2$

$\theta = \pi + \tan^{-1}\left(\frac{1}{-\sqrt{3}}\right)$

$= \pi - \frac{\pi}{6}$

$= \frac{5\pi}{6}$

So, $z = 2e^{i\frac{5\pi}{6}}$

~~(iii) $z = -1$~~

~~$r = 1$~~

~~$\theta = \pi + \tan^{-1}\left(\frac{1}{-\sqrt{3}}\right)$~~

~~$= \pi$~~

(iii) $z = -1$

$r = 1$

$\theta = \pi + \tan^{-1}\left(\frac{0}{-1}\right)$

$= \pi$

So, $z = e^{i\pi}$

(iv) $z = 1+i$

$r = \sqrt{2}$

$\theta = \tan^{-1}(1)$

$= \frac{\pi}{4}$

$z = \sqrt{2}e^{i\frac{\pi}{4}}$

(v) $z = -3i$

$r = 3$

$\theta = \frac{3\pi}{2}$

So, $z = 3e^{i\frac{3\pi}{2}}$

(vi) $z = -2 - 2\sqrt{3}i$

$r = 4$

$\theta = \pi + \tan^{-1}(\sqrt{3})$

$= \pi + \frac{\pi}{3}$

$= \frac{4\pi}{3}$

$z = 4e^{i\frac{4\pi}{3}}$

(vii) $z = 1 + i$

$r = \sqrt{2}$

$\theta = \tan^{-1}\left(\frac{1}{1}\right)$

$= \frac{\pi}{4}$

$z = \sqrt{2}\, e^{i\pi/4}$

(viii) $\frac{1+i}{(1-i)^2} = -\frac{1}{2} + \frac{1}{2}i$

$r = \frac{1}{\sqrt{2}}$

$\theta = \pi + \tan^{-1}\left(\frac{\frac{1}{2}}{-\frac{1}{2}}\right)$

$= \pi + (-\pi/4)$

$= \frac{3\pi}{4}$

$z = \frac{1}{\sqrt{2}} e^{i\frac{3\pi}{4}}$

(g) (i) $z = (-1)^{\frac{1}{4}}$

$r = 1$

$\theta = \pi + \tan^{-1}\left(\frac{0}{-1}\right)$

$= \pi$

So, $z = (e^{i\pi})^{\frac{1}{4}}$

$= e^{i\left(\frac{\pi + 2\pi n}{4}\right)}$

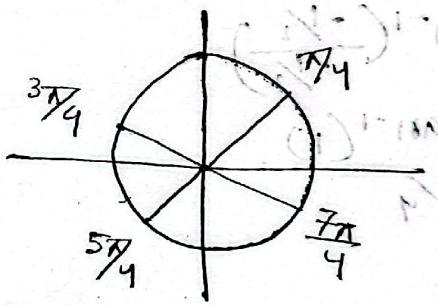
for, $n = 0, 1, 2, 3$

$z_1 = e^{i\pi/4}$

$z_2 = e^{i\,3\pi/4}$

$z_3 = e^{i\,5\pi/4}$

$z_4 = e^{i\,7\pi/4}$



(ii) $z = (-1+i)^{\frac{1}{5}}$

$r = \sqrt{2}$

$\theta = \pi + \tan^{-1}\left(\frac{1}{-1}\right)$

$= \pi - \pi/4$

$= \frac{3\pi}{4}$

So, $z = \left(\sqrt{2}\, e^{i\frac{3\pi}{4}}\right)^{\frac{1}{5}}$

$= 2^{\frac{1}{10}} e^{i\left(\frac{3\pi}{20} + 2\pi n\right)}$

$= \sqrt[10]{2}\, e^{i\left(\frac{3\pi+2\pi n}{20}\right)}$

$n = 0, 1, 2, 3, 4$

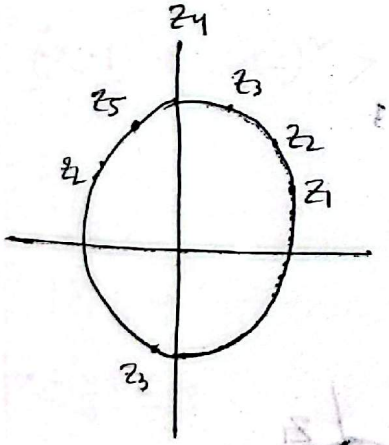
$z_1 = \sqrt[10]{2}\, e^{i\,3\pi/20}$

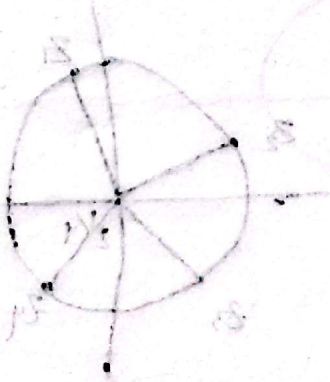
$z_2 = \sqrt[10]{2}\, e^{i\,5\pi/20}$

$z_3 = \sqrt[10]{2}\, e^{i\,7\pi/20}$

$z_4 = \sqrt[10]{2}\, e^{i\,9\pi/20}$

$z_5 = \sqrt[10]{2}\, e^{i\,11\pi/20}$





(iii) $z = (1+i)^{\frac{1}{5}}$

$r = \sqrt{2}$

$\theta = \pi/4$

$z = \left(\sqrt{2}\, e^{i\pi/4}\right)^{\frac{1}{5}}$

$= 2^{1/10}\, e^{i\left(\frac{\pi+2\pi n}{20}\right)}$

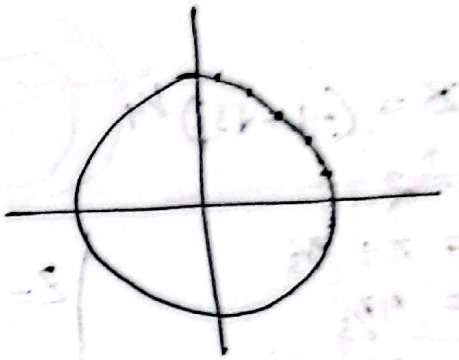
$z_1 = \frac{\pi}{20}$

$z_2 = 2^{\frac{1}{10}} e^{i\,3\pi/20}$

$z_3 = 2^{1/10} e^{i\,5\pi/20}$

$z_4 = 2^{\frac{1}{10}} e^{i\,7\pi/20}$

$z_5 = 2^{\frac{1}{10}} e^{i\,9\pi/20}$



(iv) $z = (-1+\sqrt{3}i)^{\frac{1}{2}}$

$r = 2$

$\theta = \pi + \tan^{-1}\left(\frac{\sqrt{3}}{-1}\right)$

$= \pi - \pi/3$

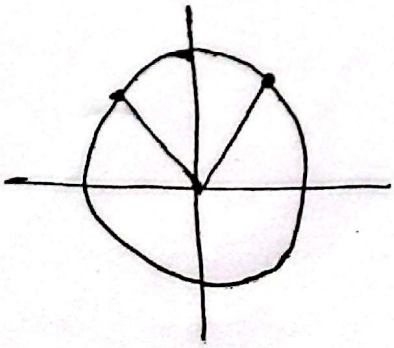
$= \frac{2\pi}{3}$

$z = \left(2e^{i\,\frac{2\pi + 2\pi n}{3}}\right)^{1/2}$

$= \sqrt{2}\, e^{i\,\frac{2\pi+2\pi n}{6}}$

$z_1 = \sqrt{2}\, e^{i\,\pi/3}$

$z_2 = \sqrt{2}\, e^{i\,2\pi/3}$



(V) $z = (8)^{\frac{1}{3}}$

$r = 8$

$\theta = 0$

$z = \{8e^{i(0+2\pi n)}\}^{\frac{1}{3}}$

$= \sqrt[3]{8}\, e^{i\frac{2\pi n}{3}}$

$z_1 = \sqrt[3]{8}\, e^{i0}$

$z_2 = \sqrt[3]{8}\, e^{2\pi/3\, i}$

$z_3 = \sqrt[3]{8}\, e^{4\pi/3\, i}$

(VI) $z = (-1-\sqrt{3}i)^{1/4}$

$r = 2$

$\theta = \pi + \pi/3$

$= 4\pi/3$

$z = (2e^{i4\pi/3})^{1/4}$

$= \sqrt[4]{2}\, e^{i\frac{4\pi+2\pi n}{12}}$

$z_1 = \sqrt[4]{2}\, e^{i\pi/3}$

$z_2 = \sqrt[4]{2}\, e^{\pi i}$

$z_3 = \sqrt[4]{2}\, e^{\frac{5\pi}{3}i}$

$z_4 = \sqrt[4]{2}\, e^{\frac{7\pi}{3}i}$

$z = (re^{i\theta})^{1/4}$

$= (2e^{i\frac{4\pi}{3}})^{1/4}$

$= 2^{1/4} e^{i\left(\frac{4\pi}{3}+2\pi n\right)^{1/4}}$

$= 2^{1/4} e^{i\left(\frac{4\pi+6\pi n}{3}\right)^{1/4}}$

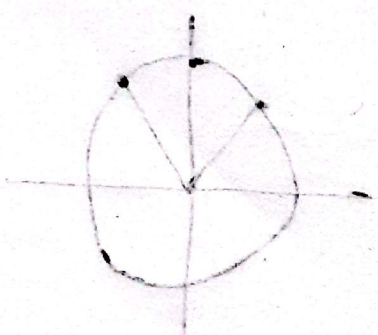
$= 2^{1/4} e^{i\left(\frac{4\pi+6\pi n}{12}\right)}$; $n = 0, 1, 2, 3$

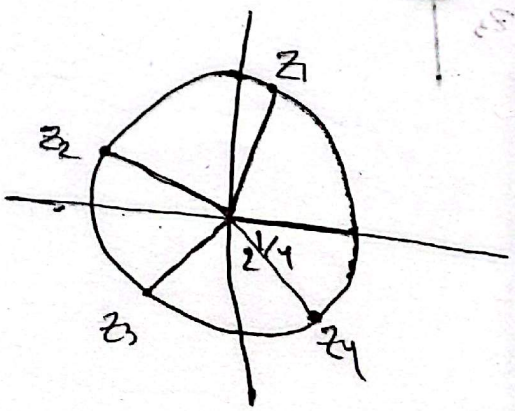
for, $n = 0$, $z_1 = 2^{1/4} e^{i\frac{4\pi}{12}}$

$n = 1$; $z_2 = 2^{1/4} \cdot e^{i\frac{10\pi}{12}}$

$n = 2$; $z_3 = 2^{1/4} \cdot e^{i\frac{16\pi}{12}}$

$n = 3$; $z_4 = 2^{1/4} \cdot e^{i\frac{22\pi}{12}}$





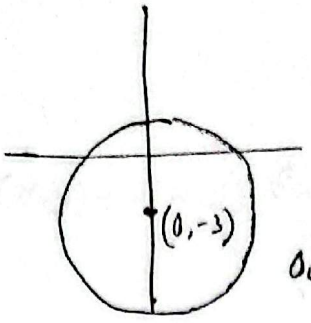
(h) locus represented.

(i) $|z+3i|>4$

So, $|x+iy+3i|$

$\Rightarrow \sqrt{x^2+(y+3)^2} > 4$

$\Rightarrow x^2+(y+3)^2 > 4^2$



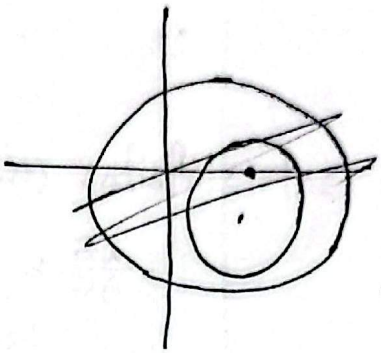
outside the circle
= region as >4

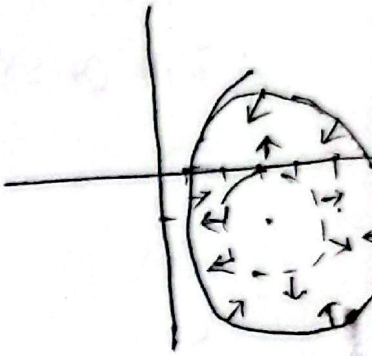
(ii) $1 < |z-5+i| \leq 2$

So, $|x+iy-5+i|$

$\Rightarrow 1 < (x-5)^2+(y+1)^2 \leq 2^2$

center $(5,-1)$

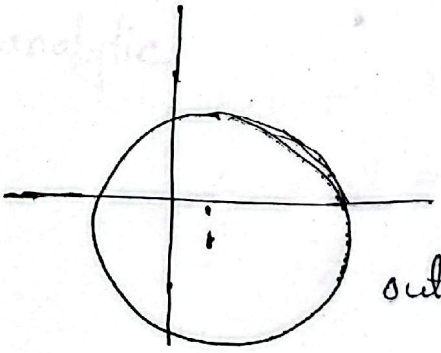




(iii) $|z-1+2i| > 5$

$\Rightarrow |x+iy-1+2i| > 5$

$\Rightarrow (x-1)^2+(y+2)^2 > 5^2$

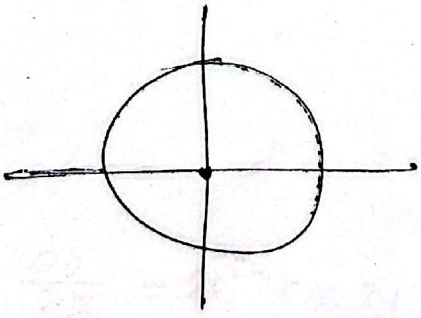


outside

(iv) $|z| \leq 4$

$\Rightarrow |x+iy| \leq 4$

$\Rightarrow x^2+y^2 \leq 4^2$



Problem set-5.1

① For Rectangular form:

If $z = x+iy$ and $f(z) = u(x,y)+v(x,y)$ satisfies the (C-R) that is

$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y}$ and $\frac{\partial u}{\partial y} = \frac{-\partial v}{\partial x}$

$f'(z) = \frac{\partial u}{\partial x} + i\frac{\partial v}{\partial x}$ or, $\frac{\partial v}{\partial y} + i\frac{-\partial u}{\partial y}$

Then $f(z)$ is said to be analytic

For Polar from:

If $z = re^{i\theta}$ and $f(z) = iu(r,\theta) + iv(r,\theta)$ satisfies the (C-R) that is,

$\frac{\partial u}{\partial r} = \frac{1}{r} \times \frac{\partial v}{\partial \theta}$ and $\frac{\partial v}{\partial r} = -\frac{1}{r} \times \frac{\partial u}{\partial \theta}$

$f'(z) = e^{i\theta}\left(\frac{\partial u}{\partial r} + i\frac{\partial v}{\partial r}\right)$

Then $f(z)$ is said to be analytic

② (i) $f(z) = e^{2z}$

$\Rightarrow f(z) = e^{2(x+iy)}$

$= e^{2x}(\cos 2y + i\sin 2y)$

So, $u = e^{2x}\cos 2y$

$v = e^{2x}\sin 2y$

(ii) C-R equations,

$\frac{\partial u}{\partial x} = 2e^{2x}\cos 2y$, $\frac{\partial v}{\partial y} = 2e^{2x}\cos 2y$

$\frac{\partial u}{\partial x} = -2e^{2x}\sin 2y$, $\frac{\partial v}{\partial y} = 2e^{2x}\sin 2y$

(iii) Derivative:

$f'(z) = u_x + iv_y$

$= 2e^{2x}\cos 2y + i(2e^{2x}\sin 2y)$

$= 2e^{2x}(\cos 2y + i\sin 2y) = 2e^{2z}$

for, $f(z) = 2z^2$

(i) $f(z) = 2(x+iy)^2$

$= 2(x^2 + 2ixy + i^2y^2)$

$= 2(x^2 + 2ixy - y^2)$

$= 2(x^2 - y^2) + i(4xy)$

$u = 2(x^2 - y^2)$.

$v = 4xy$

(ii) $u_x = \frac{\partial u}{\partial x}$

$= \frac{\partial}{\partial x}(2(x^2 - y^2))$

$= 4x$

$v_y = \frac{\partial v}{\partial y}$

$= \frac{\partial}{\partial y}(4xy)$

$= 4$

$u_y = \frac{\partial v}{\partial y}(2(x^2 - y^2))$

$= -4y$

$v_x = \frac{\partial}{\partial x}(4xy)$

$= 4y$

We know, for C-R equation,

$u_x = v_y$ and $v_x = -u_y$

So, varified C-R equation

(iii) $f'(z) = u_x + iv_x$

$= 4x + i4y$

$= 4(x+iy)$

$= 4z$

For, $f(z) = z^5$

$= (x+iy)^5$

$= x^5 + 5x^4(iy) + 10x^3(iy)^2 + 10x^2(iy)^3 + 5x(iy)^4 + (iy)^5$

$= x^5 + 5ix^4y - 10x^3y^2 - 10ix^2y^3 + 5xy^4 + iy^5$

Equating real and imaginary part,

$u = x^5 - 10x^3y^2 + 5xy^4$

$v = 5x^4y - 10x^2y^3 + y^5$

(11) $u = x^5 - 10x^3y^2 + 5xy^4$

$v = 5x^4y - 10x^2y^3 + y^5$

$u_x = 5x^4 - 30x^2y^2 + 5y^4$

$v_y = 5x^4 - 30x^2y^2 + 5y^4$

$u_y = -20x^3y + 20xy^3$

$v_x = 20x^3y - 20xy^3$

We know, for C-R equation,

$u_x = v_y$ and $v_y = -v_x$

So, varified C-R equation.

(12) $f'(z) = u_x + iv_x$

$= 5x^4 - 30x^2y^2 + 5y^4 + i(20x^3y - 20xy^3)$

$= 5[(x+iy)^4]$

$= 5z^4$

For, $f(z) = \frac{1}{z^3}$

$$\frac{1}{z^3} = \frac{1}{(x+iy)^3}$$

$$= \frac{(x-iy)^3}{(x^2+y^2)^3}$$

$$(x-iy)^3 = x^3 - 3x^2iy + 3x(iy)^2 - (iy)^3$$
$$= x^3 - 3ix^2y - 3xy^2 + iy^3$$
$$= (x^3 - 3xy^2) + i(-3x^2y + y^3)$$

So,

$$u = \frac{x^3 - 3xy^2}{(x^2+y^2)^3}$$

$$v = \frac{-3x^2y + y^3}{(x^2+y^2)^3}$$

(11) $r^2 = x^2 + y^2$

$$u = \frac{x^3 - 3xy^2}{r^6}$$

$$u_x = \frac{(3x^2 - 3y^2)r^6 - (x^3 - 3xy^2)(6r^4 \cdot x)}{r^{12}}$$

$$= \frac{(3x^2 - 3y^2)(x^2+y^2)^3 - 6x(x^3 - 3xy^2)(x^2+y^2)^2}{(x^2+y^2)^6}$$

$$v = \frac{-3x^2y + y^3}{r^6}$$

$$v_y = \frac{(-3x^2 + 3y^2)r^6 - (-3x^2y + y^3)(6r^4 y)}{r^{12}}$$

$$= \frac{3(3y^2 - 3x^2)(x^2+y^2)^3 - 6y(-3x^2y + y^3)(x^2+y^2)^2}{(x^2+y^2)^6}$$

$$u_y = \frac{(-6xy)r^6 - (x^3 - 3xy^2)(6r^4 y)}{r^{12}}$$

$$v_x = \frac{(-6xy)r^6 - (-3x^2y + y^3)(6r^4 \cdot y)}{r^{12}}$$

(11) $f'(z) = u_x + iv_y$

$$= \frac{(3x^2-3y^2)(x^2+y^2)^3 - (x^3-3xy^2)6x(x^2+y^2)^2}{(x^2+y^2)^6} + i\,\frac{(-6xy)(x^2+y^2)^3 - (-3x^2y+y^3)6x(x^2+y^2)^2}{(x^2+y^2)^6}$$

$$= -\frac{3}{z^4}$$

~~for f(z)~~

(3) $f'(z) = u_x + iv_x$

$f(z) = \bar{z}$

$\bar{z} = x - iy$

$u = x, \quad v = -y$

$u_x = 1, \quad v_y = -1$

$\therefore u_x \neq v_y$

$v_y = 0, \quad v_x = 0$

$\therefore v_y = -v_x$

But $u_x \neq v_y$

$\therefore$ not analytic

$f(z)$ does not exist.

$f(z) = 2z^2 + 3e^z$

$z^2 = (x+iy)^2$

$= x^2 - y^2 + i2xy$

$2z^2 = 2(x^2-y^2) + i(4xy)$

$\qquad\qquad u_1 \qquad\quad v_1$

$e^z = e^{x+iy}$

$= e^x(\cos y + i\sin y)$

$3e^z = 3e^x\cos y + i3e^x\sin y$

$u_2 = 3e^x\cos y, \quad v_2 = 3e^x\sin y$

$u = u_1 + u_2$

$= 2(x^2-y^2) + 3e^x\cos y$

$v = v_1 + v_2$

$= 4xy + 3e^x\sin y$

$u_x = 4x + 3e^x\cos y$

$v_y = 4x + 3e^x\cos y$

$\therefore u_x = v_y$

$u_y = -4y - 3e^x\sin y$

$v_x = 4y + 3e^x\sin y$

$\therefore v_y = -v_x$

5. (a) $f(z) = z^3$

So, $f(x+iy) = (x+iy)^3$

$= x^3 + 3x^2(iy) + 3x(iy)^2 + (yi)^3$

$= x^3 + 3x^2yi - 3xy^2 - y^3 i$

$= (x^3 - 3xy^2) + i(3x^2y - y^3)$

Now,

$u = x^3 - 3xy^2$

$v = 3x^2y - y^3$

So, $\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y}$ and $\frac{\partial v}{\partial x} = -\frac{\partial u}{\partial y}$

$\therefore \frac{\partial u}{\partial x} = 3x^2 - 3y^2$; $\frac{\partial v}{\partial y} = 3x^2 - 3y^2$

$\frac{\partial v}{\partial x} = 6xy$; $\frac{\partial u}{\partial y} = -6xy$

$\therefore \frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} = 3x^2 - 3y^2$

$\therefore \frac{\partial v}{\partial x} = -\frac{\partial u}{\partial y} = 6xy$

So, $f(t)$ satisfies the Cauchy-R equation at every point.

$f'(z) = \frac{\partial u}{\partial x} + i\frac{\partial v}{\partial u}$

$= 3x^2 - 3y^2 + i6xy$

(b) $f(z) = 3z^2 + 5z - 6i$

$f(x+iy) = 3(x+iy)^2 + 5(x+iy) - 6i$

$= 3\{x^2 + 2ixy + i^2y^2\} + 5x + 5iy - 6i$

$= 3x^2 + 6ixy - 3y^2 + 5x + 5iy - 6i$

$= (3x^2 - 3y^2 + 5x) + i(6xy + 5y - 6)$

$\therefore u = 3x^2 - 3y^2 + 5x$

$v = 6xy + 5y - 6$

We know,

$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y}$ and $\frac{\partial v}{\partial x} = -\frac{\partial u}{\partial y}$

$\frac{\partial u}{\partial x} = 6x + 5$; $\frac{\partial v}{\partial y} = 6x + 5$

$\frac{\partial v}{\partial x} = 6y$; $\frac{\partial u}{\partial y} = -6y$

$\therefore$ The equation satisfies C-R equation.

Now,

$f'(z) = \frac{\partial u}{\partial x} + i\frac{\partial v}{\partial x}$

$= 6x + 5 + i6y$

$= x^3 - 3xy^2 + 3xy^2 - iy^3 - x - iy$

$= (x^3 - 3xy^2 - x) + i(3x$

(c) $f(z) = z^3 - z$

$= (x+iy)^3 - (x+iy)$

$= x^3 + 3x^2(iy) + 3x(iy)^2 + (iy)^3 - (x+iy)$

$= x^3 + 3x^2 iy - 3xy^2 - iy^3 - x - iy$

$= (x^3 - 3xy^2 - x) + i(3x^2y - y^3 - y)$

$u = x^3 - 3xy^2 - x$

$v = 3x^2 - y^3 - y$

~~$U_x = 3x^2 - 3y^2 - 1$; $V_y = -3y^2 - 1$~~

~~$U_y = -6xy$; $V_x = 6x$~~

$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y}$ and $\frac{\partial v}{\partial x} = -\frac{\partial u}{\partial y}$

$\frac{\partial u}{\partial x} = 3x^2 - 3y^2 - 1$; $\frac{\partial v}{\partial y} = 3x^2 - 3y^2 - 1$

$\frac{\partial v}{\partial x} = 6xy$; $\frac{\partial u}{\partial y} = -6xy$

satisfies C-R equation.

Now,

$f'(z) = U_x + iV_x$

$= (3x^2 - 3y^2 - 1) + (6xy)i$

$= 3(x+iy)^2 - 1$

$= 3z^2 - 1$

(d) $f(z) = z(z^2 + z)$

$= z^3 + z^2$

$= (x+iy)^3 + (x+iy)^2$

$= \{(x^3 - 3xy^2) + i(3x^2y - y^3)\} + \{(x^2 - y^2) + i(2xy)\}$

$= (x^3 - 3xy^2 + x^2 - y^2) + i(3x^2y - y^3 + 2xy)$

$u = x^3 - 3xy^2 + x^2 - y^2$

$v = 3x^2y - y^3 + 2xy$

$u_x = 3x^2 - 3y^2 + 2x$; $v_y = 3x^2 - 3y^2 + 2$

$u_y = -6xy - 2y$; $v_x = 6xy + 2y$

$\therefore u_x = v_y$ and $-u_y = v_x$

satisfies C-R equation.

$f'(z) = u_x + iv_x$

$= (3x^2 - 3y^2 + 2x) + i(6xy + 2y)$

$= 3z^2 + 2z$

(e) $f(z) = \bar{z}$

$= x - iy$

$u = x$

$v = -y$

$u_x = 1$; $v_y = -1$

$u_y = 0$; $v_x = 0$

$\therefore$ not satisfies C-R equation

~~$f'(z) = u_x + iv_x$~~

~~$= 1 + i(-1)$~~

$\neq$

(f) $f(z) = i\bar{z}$

$= i(x - iy)$

$= ix - i^2y$

$= ix + y$

$u = y$

$v = x$

$u_x = 0 \; ; \; u_y = 0$

$u_y = 1 \; ; \; v_x = 1$

$u_x = v_y \; ; \; u_y \neq -v_x$

$\therefore$ C-R not satisfies.

(g) $f(z) = 4z - 6\bar{z} + 3$

$= 4(x + iy) - 6(x - iy) + 3$

$= (4x + 4iy) - (6x - 6iy) + 3 = (4x - 6x + 3) + i(4y + 6y)$

$= (-2x + 3) + i(10y)$

$u = -2x + 3$

$v = 10y$

$u_x = -2 \; ; \; v_y = 10$

$u_y = 0 \; ; \; v_x = 0$

$u_x \neq v_y \; ; \; v_y = -v_x$

$\therefore$ C-R not satisfied.

(h) $f(z) = \bar{z}^2$

$= (x - iy)^2$

$= x^2 - 2ixy + i^2y^2$

$= x^2 - y^2 - 2ixy$

$u = x^2 - y^2$

$v = -2xy$

$u_x = 2x \; ; \; v_y = -2x$

$u_y = -2y \; ; \; v_x = -2x$

$u_x \neq v_y \; ; \; u_y = -v_x$

$\therefore$ C-R not satisfied

(i) $f(z) = |z|^2 + iz\bar{z}$

$= z\bar{z} + iz\bar{z}$.

$= (1+i) z\bar{z}$

$= (1+i)(x^2+y^2)$

$z\bar{z} = x^2 + y^2$

$u = x^2 + y^2$

$v = x^2 + y^2$

$u_x = 2x \; ; \; v_y = 2y$

$u_y = 2y \; ; \; v_x = 2x$

$\therefore u_x \neq v_y \; ; \; v_y \neq -v_x$

$\therefore$ C-R not satisfied.

(j) $f(z) = z + \sin(z)$

$= (x+iy) + \sin(x+iy)$

$= (x+iy) + \sin x \cosh y + i \cos x \sinh y$

$= (x + \sin x \cosh y) + i(y + \cos x \sinh y)$.

$u = x + \sin x \cosh y$

$v = y + \cos x \sinh y$

$u_x = 1 + \cos x \cosh y$; $u_y = \sin x \sinh y$

$v_x = -\sin x \sinh y$; $v_y = 1 + \cos x \cosh y$

$\therefore u_x = v_y$

$u_y = -v_x$

C-R satisfied.

$f'(z) = u_x + i v_x$

$= 1 + \cos x \cosh y + i(-\sin x \sinh y)$

$= 1 + \cos z$

(k) $f(z) = e^z$

$= e^{x+iy}$

$= e^x (\cos y + i \sin y)$

$= e^x \cos y + i e^x \sin y$

$u = e^x \cos y$, $v = e^x \sin y$

$u_x = e^x \cos y$; $v_y = e^x \cos y$

$u_y = -e^x \sin y$; $v_x = -e^x \sin y$

$\therefore$ C-R satisfied.

$f'(z) = u_x + i v_x$

$= (e^x \cos y) + i(-e^x \sin y)$

$= e^z$

(l) $f(z) = 4z^2 + 5z + 9 - i$

$= 4(x+iy)^2 + 5(x+iy) + 9 - i$

$= 4(x^2 + 2xyi + i^2y^2) + 5x + 5iy + 9 - i$

$= 4x^2 + 8xyi - 4y^2 + 5x + 5iy + 9 - i$

$= (4x^2 - 4y^2 + 5x + 9) + i(8xy + 5y - 1)$

$u = 4(x^2 - y^2) + 5x + 9$

$v = 8xy + 5y - 1$

$u_x = 8x + 5 \; ; \; v_y = 8x + 5$

$u_y = -8y \; ; \; v_x = 8y.$

$\therefore$ C-R satisfied

$f'(z) = u_x + iv_x$

$= (8x + 5) + i(8y)$

$= 8(x + iy) + 5$

$= 8z + 5$

(m) $f(z) = z^3 + 6z$

$= (x+iy)^3 + 6(x+iy)$

$= x^3 - 3xy^2 + i(3x^2y - y^3) + 6x + 6iy$

$= (x^3 - 3xy^2 + 6x) + i(3x^2y - y^3 + 6y)$

$u = x^3 - 3xy^2 + 6x$

$v = 3x^2y - y^3 + 6y$

$u_x = 3x^2 - 3y^2 + 6 \; ; \; v_y = 3x^2 - 3y^2 + 6$

$u_y = -6xy \; ; \; v_x = 6xy$

$\therefore$ C-R satisfied.

$f'(z) = u_x + iv_x$

$= (3x^2 - 3y^2 + 6) + i(6xy)$

$= 3z^2 + 6$

(82) $f(z) = e^{\bar{z}} = \exp(\bar{z})$

$= e^{x-iy}$

$= e^{x}(\cos(-y) + i\sin(-y))$

$= e^{x}(\cos y - i\sin y)$

$u = e^{x}\cos y$

$v = -e^{x}\sin y$

$u_x = e^{x}\cos y \; ; \; v_y = -e^{x}\cos y$

$u_y = -e^{x}\sin y \; ; \; v_x = -e^{x}\sin y$

$\therefore$ C-R not satisfied.

$\therefore f'(z)$ does not exist

(c) $f(z) = e^{iz}$

$= e^{-i(x+iy)}$

$= e^{-ix+y}$

$= e^{y} \cdot e^{-ix}$

$= e^{y}(\cos(-x) + i\sin(-x))$

$= e^{y}(\cos x - i\sin x)$

$u = e^{y}\cos x, \; v = -e^{y}\sin x$

$u_x = -e^{y}\sin x \; ; \; v_y = -e^{y}\sin x$

$u_y = e^{y}\cos x \; ; \; v_x = -e^{y}\cos x$

$\therefore$ C-R satisfied.

$f'(z) = -e^{y}\sin x + i(-e^{y}\cos x)$

$= -ie^{-iz}$

(P) $f(z) = e^{z^2}$

$= e^{x^2-y^2+2xiy}$

$= e^{x^2-y^2}(\cos 2xy + i\sin 2xy)$

$u = e^{x^2-y^2}\cos(2xy)$

$v = e^{x^2-y^2}\sin(2xy)$

$u_x = e^{x^2-y^2}(2x\cos 2xy - 2y\sin 2xy)$

$v_x = e^{x^2-y^2}(2x\sin 2xy + 2y\cos 2xy)$

$\therefore$ C-R satisfied.

$f'(z) = e^{x^2-y^2}(2x\cos 2xy - 2y\sin 2xy) + i\, e^{x^2-y^2}(2x\sin 2xy + 2y\cos 2xy)$

$= 2z e^{z^2}$

(q) $f(z) = e^{-z}$

$= e^{-(x+iy)}$

$= e^{-x-iy}$

$= e^{-x}(\cos(-y) + i\sin(-y))$

$= e^{-x}(\cos y - i\sin y)$

$u = e^{-x}\cos y$

$v = -e^{-x}\sin y$

$u_x = -e^{-x}\cos y$; $u_y = e^{-x}\sin y$

$u_y = -e^{-x}\sin y$; $v_x = e^{-x}\sin y$

$\therefore$ C-R satisfied.

$f'(z) = (-e^{-x}\cos y) + i(e^{-x}\sin y)$

$= -e^{-z}$

(γ) $f(z) = e^{iz}$

$= e^{i(x+iy)}$

$= e^{ix-y}$

$= e^{-y} \cdot e^{ix}$

$= e^{-y}(\cos x + i \sin x)$

$u = e^{-y}\cos x$

$v = e^{-y}\sin x$

$u_x = -e^{-y}\sin x$; $v_y = e^{-y}\cos x - e^{-y}\sin x$

$v_x = -e^{-y}\cos x$; $u_y = -e^{-y}\cos x$

$\therefore$ C-R satisfied.

$f'(z) = (-e^{-y}\sin x) + i(-e^{-y}\cos x)$

$= ie^{iz}$

Polar form

(α) $f(z) = \frac{z}{|z|^2}$

$z = re^{i\theta}$

$\Rightarrow |z| = r$

$f(z) = \frac{z}{|z|} = \frac{re^{i\theta}}{r^2} = \frac{1}{r}e^{i\theta}$

$= \frac{1}{r}(\cos\theta + i\sin\theta)$

$u = \frac{\cos\theta}{r}$, $v = \frac{\sin\theta}{r}$

$u_r = -\frac{\cos\theta}{r^2}$, $v_r = -\frac{\sin\theta}{r^2}$

$u_\theta = -\frac{\sin\theta}{r}$, $v_\theta = \frac{\cos\theta}{r}$

Now, $u_r = \frac{1}{r}v_\theta$

$= \frac{1}{r} \cdot \frac{\cos\theta}{r} = \frac{\cos\theta}{r^2}$

and, $v_r = -\frac{1}{r}u_\theta$

$= -\frac{1}{r} \cdot -\frac{\sin\theta}{r} = \frac{\sin\theta}{r^2}$

$\therefore$ C-R not satisfied

$u = x^5 - 10x^3y^2 + 5xy^4$

$v = 5x^4y - 10x^2y^3 + y^5$

$u_x = 5x^4 - 30x^2y^2 + 5y^4$; $v_x = 20x^3y - 20xy^3$

$u_y = -20x^3y + 20xy^3$; $v_y = 5x^4 - 30x^2y^2 + 5y^4$

Now,

$u_r = \frac{1}{r} v_\theta$

$= \frac{1}{r}$.

(b) $f(z) = z^5$

$z = re^{i\theta}$

$z^5 = r^5 e^{i5\theta}$

$f(z) = r^5(\cos 5\theta + i\sin 5\theta)$

$u = r^5 \cos 5\theta$, $v = r^5 \sin 5\theta$

~~$u_r = \frac{1}{r} \cdot v_\theta$~~ $u_r = 5r^4 \cos 5\theta$; $v_r = 5r^4 \sin 5\theta$

~~$= \frac{1}{r}$~~ $v_\theta = 5r^5 \cos 5\theta$; $v_\theta = -5r^5 \sin 5\theta$

$v_r = \frac{1}{r} \cdot v_\theta$

$= \frac{1}{r} \cdot 5r^5 \cos 5\theta$

$= 5r^4 \cos 5\theta$

$v_r = -\frac{1}{r} \cdot u_\theta$

$= -\frac{1}{r} \cdot -5r^5 \sin 5\theta$

$= 5r^4 \sin 5\theta$

So, varified C-R equation.

$f'(z) = e^{i\theta}(u_r + iv_r)$

$= e^{i\theta}(5r^4 \cos 5\theta + i5r^4 \sin 5\theta)$

$= 5r^4 e^{-i\theta} e^{i5\theta} = 5r^4 e^{i4\theta} = 5z^4$

© $f(z) = \frac{1}{z^7} = z^{-7}$

$= r^{-7} e^{i7\theta}$

$= r^{-7}(\cos 7\theta - i \sin 7\theta)$

Now,

$u = r^{-7} \cos 7\theta$

$v = -r^{-7} \sin 7\theta$

$u_r = -7r^{-8} \cos 7\theta$; $v_r = 7r^{-8} \sin 7\theta$

$u_\theta = -7r^{-7} \sin 7\theta$; $v_\theta = -7r^{-7} \cos 7\theta$

Now,

$u_r = \frac{1}{r} \cdot (-7r^{-7} \sin 7\theta)$

$= -7r^{-8} \sin 7\theta$

and,

$v_r = -\frac{1}{r} \cdot (-7r^{-7} \cos 7\theta)$

$= 7r^{-8} \cos 7\theta$

$\therefore$ C-R satisfied.

$f'(z) = e^{-i\theta}(u_r + iv_r)$

$= e^{i\theta}(-7r^{-8} \sin 7\theta + i\, 7r^{-8} \cos 7\theta)$

$= e^{i\theta}(-7r^{-8} e^{-i7\theta})$

$= -7r^{-8} e^{-i8\theta}$

$= -7z^{-8}$

(d) $f(z) = \frac{\bar{z}}{|z|^2}$

$= \frac{re^{-i\theta}}{r^2}$

$= \frac{1}{r} e^{-i\theta}$

$= \frac{1}{r} (\cos(-\theta) + i\sin(-\theta))$

$= \frac{1}{r} (\cos\theta - i\sin\theta)$

$U = \frac{\cos\theta}{r}, \quad V = -\frac{\sin\theta}{r}$

$U_r = -\frac{\cos\theta}{r^2}; \quad V_r = \frac{\sin\theta}{r^2}$

$U_\theta = -\frac{\sin\theta}{r}; \quad V_\theta = -\frac{\cos\theta}{r}$

Now,

$U_r = \frac{1}{r} \cdot V_\theta$

$= \frac{1}{r} \cdot \left(-\frac{\cos\theta}{r}\right)$

$= -\frac{\cos\theta}{r^2}$

and,

$V_r = -\frac{1}{r} \cdot U_\theta$

$= -\frac{1}{r} \cdot \left(-\frac{\sin\theta}{r}\right)$

$= \frac{\sin\theta}{r^2}$

$\therefore$ C-R satisfied

$f'(z) = e^{i\theta} (U_r + iV_r)$

$= e^{-i\theta} \left(-\frac{\cos\theta}{r^2} + i\frac{\sin\theta}{r^2}\right)$

$= e^{-i\theta} \left(-\frac{1}{r^2} (\cos\theta - i\sin\theta)\right)$

$= e^{-i\theta} \left(-\frac{1}{r^2} e^{i\theta}\right)$

$= -\frac{1}{r^2} e^{-i2\theta}$

$= -\frac{1}{z^2}$

(e) $f(z) = \ln(z)$

$= \ln(re^{i\theta})$

$= \ln r + i\theta$

$u = \ln r$

$v = \theta$

$u_r = \frac{1}{r}$; $v_r = 0$

$u_\theta = 0$; $v_\theta = 1$

Now,

$u_r = \frac{1}{r} \cdot (1)$

$= \frac{1}{r}$

$v_r = -\frac{1}{r} \cdot (0)$

$= 0$

$\therefore$ C-R satisfied

$f'(z) = e^{-i\theta}(u_r + iv_r)$

$= e^{i\theta}\left(\frac{1}{r} + i.0\right)$

$= \frac{1}{r} e^{-i\theta}$

$= \frac{1}{z}$

(f) $f(z) = z^{1/2}$

$= (re^{i\theta})^{1/2}$

$= r^{1/2} e^{i\theta/2}$

$= r^{1/2}(\cos\theta/2 + i\sin\theta/2)$

Now,

$u = r^{1/2}\cos\theta/2$

$v = r^{1/2}\sin\theta/2$

$u_r = \frac{1}{2} r^{-1/2}\cos(\theta/2)$

$v_r = \frac{1}{2} r^{-1/2}\sin(\theta/2)$

$u_\theta = -\frac{1}{2} r^{1/2}\sin(\theta/2)$

$v_\theta = \frac{1}{2} r^{1/2}\cos(\theta/2)$

$u_r = \frac{1}{r} \cdot (-1/2\, r^{1/2}\sin\theta/2)$

$= \frac{1}{2} r^{-1/2}\cos\theta/2$

$v_r = \frac{1}{r}(-1/2\, r^{1/2}\sin\theta/2)$

$= \frac{1}{2} r^{-1/2}\sin\theta/2$

C-R satisfied.

$f'(z) = e^{-i\theta}\left(\frac{1}{2} r^{-1/2}\cos\theta/2 + i\,1/2\, r^{-1/2}\sin\theta/2\right)$

$= e^{-i\theta} \cdot \frac{1}{2} r^{-1/2} e^{i\theta/2}$

$= \frac{1}{2} r^{-1/2} e^{-i\theta/2}$

$= \frac{1}{2} \frac{1}{r^{1/2} e^{i\theta/2}}$

$= \frac{1}{2} z^{-1/2}$

Problem set - 3.2

1. (a) $x' = y$; —(i)
$y' = 16x$; —(ii)
$x(0) = 0, y(0) = 4$

$x' = y$
$\Rightarrow \mathcal{L}\{x'\} = \mathcal{L}\{y\}$
$\Rightarrow sX(s) - x(0) = Y(s)$
$\Rightarrow sX(s) = Y(s)$
$\therefore sX(s) - Y(s) = 0$ —(iii)

$y' = 16x$
$\Rightarrow \mathcal{L}\{y'\} = \mathcal{L}\{16x\}$
$\Rightarrow sY(s) - Y(0) = 16X(s)$
$\Rightarrow sY(s) - 4 - 16X(s) = 0$
$\Rightarrow 16X(s) - sY(s) = -4$ —(iv)

Using Cramer's rule,

$$X(s) = \frac{\begin{vmatrix} 0 & -1 \\ -4 & -s \end{vmatrix}}{\begin{vmatrix} s & -1 \\ 16 & -s \end{vmatrix}} = \frac{0-4}{-s^2+16} = \frac{-4}{-s^2+16} = \frac{4}{s^2-16}$$

So, $\mathcal{L}^{-1}\{X(s)\} = \mathcal{L}^{-1}\left\{\frac{4}{s^2-16}\right\} = \sinh(4t)$

$\therefore x(t) = \sinh(4t)$

Again,

$$Y(s) = \frac{\begin{vmatrix} s & 0 \\ 16 & -4 \end{vmatrix}}{\begin{vmatrix} s & -1 \\ 16 & -s \end{vmatrix}} = \frac{(-4s-0)}{-s^2+16} = \frac{4s}{s^2-16}$$

So, $\mathcal{L}^{-1}\{Y(s)\} = \mathcal{L}^{-1}\left\{\frac{4s}{s^2-16}\right\} = 4\cosh(4t)$

$\therefore y(t) = 4\cosh(4t)$

(b) $x' = -4y$ — (i)

$y' = x$ — (ii) $x(0) = 2,\ y(0) = 0$

$x' = -4y$

$\Rightarrow \mathcal{L}\{x'\} = \mathcal{L}\{-4y\}$

$\Rightarrow sX(s) - x(0) = -4Y(s)$

$\Rightarrow sX(s) - 2 = -4Y(s)$

$\therefore sX(s) + 4Y(s) = 2$ — (iii)

$y' = x$

$\Rightarrow \mathcal{L}\{y'\} = \mathcal{L}\{x\}$

$\Rightarrow sY(s) - y(0) = X(s)$

$\Rightarrow sY(s) - 0 = X(s)$

$\Rightarrow sY(s) - X(s) = 0$ — (iv)

Using cramer rule,

$$X(s) = \frac{\begin{vmatrix} 2 & 4 \\ 0 & s \end{vmatrix}}{\begin{vmatrix} s & 4 \\ -1 & s \end{vmatrix}} = \frac{2s - 0}{s^2 + 4} = \frac{2s}{s^2 + 4}$$

So, $\mathcal{L}^{-1}\{X(s)\} = \mathcal{L}^{-1}\left\{\frac{2s}{s^2+4}\right\} = 2\cos(2t)$

Again,

$$Y(s) = \frac{\begin{vmatrix} 4 & 2 \\ -1 & 0 \end{vmatrix}}{\begin{vmatrix} s & 4 \\ -1 & s \end{vmatrix}} = \frac{0 + 2}{s^2 + 4} = \frac{2}{s^2 + 2^2}$$

So, $\mathcal{L}^{-1}\{Y(s)\} = \mathcal{L}^{-1}\left\{\frac{2}{s^2+2^2}\right\} = \sin(2t)$

$\therefore y(t) = \sin(2t)$

(c) $x' = 2x + y$ — (i) $y' = 4x + 2y$ — (ii) ; $x(0) = -1$, $y(0) = 6$

$\mathcal{L}\{x'\} = \mathcal{L}\{2x + y\}$

$\Rightarrow sX(s) - x(0) = 2X(s) + Y(s)$

$\Rightarrow sX(s) - (-1) = 2X(s) + Y(s)$

$\Rightarrow sX(s) + 1 = 2X(s) + Y(s)$

$\Rightarrow sX(s) - 2X(s) - Y(s) = -1$

$\Rightarrow X(s)(s-2) - Y(s) = -1$ — (iii)

$\mathcal{L}\{y'\} = \mathcal{L}\{4x + 2y\}$

$\Rightarrow sY(s) - y(0) = 4X(s) + 2Y(s)$

$\Rightarrow sY(s) - 6 = 4X(s) + 2Y(s)$

$\Rightarrow sY(s) - 2Y(s) - 4X(s) = 6$

$\therefore Y(s)(s-2) - 4X(s) = 6$ — (iv)

Using cramer's rule,

$$X(s) = \frac{\begin{vmatrix} -1 & 1 \\ 6 & (s-2) \end{vmatrix}}{\begin{vmatrix} (s-2) & -1 \\ -4 & (s-2) \end{vmatrix}} = \frac{-6 - s + 2}{s^2 - 4s + 4 - 4} = \frac{-s - 4}{s^2 - 4} = \frac{-(s+4)}{s(s-4)}$$

So, $\mathcal{L}^{-1}\{X(s)\} = \mathcal{L}^{-1}\left\{\frac{A}{s} + \frac{B}{s-4}\right\}$

$\Rightarrow x(t) = \mathcal{L}^{-1}\left\{\frac{1}{s}\right\} + \mathcal{L}^{-1}\left\{\frac{-2}{s-4}\right\}$

$= 1 - 2e^{4t}$

$A = \frac{-(0+4)}{-4} = 1$

$B = \frac{-(4+4)}{4} = -2$

$$Y(s) = \frac{\begin{vmatrix} (s-2) & 1 \\ -4 & 6 \end{vmatrix}}{\begin{vmatrix} (s-2) & -1 \\ -4 & (s-2) \end{vmatrix}} = \frac{6s - 12 + 4}{s^2 - 4s + 4 - 4} = \frac{6s - 8}{s^2 - 4s} = \frac{6s - 8}{s(s-4)}$$

So, $\mathcal{L}^{-1}\{Y(s)\} = \mathcal{L}^{-1}\left\{\frac{A}{s}\right\} + \mathcal{L}^{-1}\left\{\frac{B}{s-4}\right\}$

$= \mathcal{L}^{-1}\left\{\frac{2}{s}\right\} + \mathcal{L}^{-1}\left\{\frac{4}{s-4}\right\}$

$= 2 + 4e^{4t}$

$A = \frac{-8}{-4} = 2$

$B = \frac{24 - 8}{4} = 4$

(1) $x' = 3x + y$; — (i) $y' = 7x + 3y$ — (ii) $x(0) = 3, y(0) = 2$

$\Rightarrow \mathcal{L}\{x'\} = \mathcal{L}\{3x + y\}$

$\Rightarrow sX(s) - x(0) = 3X(s) + Y(s)$

$\Rightarrow sX(s) - 3 = 3X(s) + Y(s)$

$\Rightarrow sX(s) - 3X(s) - Y(s) = 3$

$\therefore X(s)(s-3) - Y(s) = 3$ — (iii)

$\Rightarrow \mathcal{L}\{y'\} = \mathcal{L}\{7x + 3y\}$

$\Rightarrow sY(s) - Y(0) = 7X(s) + 3Y(s)$

$\Rightarrow sY(s) - 2 = 7X(s) + 3Y(s)$

$\Rightarrow sY(s) - 7X(s) - 3Y(s) = 2$

$\Rightarrow Y(s)(s-3) - 7X(s) = 2$ — (iv)

Using cramer's rule,

$$X(s) = \frac{\begin{vmatrix} 3 & -1 \\ 2 & (s-3) \end{vmatrix}}{\begin{vmatrix} (s-3) & -1 \\ -7 & (s-3) \end{vmatrix}} = \frac{3s - 9 + 2}{(s-3)^2 - 7} = \frac{3s-7}{(s-3)^2 - 4} = \frac{3(s-3) + 2}{(s-3)^2 - 4}$$

So, $\mathcal{L}^{-1}\{X(s)\} = 3\mathcal{L}^{-1}\left\{\frac{s-3}{(s-3)^2 - 7}\right\} + \mathcal{L}^{-1}\left\{\frac{2}{(s-3)^2 - 7}\right\}$

$= 3e^{3t}\mathcal{L}^{-1}\left\{\frac{s}{s^2 - 2^2}\right\} + e^{3t}\mathcal{L}^{-1}\left\{\frac{2}{s^2 - 2^2}\right\}$

$= 3e^{3t}\cosh(2t) + e^{3t}\sinh(2t)$

$$Y(s) = \frac{\begin{vmatrix} (s-3) & 3 \\ -7 & 2 \end{vmatrix}}{\begin{vmatrix} (s-3) & -1 \\ -7 & (s-3) \end{vmatrix}} = \frac{2s - 6 + 12}{(s-3)^2 - 7} = \frac{2s + 6}{(s-3)^2 - 4} = \frac{2(s-3) + 12}{(s-3)^2 - 4}$$

So, $\mathcal{L}^{-1}\{Y(s)\} = 2\mathcal{L}^{-1}\left\{\frac{s-3}{(s-3)^2 - 4}\right\} + \mathcal{L}^{-1}\left\{\frac{12}{(s-3)^2 - 4}\right\}$

$= 2e^{3t}\mathcal{L}^{-1}\left\{\frac{s}{s^2 - 2^2}\right\} + 6e^{3t}\mathcal{L}^{-1}\left\{\frac{2}{s^2 - 2^2}\right\}$

$= 2e^{3t}\cosh(2t) + 6e^{3t}\cosh(2t)$

2. @ $y_1' = -y_1 + 4y_2$ — (i) $y_2' = 3y_1 - 2y_2$ — (ii) $y_1(0) = 3, y_2(0) = 4$

$\Rightarrow \mathcal{L}\{y'\} = \mathcal{L}\{-y_1 + 4y_2\}$

$\Rightarrow s Y_1(s) + y_1(0) = -Y_1(s) + 4Y_2(s)$

$\Rightarrow s Y_1(s) + Y_1(s) - 4Y_2(s) = 3$

$\therefore Y_1(s)(s+1) - 4Y_2(s) = 3$ — (iii)

$\Rightarrow \mathcal{L}\{y_2'\} = \mathcal{L}\{3y_1 - 2y_2\}$

$\Rightarrow s Y_2(s) - y_2(0) = 3Y_1(s) - 2Y_2(s)$

$\Rightarrow s Y_2(s) + 2Y_2(s) - 3Y_1(s) = 4$

$\therefore Y_2(s)(s+2) - 3Y_1(s) = 4$ — (iv)

Using cramer rule,

$$Y_1(s) = \frac{\begin{vmatrix} 3 & -4 \\ 4 & s+2 \end{vmatrix}}{\begin{vmatrix} s+1 & -4 \\ -3 & s+2 \end{vmatrix}} = \frac{3s+6+16}{s^2+2s+s+2-12} = \frac{3s+22}{s^2+3s-10} = \frac{3s+22}{s^2+5s-2s-10} = \frac{3s+22}{(s+5)(s-2)}$$

So,

$\mathcal{L}^{-1}\{Y_1(s)\} = \mathcal{L}^{-1}\left\{\frac{A}{s+5} + \frac{B}{s-2}\right\}$ $\therefore$

$\Rightarrow y(t) = \mathcal{L}^{-1}\left\{\frac{-1}{s+5}\right\} + \mathcal{L}^{-1}\left\{\frac{4}{s-2}\right\}$

$= -e^{-5t} + 4e^{2t}$

$A = \frac{15+22}{-7}$

$A(s-2) + B(s+5) = 3s+22$

$= As - 2A + Bs + 5B = 3s + 22$

$\Rightarrow (A+B)s + (5B-2A) = 3s+22$

$\Rightarrow A + B = 3$

$\therefore B = 3 - A$

$= 3 - (-1)$

$= 4$

$5B - 2A = 22$

$\Rightarrow 5(3-A) - 2A = 22$

$\Rightarrow 15 - 5A - 2A = 22$

$\Rightarrow 15 - 7A = 22$

$\therefore A = -1$

Again,

$$Y_2(s) = \frac{\begin{vmatrix} (s+1) & 3 \\ -3 & 4 \end{vmatrix}}{\begin{vmatrix} (s+1) & -4 \\ -3 & (s+2) \end{vmatrix}} = \frac{4s+4+9}{s^2+2s+s+2-12} = \frac{4s+13}{s^2+3s-10}$$

$$= \frac{4s+13}{s^2+5s-2s-10} = \frac{4s+13}{(s+5)(s-2)}$$

So,

$\mathcal{L}^{-1}\{Y_2(s)\} = \mathcal{L}^{-1}\left\{\frac{A}{(s+5)} + \frac{B}{(s-2)}\right\}$

$= \mathcal{L}^{-1}\left\{\frac{1}{s+5} + \frac{3}{s-2}\right\}$

$= e^{-5t} + 3e^{2t}$

$4s+13 = A(s-2) + B(s+5)$

$\Rightarrow 4s+13 = As - 2A + Bs + 5B$

$\Rightarrow 4s+13 = (A+B)s + (5B-2A)$

$A + B = 4$

$\therefore B = 4 - A$

$= 4 - 1$

$= 3$

$5B - 2A = 13$

$\Rightarrow 5(4-A) - 2A = 13$

$\Rightarrow 20 - 5A - 2A = 13$

$\Rightarrow 20 - 7A = 13$

$\Rightarrow -7A = 13 - 20$

$\Rightarrow -7A = -7$

$\therefore A = 1$

(1) $y_1' = 5y_1 + y_2$ — (i) $y_2' = y_1 + 5y_2$ — (ii) $y_1(0) = 1, y_2(0) = 3$

$\mathcal{L}\{y_1'\} = \mathcal{L}\{5y_1 + y_2\}$

$\Rightarrow sY_1(s) - y_1(0) = 5Y_1(s) + Y_2(s)$

$\Rightarrow sY_1(s) - 1 = 5Y_1(s) + Y_2(s)$

$\Rightarrow sY_1(s) - 5Y_1(s) + Y_2(s) = 1$

$\therefore Y_1(s)(s-5) - Y_2(s) = 1$

$\mathcal{L}\{y_2'\} = \{y_1 + 5y_2\}$

$\Rightarrow sY_2(s) - y_2(0) = Y_1(s) + 5Y_2(s)$

$\Rightarrow sY_2(s) - 3 = Y_1(s) + 5Y_2(s)$.

$\Rightarrow sY_2(s) - Y_1(s) - 5Y_2(s) = 3$

$\Rightarrow Y_2(s)(s-5) - Y_1(s) = 3$

Using cramer rule,

$$Y_1 = \frac{\begin{vmatrix} 1 & -1 \\ -3 & (s-5) \end{vmatrix}}{\begin{vmatrix} (s-5) & -1 \\ -1 & (s-5) \end{vmatrix}} = \frac{s-5-3}{s^2-10s+25-1} = \frac{s-8}{s^2-10s+24} = \frac{s-8}{(s-6)(s-4)}$$

$\mathcal{L}^{-1}\{Y_1(s)\} = \mathcal{L}^{-1}\left\{\frac{A}{s-6} + \frac{B}{s-4}\right\}$

$\Rightarrow y_1(t) = \mathcal{L}^{-1}\left\{\frac{-1}{s-6} + \frac{2}{s-4}\right\}$

$= -e^{6t} + 2e^{4t}$

$s-8 = A(s-4) + B(s-6)$

$\Rightarrow s-8 = As - 4A + Bs - 6B$

$\Rightarrow s-8 = (A+B)s + (-4A-6B)$

$A+B = 1$

$\therefore B = 1-A$

$= 1-(-1)$

$= 2$

$-4A - 6B = -8$

$\Rightarrow -4A - 6(1-A) = -8$

$\Rightarrow -4A - 6 + 6A = -8$

$\Rightarrow 2A = -2$

$\therefore A = -1$

$$Y_2 = \frac{\begin{vmatrix} (s-5) & 1 \\ -1 & -3 \end{vmatrix}}{\begin{vmatrix} (s-5) & -1 \\ -1 & (s-5) \end{vmatrix}} = \frac{-3s+15+1}{s^2-10s+25-1} = \frac{16-3s}{(s-6)(s-4)}$$

$\Rightarrow \mathcal{L}^{-1}\{Y_2(s)\} = \mathcal{L}^{-1}\left\{\frac{A}{(s-6)} + \frac{B}{(s-4)}\right\}$

$\Rightarrow y_2(t) = \mathcal{L}^{-1}\left\{\frac{-1}{s-6} + \frac{-2}{s-4}\right\}$

$= -e^{6t} - 2e^{4t}$

$16-3s = A(s-4) + B(s-6)$

$\Rightarrow 16-3s = As - 4A + Bs - 6B$

$\Rightarrow 16-3s = (A+B)s + (-4A-6B)$

$A+B = -3$

$\therefore B = -3-A$

$= -3-(-1)$

$= -2$

$-4A - 6B = 16$

$\Rightarrow -4A - 6(-3-A) =$

$\Rightarrow -4A + 18 + 6A =$

$\Rightarrow 2A = -2$

$\therefore A = -1$

(c) $y_1' = 4y_1 + y_2$ — (I) $y_2' = -y_1 + 2y_2$ — (II) $y_1(0) = 3$, $y_2(0) = 1$

$\Rightarrow \mathcal{L}\{y_1'\} = \mathcal{L}\{4y_1 + y_2\}$

$\Rightarrow sY_1(s) - y_1(0) = 4Y_1(s) + Y_2(s)$

$\Rightarrow sY_1(s) - 3 = 4Y_1(s) + Y_2(s)$

$\Rightarrow sY_1(s) - 4Y_1(s) - Y_2(s) = 3$

$\Rightarrow Y_1(s)(s-4) - Y_2(s) = 3$ — (III)

$\mathcal{L}\{y_2'\} = \mathcal{L}\{-y_1 + 2y_2\}$

$\Rightarrow sY_2(s) - y_2(0) = -Y_1(s) + 2Y_2(s)$

$\Rightarrow sY_2(s) - 1 = -Y_1(s) + 2Y_2(s)$

$\Rightarrow sY_2(s) + Y_1(s) - 2Y_2(s) = 1$

$\Rightarrow Y_2(s)(s-2) + Y_1(s) = 1$ — (IV)

Using cramer rules,

$$Y_1 = \frac{\begin{vmatrix} 3 & -1 \\ 1 & (s-2) \end{vmatrix}}{\begin{vmatrix} (s-4) & -1 \\ 1 & (s-2) \end{vmatrix}} = \frac{3s-6+1}{s^2-6s+8+1} = \frac{3s-5}{s^2-6s+9} = \frac{3s-5}{s^2-3s-3s+9} = \frac{3s-5}{(s-3)(s-3)}$$

$$\mathcal{L}^{-1}\{Y_1(s)\} = \mathcal{L}^{-1}\left\{\frac{A}{(s-3)} + \frac{B}{(s-3)^2}\right\}$$

$\Rightarrow y_1(t) = \mathcal{L}^{-1}\left\{\frac{3}{s-3} + \frac{4}{(s-3)^2}\right\}$

$= 3e^{3t} + 4\mathcal{L}^{-1}\left\{\frac{1}{(s-3)^2}\right\}$

$= 3e^{3t} + 4te^{3t}$

$3s-5 = A(s-3) + B$

$\Rightarrow 3s-5 = As - 3A + B$

$\Rightarrow 3s-5 = As + (B-3A)$

$A = 3$

$B - 3A = -5$

$\Rightarrow B - 9 = -5$

$\therefore B = 4$

$3s-5 = A(s-3) + B(s-3) + B$

$\Rightarrow 3s-5 = As - 3A + Bs - 3B + B$

$\Rightarrow 3s-5 = (A+B)s + (-3A-3B)$

$A + B = 3$

$\therefore B = 3 - A$

$-3A - 3B = -5$

$\Rightarrow -3A - 3(3-A) = -5$

$\Rightarrow -3A - 9 + 3A = 5$

$\Rightarrow$

Again,

$$Y_2(s) = \frac{\begin{vmatrix} (s-4) & 3 \\ 1 & 1 \end{vmatrix}}{\begin{vmatrix} (s-4) & -1 \\ 1 & (s-2) \end{vmatrix}} = \frac{s-4-3}{s^2-6s+8+1} = \frac{s-7}{(s-3)^2}$$

$\Rightarrow \mathcal{L}^{-1}\{Y_2(s)\} = \mathcal{L}^{-1}\left\{\frac{A}{s-3} + \frac{B}{(s-3)^2}\right\}$

$\Rightarrow y_2(t) = \mathcal{L}^{-1}\left\{\frac{1}{s-3} + \frac{-4}{(s-3)^2}\right\}$

$= e^{3t} - 4e^{3t}t$

$s-7 = A(s-3) + B$

$\Rightarrow s-7 = As - 3A + B$

$\Rightarrow s-7 = As + (B-3A)$

$A = 1$

$B - 3A = -7$

$\Rightarrow B - 3 = -7$

$\therefore B = -4$

(e) $y_1'' = y_1 + 3y_2$ — (i) $y_2'' = 4y_1 - 4e^t$ — (ii) $y_1(0) = 2, y_1'(0) = 3, y_2(0) = 1, y_2'(0)$

(f) $y_1' = y_1$ — (i) $y_2' = 4$

(g) $y_1' = 2y_1 - 4y_2$ — (i) $y_2' = y_1 - 3y_2$ — (ii) $y_1(0) = 3 ; y_2(0) = 0$

$\mathcal{L}\{y_1'\} = \mathcal{L}\{2y_1 - 4y_2\}$

$\Rightarrow sY_1(s) - y_1(0) = 2Y_1(s) - 4Y_2(s)$

$\Rightarrow sY_1(s) - 2Y_1(s) + 4Y_2(s) = 3$

$\Rightarrow Y_1(s)(s-2) + 4Y_2(s) = 3$ — (iii)

$\mathcal{L}\{y_2'\} = \mathcal{L}\{y_1 - 3y_2\}$

$\Rightarrow sY_2(s) - y_2(0) = Y_1(s) - 3Y_2(s)$

$\Rightarrow sY_2(s) + 3Y_2(s) - Y_1(s) = 0$

$\Rightarrow Y_2(s)(s+3) - Y_1(s) = 0$ — (iv)

$$Y_1(s) = \frac{\begin{vmatrix} 3 & 4 \\ 0 & s+3 \end{vmatrix}}{\begin{vmatrix} s-2 & 4 \\ -1 & s+3 \end{vmatrix}} = \frac{3s+9-0}{s^2+3-6+4} = \frac{3s+9}{s^2+s-2} = \frac{3s+9}{s^2+2s-s-2} = \frac{3s+9}{(s-1)(s+2)}$$

$$\mathcal{L}^{-1}\{Y_1(s)\} = \mathcal{L}^{-1}\left\{\frac{A}{s-1} + \frac{B}{s+2}\right\}$$

$\Rightarrow y_1(t) = \mathcal{L}^{-1}\left\{\frac{4}{s-1} + \frac{-1}{s+2}\right\}$

$= 4e^t - e^{-2t}$

$3s+9 = A(s+2) + B(s-1)$

$\Rightarrow 3s+9 = As + 2A + Bs - B$

$\Rightarrow 3s+9 = (A+B)s + (2A-B)$

$A+B = 3$

$\therefore B = 3 - A$

$= 3 - 4$

$= -1$

$2A - B = 9$

$2A - (3-A) = 9$

$\Rightarrow 2A - 3 + A = 9$

$\Rightarrow 3A - 3 = 9$

$\Rightarrow A = 4$

$$Y_2(s) = \frac{\begin{vmatrix} s+2 & 3 \\ -1 & 0 \end{vmatrix}}{\begin{vmatrix} s+2 & 4 \\ -1 & s+3 \end{vmatrix}} = \frac{3}{s^2+s-2} = \frac{3}{(s-1)(s+2)}$$

$$\mathcal{L}^{-1}\{Y_2(s)\} = \mathcal{L}^{-1}\left\{\frac{A}{s-1} + \frac{B}{s+2}\right\}$$

$= e^t - e^{-2t}$

$3 = A(s+2) + B(s-1)$

$\Rightarrow 3 = As + 2A + Bs - B$

$= (A+B)s + (2A-B)$

$2A - B = 3$

$\Rightarrow 2A - (1-A) = 3$

$\Rightarrow 2A - 1 + A = 3$

$\Rightarrow 3A = 4$

$\therefore A =$

$A + B = 1$

$\therefore B = 1 - A$

2. (h) $y_1' = 2y_1 + 4y_2$ — (i), $y_2' = y_1 + 2y_2$ — (ii); $y_1(0) = -4,\ y_2(0) = -4$

$\mathcal{L}\{y_1'\} = \mathcal{L}\{2y_1 + 4y_2\}$.

$\Rightarrow sY_1(s) - y_1(0) = 2Y_1(s) + 4Y_2(s)$

$\Rightarrow sY_1(s) + 4 = 2Y_1(s) + 4Y_2(s)$

$\Rightarrow sY_1(s) - 2Y_1(s) - 4Y_2(s) = -4$

$\therefore Y_1(s)(s-2) - 4Y_2(s) = -4$ — (iii)

$\mathcal{L}\{y_2'\} = \mathcal{L}\{y_1 + 2y_2\}$

$\Rightarrow sY_2(s) - y_2(0) = Y_1(s) + 2Y_2(s)$

$\Rightarrow sY_2(s) + 4 = Y_1(s) + 2Y_2(s)$

$\Rightarrow Y_2(s)(s-2) - Y_1(s) = -4$ — (iv)

Using Cramer's rule,

$$Y_1(s) = \frac{\begin{vmatrix} -4 & -4 \\ -4 & (s-2) \end{vmatrix}}{\begin{vmatrix} (s-2) & -4 \\ -1 & (s-2) \end{vmatrix}} = \frac{-4s+8-16}{s^2-2s-2s+4-4} = \frac{-4s-8}{s^2-4s} = \frac{-4s-8}{s(s-4)}$$

$\mathcal{L}^{-1}\{Y_1(s)\} = \mathcal{L}^{-1}\left\{\frac{A}{s} + \frac{B}{s-4}\right\}$

$\Rightarrow y_1(t) = \mathcal{L}^{-1}\left\{\frac{2}{s} + \frac{-6}{s-4}\right\}$

$= -6e^{4t} + 2$

$-4s-8 = A(s-4) + Bs$

$\Rightarrow -4s-8 = As - 4A + Bs$

$\Rightarrow -4s-8 = (A+B)s - 4A$

$A + B = -4$ | $-4A = -8$

$\therefore B = -4 - A$ | $\therefore A = 2$

$= -4 - 2$

$= -6$

$$Y_2(s) = \frac{\begin{vmatrix} s-2 & -4 \\ -1 & -4 \end{vmatrix}}{\begin{vmatrix} (s-2) & -4 \\ -1 & (s-2) \end{vmatrix}} = \frac{-4s+8-4}{s^2-4s} = \frac{-4s+4}{s(s-4)}$$

$\mathcal{L}^{-1}\{Y_2(s)\} = \mathcal{L}^{-1}\left\{\frac{A}{s} + \frac{B}{s-4}\right\}$

$\Rightarrow y_2(t) = \mathcal{L}^{-1}\left\{\frac{-1}{s} + \frac{-3}{s-4}\right\}$

$= -3e^{4t} - 1$

$-4s+4 = A(s-4) + Bs$

$\Rightarrow -4s+4 = As - 4A + Bs$

$\Rightarrow -4s+4 = (A+B)s - 4A$

$A + B = -4$ | $-4A = 4$

$\therefore B = -4 - A$ | $\therefore A = -1$

$= -4 - (-1)$

$= -3$